REDUCTION OF THE WAGSTAFF CHEBYSHEV SUFFICIENCY CONJECTURE TO THE PELL PRIMITIVE PAIR CONJECTURE

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Abstract. Let $W_p = (2^p + 1)/3$ for an odd prime $p \geq 5$, and let $\omega_3 = 3 + 2\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$. The congruence $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$ — the $a = 3$ case of the Chua $N + 1$ test — is known to hold for every prime W_p; whether it suffices to *imply* primality of W_p is an open question, referred to here as the *Wagstaff Chebyshev sufficiency conjecture* (Conjecture 1.1) — the Chua-form counterpart of the Vrba–Reix primality test for Wagstaff numbers [20]. We reduce this conjecture to a quantitative residual: a principal pair-counting conjecture, an auxiliary conjecture on a separate branch, and a third input that is mild and already discharged in the entire computational range.

Writing r for a prime divisor of a hypothetical composite W_p passing Condition (II), we split the analysis by the residue class $r \bmod 8$ (the only two possibilities being $r \equiv 1$ and $r \equiv 3$). At split factors $r \equiv 1 \pmod 8$ we reduce to the *nonexistence of compatible triples* (NCT), proved unconditionally for every odd $d \leq 200$ (Theorem 4.18 and complete primitive-part certificates) and at every parity-unblocked value in $(200, 400]$ — hence for every odd $d \leq 400$. At inert factors $r \equiv 3 \pmod 8$ we prove a sequence of unconditional cross-case reductions — the Order-Pinning Lemma, the Multi-Factor Pinning Theorem, and the Second-Moment Reduction — together with one computational input, the Platinum Lemma (a 684,965,381-row enumeration showing that each p has at most one *small* local-pass inert factor $r \leq 10^{12}$); these collapse the inert branch to a single pair-counting claim above the cutoff, modulo the stated Platinum enumeration.

The principal residual is named the **Pell Primitive Pair Conjecture (PPPC)**:

> For every prime $p \geq 5$, at most one prime r exists satisfying all of $r \equiv 3 \pmod 8$, $\mathrm{ord}_r(2) = 2p$, $r > 10^{12}$, $d_r := \mathrm{ord}_r(\omega_3)/4 > 43$, and $d_r \mid W_{p-2}$.

Each such r is a primitive divisor of the Pell–Lucas number V_{2d_r} and lies in the arithmetic progression $r \equiv 1 \pmod{2p}$; PPPC asserts that two such primitive divisors cannot simultaneously pair at a single Wagstaff exponent. The Second-Moment Reduction proves $E_3 \leq \Pi + W$, where $\Pi = 0$ is PPPC and W counts the composite all-inert Condition-(II)-passing W_p that are *not squarefree*; PPPC together with NCT for general d and the third input $W = 0$ implies the full conjecture. This

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is a conditional reduction, not an unconditional proof. The Wieferich–Wagstaff term $W = 0$ holds unconditionally for every W_p all of whose prime factors lie below 2^{64} (a repeated factor would be a base-2 Wieferich prime $\equiv 3 \pmod 8$), of which there are none in that range) and is the mildest input; the 684,965,381-row Platinum enumeration ($r \le 10^{12}$) is finite-range evidence for PPPC, not a proof of it. Two unconditional constraints on the forbidden configuration are also proved: a pair-separation theorem, and the vanishing of the diagonal pair count for every admissible $d \le 400$ — the latter via a complete census of the danger triples with $d \le 400$ (exactly two real ones exist, both at exponents closed by secondary factors).

Sharper forms of the chain are also established (§7.7, §7.8): the split-branch residual weakens to a *Mixed Pair Conjecture* implied by NCT for general d; the term W decomposes into an explicit super-Wieferich bucket — closed unconditionally via the Nagell–Ljunggren equation at $x = -2$ — and a single-exponent bucket; and a triple-count erosion (PPPC3) reduces the role of PPPC from a pair count to a triple count.

This paper is a companion to [22], which gives three unconditional ECPP-independent primality certificates (Brillhart–Lehmer–Selfridge with a Chua $N + 1$ side) for $W_{2617}, W_{10501}, W_{12391}$.

1. Introduction

1.1. The Wagstaff Chebyshev sufficiency conjecture. For an odd prime p, the Wagstaff number is $W_p = (2^p+1)/3$. The $+1$ companion of Mersenne, W_p lacks a standalone primality criterion of Lucas–Lehmer's strength [21]. The most natural candidate is the base $a = 3$ case of the Chua framework [8]: set $\omega_3 = 3 + 2\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$ and ask whether the single congruence

$$\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p} \qquad \text{(Condition (II))}$$

implies primality of W_p. One half of this is straightforward: if W_p is prime, then Condition (II) holds (Proposition 2.2 below, a direct consequence of Chua's framework [8]). The converse is open.

Conjecture 1.1 (Wagstaff Chebyshev sufficiency conjecture)**.** No composite W_p satisfies Condition (II).

The conjecture restates an open problem of some standing. Its recurrence form is the Chua iteration $S_{i+1} = S_i^2 - 2 \pmod{W_p}$ in the base ω_3, whose return test encodes Condition (II). It is closely related to — but, as literally stated, distinct from — the *Vrba–Reix primality test* for Wagstaff numbers [20]: the latter iterates the same Chebyshev map started at $S_0 = 1/4 \pmod{2^p + 1}$, and this iteration lives in $\mathbb{Q}(\sqrt{-7})$ rather than $\mathbb{Q}(\sqrt{2})$ — its base is $\theta = (1 + 3\sqrt{-7})/8 = (\bar{\pi}/\pi)^2$, where $2 = \pi\bar{\pi}$ splits in $\mathbb{Q}(\sqrt{-7})$. The two return conditions decouple at the level of individual prime factors, in both directions, and the auxiliary no-early-return hypothesis of the Vrba–Reix prize statement holds vacuously; the exact relation is worked out in the companion note [23]. The forward direction (prime $\Rightarrow$ the test passes) is classical for both forms; sufficiency is open for both, has been since 2008, and for the Vrba–Reix form carries a standing 500-euro reward for a published proof [20]. Both tests are implemented in standard software, agree with Condition (II) empirically at every tested exponent, and Conjecture 1.1 is the sufficiency question for the Chua/Condition-(II) form — the canonical order-forcing statement that this paper reduces.

Every known fully resolved composite W_p tested in §B.4 is consistent with Conjecture 1.1, and no global composite pass is known. The current record/probable-prime status for Wagstaff numbers is maintained externally [17] and is not used as an input to the reduction. The purpose of this paper is to reduce Conjecture 1.1 to a quantitative residual whose principal component is named the *Pell Primitive Pair Conjecture* (§1.3 and §7); the reduction carries two further inputs — NCT for general d on the split branch, and a mild Wieferich-type exclusion $W = 0$ that is already unconditionally discharged for all factors below 2^{64} [16]. Thus the main theorem is a conditional reduction theorem; the unconditional sufficiency conjecture remains open.

1.2. **Reduction chain (summary).** The reduction proceeds in three stages. Throughout, r denotes a prime divisor of a hypothetical composite W_p passing Condition (II).

Stage A (case analysis). The mod-8 exclusion (Lemma 2.4) forces $r \equiv 1$ or $r \equiv 3 \pmod 8$. Each class is treated separately.

- *Split case* $r \equiv 1 \pmod 8$. A descent to the split algebra $\mathbb{F}_r \times \mathbb{F}_r$ (Section 4) identifies an odd parameter $d \mid W_{p-2}$ with $\operatorname{ord}(\omega_3) = 4d$. If Condition (II) holds at r, Proposition 4.14 produces a compatible triple (d, r, p); the **nonexistence of compatible triples** (NCT) rules out this situation. NCT is proved unconditionally for every odd $d \le 200$: for $d \le 99$ by Theorem 4.18, and at all nine parity-unblocked values $d = 107$, 121, 129, 131, 139, 163, 171, 177, 179 of $100 \le d \le 200$ by complete fixed-d certificates (§B.3); the same certificate closes all ten parity-unblocked values of $(200, 300]$ ($d = 201$, 209, 211, 227, 243, 249, 251, 281, 283, 297) and all nine of $(300, 400]$ ($d = 307$, 321, 331, 347, 361, 363, 379, 387, 393) (§B.3); NCT therefore holds unconditionally for every odd $d \le 400$. The related Pell-rank divisibility Conjecture R3 is a stronger sufficient split-branch obstruction and holds at all 185 known split factors (§B.2).
- *Inert case* $r \equiv 3 \pmod 8$. A descent in the inert algebra $\mathbb{F}_{r^2}$ (Section 5) gives an odd parameter $g = \operatorname{ord}(\omega_3)/4$ dividing $G_r = \gcd((r+1)/4, W_{p-2})$; Condition (II) at r holds iff $g \mid W_{p-2}$. The **Pell-sequence exclusion** (Theorem 5.10) excludes $g \le 43$ unconditionally; the **parity obstruction** (Theorem 5.3) blocks every g with a non-Class III prime factor; together these close the inert branch at 845 of 869 known inert factors (§B.4).

Stage B (cross-case pinning, Section 6). Uniformly across all candidate composite W_p:

(1) **Order-Pinning Lemma** (Theorem 6.1): every prime factor $r > 3$ of W_p has $\mathrm{ord}_r(2) = 2p$ *exactly*.
(2) **Multi-Factor Pinning** (Theorem 6.2): all prime factors $r > 3$ of W_p share the pinned order parameter.
(3) **Platinum Lemma** (Theorem 6.6): a complete enumeration of inert prime factors $r \le 10^{12}$ shows that every prime p has at most one such factor passing Condition (II).
(4) **Exact-AP characterization** (Theorem 6.7): for a prime $q \in A$ with $q > 3$, the divisibility $q \mid W_{p-2}$ holds iff $p \equiv m_q + 2 \pmod{2m_q}$, where $m_q = \mathrm{ord}_q(2)/2$ and A is the set of Class III primes. Prime powers require the valuation-adjusted LTE modulus recorded after the theorem; the prime $q = 3$ is governed by Theorem 5.4.
(5) **Second-Moment Reduction** (Theorem 6.15): the cardinality E_3 of p-values contributing to the all-inert branch of Conjecture 1.1 satisfies $E_3 \le \Pi + W$, where $\Pi = \sum_p \binom{|T_p^>|}{2}$, $T_p^>$ is the set of inert factors $r > 10^{12}$ with $d_r > 43$ and $d_r \mid W_{p-2}$ at that p, and W counts the composite all-inert Condition-(II)-passing W_p that are *not squarefree*. This step takes the Platinum enumeration (item 3) as a computational input; a computational-input-free variant of the bound, with a triple-count residual in place of the pair-count Π, is recorded in Remark 6.16.

Stage C (residual). The right-hand side of the Second-Moment Reduction splits into two residuals. The pair-count Π is the *Pell Primitive Pair Conjecture* (PPPC, Conjecture 1.2 below), the principal residual for the all-inert branch. The term W is the **(E3) Wieferich–Wagstaff exclusion**: a composite all-inert Condition-(II)-passing W_p with a repeated prime factor r would make r a base-2 Wieferich prime $\equiv 3 \pmod 8$ (Remark 6.17); the only base-2 Wieferich primes below 2^{64} are 1093 and 3511, congruent to 5 and 7 (mod 8) respectively, so the exhaustive search below 2^{64} excludes every base-2 Wieferich prime $\equiv 3 \pmod 8$ in that range [16]. Therefore the condition $W = 0$ holds *unconditionally* for all W_p with prime factors below 2^{64} and is conjectured to hold always — by far the mildest of the three residuals. Conjecture 1.1 follows from PPPC *together* with NCT for general d *and* the Wieferich–Wagstaff exclusion $W = 0$ (Theorem 1.3); the first two gaps are structurally and technically independent of each other, and $W = 0$ is the mild, mostly-discharged third. Section 7.7 additionally proves a sharper form of the reduction: the split-branch residual weakens to the pair-type *Mixed Pair Conjecture* (MPC, §4.7), implied by NCT for general d, and the W-residual decomposes into a super-Wieferich bucket W_1 — now closed unconditionally via the Nagell–Ljunggren equation (Lemma 6.26) — and a witness-exponent bucket W_2 pinned to one explicit exponent (Theorem 6.24, Theorem 7.4); the sharpened chain reads PPPC + MPC$^>$ + X1.

A triple form of the sharpened chain is also established (Corollary 7.9, §7.8): Conjecture 1.1 follows from MPC$^{>}$, the triple-count conjecture PPPC3 (Conjecture 7.6, implied by PPPC), the corner exclusion $W_2' = 0$ (Conjecture 7.7, implied by either PPPC or $W = 0$), and X1 — eroding the role of PPPC from the pair count Π to the triple count Π_3.

Table 1 summarises the algebraic and certified inputs, finite-range evidence, and remaining open part of each residual.

Residual	Algebraic / certified input	Finite-range evidence	Remaining gap
NCT, general d (split, $r \equiv 1$)	no compatible triple for any odd $d \leq 200$, nor at any parity-unblocked value of $(200, 400]$ (Thm 4.18 + twenty-eight complete fixed-d closures, §B.3)	all 185 known split factors satisfy R3	general odd $d > 400$
PPPC, $\Pi = 0$ (inert, $r \equiv 3$)	Platinum small-factor bound (Thm 6.6, certified input); pair separation (Thm 6.10); diagonal $S_{\mathrm{diag}}(d) = 0$ for admissible $d \leq 400$ (Props A.9, 5.18); inert floor $d_r > 400$ (Cor 5.19)	at most one local-pass inert factor $r \leq 10^{12}$ per p; no above-cutoff pair found in the p-window scan	general p, off-diagonal and diagonal $d > 400$
$W = 0$ (Wieferich–Wagstaff)	$W = 0$ for all W_p with prime factors $< 2^{64}$	no base-2 Wieferich prime $\equiv 3 \pmod 8$ below 2^{64}	conjectural for all p (mildest)

TABLE 1. State of the three residuals of Theorem 1.3. Conjecture 1.1 follows from all three. NCT-general and PPPC are the two principal gaps; $W = 0$ is the mild third, discharged unconditionally throughout the computational range. The sharpened residuals MPC$^{>}$, X1 of Theorem 7.4 are implied by these three as a conjunction (§7.6); the former fourth residual $W_1 = 0$ is closed unconditionally (Lemma 6.26).

1.3. **The Pell Primitive Pair Conjecture.**

Conjecture 1.2 (Pell Primitive Pair Conjecture, PPPC)**.** For every prime $p \geq 5$, at most one prime r exists satisfying all of:

$$r \equiv 3 \pmod 8; \qquad \mathrm{ord}_r(2) = 2p; \qquad r > 10^{12};$$

$$d_r := \mathrm{ord}_r(\omega_3)/4 > 43; \qquad d_r \mid W_{p-2}.$$

Equivalently: $|T_p^>| \le 1$ for every prime $p \ge 5$, where $T_p^>$ denotes the set of primes r satisfying the five bullet conditions.

PPPC is genuinely Pell-theoretic. By Theorem 5.5 and Lemma 5.8, any such r is a primitive divisor of the Pell–Lucas number V_{2d_r}, where $V_n = (1+\sqrt{2})^n + (1-\sqrt{2})^n$; by Order-Pinning it simultaneously lies in the arithmetic progression $r \equiv 1 \pmod{2p}$; by the Exact-AP characterization and its valuation-adjusted prime-power extension, the condition $d_r \mid W_{p-2}$ is a strict congruence condition on p. PPPC asserts that two such primitive divisors — not necessarily attached to different values of d_r — cannot simultaneously be pinned to the same Wagstaff exponent p.

The name reflects three properties of the forbidden configuration:

- **Pell**: r divides V_{2d}, the Pell–Lucas sequence associated to the fundamental unit $\alpha = 1+\sqrt{2}$ of $\mathbb{Z}[\sqrt{2}]$.
- **Primitive**: r is primitive for the d-indexed Pell–Lucas family, i.e., it divides V_{2d_r} but not V_{2d} for any $d < d_r$ (Lemma 5.8); Bilu–Hanrot–Voutier [3] is used only as general background on primitive divisors of Lucas sequences.
- **Pair**: the conjecture forbids the *coexistence* of two such primitive divisors at a single p; a single such divisor per p is entirely compatible with PPPC and occurs in three identified “local pass” witnesses (§B.4).

The Platinum enumeration of §B.4 verifies the small-factor input used by the reduction: at most one local-pass inert factor $r \le 10^{12}$ occurs at any pinned exponent p in the enumerated range. PPPC itself concerns the above-cutoff set $T_p^>$, and its unconditional resolution is the main open problem of this paper. Two unconditional constraints on the forbidden configuration are established below: the Pair Separation Theorem (§6.6) forces $8p\gcd(d_{r_1}, d_{r_2}) \mid r_2 - r_1$ for any two members of $T_p^>$, and the diagonal $(d_{r_1} = d_{r_2})$ contribution to the pair count vanishes for every admissible order parameter $d \le 400$ (Propositions A.9 and 5.18), in particular for every pair with both indices $(r_i - 1)/2p \le 1604$ (Corollary A.10).

Theorem 1.3 (Main reduction). *Assume PPPC (Conjecture 1.2), the general Nonexistence of Compatible Triples (Conjecture 4.20 below), and the Wieferich–Wagstaff exclusion $W = 0$ — equivalently, no composite W_p whose prime factors are all inert and all pass Condition (II) has a repeated prime factor. Then Conjecture 1.1 holds. The reduction additionally takes the Platinum enumeration (Theorem 6.6) as a stated computational input.*

The proof is Theorem 7.2. The third hypothesis $W = 0$ is the (E3) Wieferich–Wagstaff exclusion (§1.2, §7.6, §A.2): it holds *unconditionally* for every W_p all of whose prime factors lie below 2^{64} — a repeated prime factor would be a base-2 Wieferich prime $\equiv 3 \pmod 8$, and the exhaustive search below 2^{64} finds only $1093 \equiv 5 \pmod 8$ and $3511 \equiv 7 \pmod 8$ [16] — and is conjectured to hold for all p; it is the mildest of the three inputs. Theorem 7.3 proves the same conclusion from the formally weaker split-branch hypothesis

MPC (Conjecture 4.22), and Theorem 7.4 refines the residuals further to PPPC + MPC$^{>}$ + X1, a conjunction implied by the three hypotheses of Theorem 1.3.

Unconditional contributions. Independently of the conditional reduction, the following are proved with no conjectural input and constitute the technical core of the paper: the nonexistence of compatible triples for every odd $d \le 200$ and at every parity-unblocked value in $(200, 400]$ (Theorem 4.18 and complete fixed-d certificates, §B.3); the Order-Pinning Lemma, the Multi-Factor Pinning Theorem, and the Second-Moment Reduction (Section 6); the Pair Separation Theorem and the vanishing of the diagonal pair count for every admissible $d \le 400$ (§6.6, Proposition 5.18); the complete danger census through $d = 400$ with the inert order-parameter floor $d_r > 400$ (Proposition 5.18, Corollary 5.19); the quartic residuosity of 2 at primitive Pell divisors (Theorem 4.17); and the Exact-AP characterization of the divisibility $q \mid W_{p-2}$ (Theorem 6.7).

1.4. Relation to the companion paper.

- [22] Three unconditional Brillhart–Lehmer–Selfridge [4] primality proofs for $W_{2617}, W_{10501}, W_{12391}$, independent of ECPP [1]. Uses the same Chua $N + 1$ criterion (Proposition 2.2 here, Proposition 2.4 of [22]) on the necessity side, but is otherwise disjoint from this paper: it does not use, and does not prove, Conjecture 1.1.

This paper does not depend on [22]; the two may be read independently.

1.5. **Notation.** Throughout, $p \ge 5$ denotes an odd prime and $W_p = (2^p + 1)/3$; r denotes a prime divisor of W_p. For x coprime to m, $\mathrm{ord}_m(x)$ is the multiplicative order modulo m; v_q is the q-adic valuation. We work in $\mathbb{Z}[\sqrt{2}]$ with fundamental unit $\alpha = 1 + \sqrt{2}$ and Chebyshev base $\omega_3 = \alpha^2 = 3 + 2\sqrt{2}$. The Pell sequences $(U_n), (V_n)$ satisfy $\alpha^n = V_n/2 + U_n\sqrt{2}$ with the recurrence $X_{n+1} = 2X_n + X_{n-1}$, $U_0 = 0$, $U_1 = 1$, $V_0 = V_1 = 2$; the identity $V_n^2 - 8U_n^2 = 4(-1)^n$ recurs throughout.

For a prime $r \ne 2$ with $(2/r) = 1$, $\rho(r)$ is the Pell rank of apparition: the least $n \ge 1$ with $r \mid U_n$. For an inert prime r with $r \mid W_p$ we write

$$G_r \;=\; \gcd\!\left(\frac{r+1}{4},\; W_{p-2}\right), \qquad d_r \;=\; g \;:=\; \mathrm{ord}_r(\omega_3)/4$$

for the order parameter; we use d_r when emphasising dependence on r and g in local proofs. $\Phi_d(x)$ is the d-th cyclotomic polynomial; $A = \{q \text{ odd prime} : v_2(\mathrm{ord}_q(2)) = 1\}$ is the set of *Class III* primes (Definition 5.2).

2. Preliminaries

2.1. **Chua framework and the base $a = 3$.** For an integer $a \ne 0, \pm 1$, set $\omega_a = a + \sqrt{a^2 - 1}$ so that $\omega_a \bar{\omega}_a = 1$ and $\omega_a^n + \omega_a^{-n} = 2T_n(a)$. The Chua identity is:

Theorem 2.1 (Chua [8]). *Let Q be an odd prime with* $\gcd(a^2-1, Q) = 1$. *With $D = a^2 - 1$, $\varepsilon = (D/Q)$, $\delta = (2(a+1)/Q)$:*

$$\omega_a^{(Q-\varepsilon)/2} \equiv \delta \pmod{Q}$$

in $\mathbb{Z}[\sqrt{D}]/(Q)$.

For $a = 3$: $D = 8$, $2(a+1) = 8$, so the two Legendre symbols coincide and become $(8/Q) = (2/Q)^3 = (2/Q)$. This gives the $a = 3$ specialization:

Proposition 2.2 (Base $a = 3$ necessity). *For every Wagstaff prime W_p with $p \geq 5$: $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$.*

Proof. $W_p \equiv 3 \pmod 8$ (Lemma 2.4 below) gives $(2/W_p) = -1$ by the second supplement, hence $\varepsilon = \delta = (8/W_p) = -1$; apply Theorem 2.1. $\square$

The base $a = 3$ is uniquely distinguished among integer Chebyshev bases by a^2-1 being a power of 2; $(a-1)(a+1) = 2^k$ forces $\{a-1, a+1\} = \{2, 4\}$ (Mihailescu [14]). This is the source of the symmetric role played by ω_3 in everything that follows.

2.2. The order of 2 and the mod-8 dichotomy.

Lemma 2.3 (Order of 2 at every factor). *For every prime $p \geq 5$ and every prime $r \mid W_p$: $\mathrm{ord}_r(2) = 2p$, $r \equiv 1 \pmod{2p}$.*

Proof. $2^p \equiv -1 \pmod r$, so $\mathrm{ord}_r(2) \mid 2p$ but $\nmid p$. Since p is prime, $\mathrm{ord}_r(2) \in \{2, 2p\}$. If $\mathrm{ord}_r(2) = 2$, then $r \mid 3$. But $3 \nmid W_p$ for prime $p \geq 5$: otherwise $9 \mid 2^p + 1$, while $\mathrm{ord}_9(2) = 6$ forces $p \equiv 3 \pmod 6$, impossible. Hence $\mathrm{ord}_r(2) = 2p$, and $r \equiv 1 \pmod{2p}$. $\square$

Lemma 2.4 (mod-8 classification). *Let $p \geq 5$ be prime. Then $W_p \equiv 3 \pmod 8$; every prime factor of W_p is $\equiv 1$ or $3 \pmod 8$; and the total multiplicity of prime factors $\equiv 3 \pmod 8$ is odd.*

Proof. $W_p \equiv 3 \pmod 8$: for $p \geq 5$, $2^p \equiv 0 \pmod{32}$, so $2^p + 1 \equiv 1 \pmod{32}$, i.e. $3W_p \equiv 1 \pmod{32}$; since $3 \cdot 11 \equiv 1 \pmod{32}$, this gives $W_p \equiv 11 \pmod{32}$, hence $W_p \equiv 3 \pmod 8$ (consistent with Theorem 2.6). Factor classes: By Lemma 2.3, $v_2(\mathrm{ord}_r(2)) = 1$. Suppose $r \equiv 7 \pmod 8$: then $(2/r) = +1$ by the second supplement, so 2 is a quadratic residue and $\mathrm{ord}_r(2) \mid (r-1)/2$; since $v_2(r-1) = 1$ here, $(r-1)/2$ is odd, so $\mathrm{ord}_r(2)$ is odd and $v_2(\mathrm{ord}_r(2)) = 0 \neq 1$. Suppose $r \equiv 5 \pmod 8$: then $(2/r) = -1$, so $2^{(r-1)/2} \equiv -1 \pmod r$ and $\mathrm{ord}_r(2) \nmid (r-1)/2$, forcing $v_2(\mathrm{ord}_r(2)) > v_2((r-1)/2) = v_2(r-1) - 1 = 1$; with $\mathrm{ord}_r(2) \mid r-1$ this gives $v_2(\mathrm{ord}_r(2)) = v_2(r-1) = 2 \neq 1$. Both classes are excluded. Parity: modulo 8 the product $W_p \equiv 3^{\sum'} \pmod 8$ where $\sum'$ counts factors $\equiv 3 \pmod 8$ with multiplicity, and $W_p \equiv 3 \pmod 8$ forces this sum odd. $\square$

Corollary 2.5. *No composite W_p passing Condition (II) has a prime factor $r \equiv 5$ or $r \equiv 7 \pmod 8$; every prime factor is split ($r \equiv 1 \pmod 8$) or inert ($r \equiv 3 \pmod 8$).*

2.3. **Refined mod-16 parity.**

Theorem 2.6 (Refined SFP mod 16). *For every odd prime $p \geq 5$:*

(a) $W_p \equiv 11 \pmod{16}$.

(b) Every prime factor $r > 3$ of W_p with $r \equiv 3 \pmod 8$ satisfies $r \equiv 3$ or $r \equiv 11 \pmod{16}$.

(c) Assume Conjecture 4.6, and suppose W_p is composite and passes Condition (II) at every factor. Then by Theorem 4.8 below every factor is inert; writing t for the total number of inert factors (with multiplicity) and k_{11} for the number of those $\equiv 11 \pmod{16}$:

$$t + 2k_{11} \;\equiv\; 3 \pmod 4.$$

Proof. (a) For odd prime $p \geq 5$: $3W_p = 2^p + 1 \equiv 1 \pmod{32}$ (since $p \geq 5$ forces $2^p \equiv 0 \pmod{32}$). The inverse of 3 modulo 32 is 11 (as $3 \cdot 11 = 33 \equiv 1 \pmod{32}$), so $W_p \equiv 11 \pmod{32}$, hence $W_p \equiv 11 \pmod{16}$.

(b) By Lemma 2.4, $r \equiv 1$ or $3 \pmod 8$; reduce mod 16: $r \equiv 3 \pmod 8$ splits into 3 or $11 \pmod{16}$.

(c) Under the hypotheses every factor is $\equiv 3 \pmod 8$; so $r_i \in \{3, 11\} \pmod{16}$. In $(\mathbb{Z}/16)^\times$, $\operatorname{ord}(3) = 4$ with $\{3^0, 3^1, 3^2, 3^3\} = \{1, 3, 9, 11\}$; so $11 \equiv 3^3 \pmod{16}$. Thus $\prod r_i \equiv 3^{k_3 + 3k_{11}} \pmod{16}$. Using (a) and $11 \equiv 3^3$: $k_3 + 3k_{11} \equiv 3 \pmod 4$; substitute $k_3 = t - k_{11}$. ☐

Corollary 2.7 (Mod-16 sub-configurations for $t = 3$). *In the setting of Theorem 2.6(c), if $t = 3$ then $k_{11} \in \{0, 2\}$.*

Remark. Theorem 2.6 and Corollary 2.7 are structural observations on the mod-16 refinement of the all-inert configuration; they are not invoked in the reduction chain of §§4–7, and a reader interested only in the main result may skip them. They are recorded because the parity datum $t + 2k_{11} \equiv 3 \pmod 4$ constrains any future enumeration of small-t inert configurations.

2.4. **The v_2 classification of $\operatorname{ord}_r(\omega_3)$.**

Lemma 2.8 (v_2 classification). *Let $r \geq 5$ be prime. Then*

$$v_2(\operatorname{ord}_r(\omega_3)) \;=\; \begin{cases} 2 & r \equiv 3 \pmod 8, \\ 1 & r \equiv 5 \pmod 8, \\ 0 & r \equiv 7 \pmod 8, \\ \leq v_2(r-1) - 1 & r \equiv 1 \pmod 8. \end{cases}$$

Proof. Set $\alpha = 1 + \sqrt{2}$, so $\omega_3 = \alpha^2$ and $v_2(\operatorname{ord}(\omega_3)) = \max(v_2(\operatorname{ord}(\alpha)) - 1, 0)$.

Split cases ($r \equiv 1, 7 \pmod 8$): $(2/r) = 1$, so $\sqrt{2} \in \mathbb{F}_r$ and $\alpha \in \mathbb{F}_r^*$ with $\operatorname{ord}(\alpha) \mid r - 1$, so $v_2(\operatorname{ord}(\alpha)) \leq v_2(r-1)$. For $r \equiv 7$: $v_2(r-1) = 1$, giving $v_2(\operatorname{ord}(\omega_3)) \leq 0$. For $r \equiv 1$: $v_2(r-1) \geq 3$, giving the bound.

Inert cases ($r \equiv 3, 5 \pmod 8$): $\sqrt{2} \notin \mathbb{F}_r$, $\alpha \in \mathbb{F}_{r^2}^*$. Frobenius sends $\sqrt{2} \mapsto -\sqrt{2}$, so $\alpha^r = \bar{\alpha}$, hence $\alpha^{r+1} = \alpha\bar{\alpha} = -1$. Thus $\operatorname{ord}(\alpha) \mid 2(r+1)$ but $\nmid (r+1)$, which forces $v_2(\operatorname{ord}(\alpha)) = v_2(r+1) + 1$. For $r \equiv 3$: $v_2(r+1) = 2$, so $v_2(\operatorname{ord}(\omega_3)) = 2$; for $r \equiv 5$: $v_2(r+1) = 1$, so $v_2(\operatorname{ord}(\omega_3)) = 1$. ☐

The exact value $v_2(\mathrm{ord}(\omega_3)) = 2$ at $r \equiv 3 \pmod 8$ is used in §5.1 to establish $\mathrm{ord}(\omega_3) = 4g$ with g odd; the upper bound at $r \equiv 1$ is used in §4.

2.5. **The fundamental identity and coprimality.**

Lemma 2.9. *For every $p \geq 5$: $W_p + 1 = 4\,W_{p-2}$, and $\gcd(W_p, W_{p-2}) = 1$.*

Proof. $W_p + 1 = (2^p + 4)/3 = 4(2^{p-2} + 1)/3 = 4W_{p-2}$. For coprimality: $W_p - 4W_{p-2} = -1$. □

Lemma 2.10 ($p \nmid W_{p-2}$)**.** *$p \nmid W_{p-2}$ for every prime $p \geq 5$.*

Proof. $4W_{p-2} = W_p + 1 = (2^p + 4)/3 \equiv (2+4)/3 = 2 \pmod p$ by Fermat. □

These two lemmas are used repeatedly: Lemma 2.9 converts Condition (II) into a divisibility statement on W_{p-2}, and Lemma 2.10 prevents certain orders from accommodating p in their W_{p-2}-divisor.

3. Condition (I) and the sufficiency conjecture

3.1. **Condition (I) at every factor.** *Condition (I)* is the companion Euler congruence on the $N - 1$ side,

$$2^{(W_p - 1)/2} \equiv -1 \pmod{W_p},$$

which together with Condition (II) constitutes the two-sided $(N-1)/(N+1)$ primality test. We show that Condition (I) holds automatically at every prime factor of W_p, so it is non-discriminating and all arithmetic content of the test rests in Condition (II).

Theorem 3.1 (Condition (I) at every factor)**.** *For every odd prime $p \geq 5$ and every prime $r \mid W_p$: $2^{(W_p-1)/2} \equiv -1 \pmod r$.*

Proof. By Lemma 2.3, $\mathrm{ord}_r(2) = 2p$. We show $(W_p - 1)/(2p)$ is odd: $(W_p - 1)/2 = (2^{p-1} - 1)/3$, and $(W_p - 1)/(2p) = (2^{p-1} - 1)/(3p)$ is an integer (by Fermat and p odd) and odd (denominator $3p$ odd; numerator $2^{p-1} - 1$ odd for $p \geq 2$). Therefore $2^{(W_p-1)/2} = (2^p)^{(W_p-1)/(2p)} \equiv (-1)^{\mathrm{odd}} = -1$. □

Corollary 3.2 (CRT consequence)**.** *If W_p is squarefree, then $2^{(W_p-1)/2} \equiv -1 \pmod{W_p}$, i.e., Condition (I) holds for W_p.*

Squarefreeness is not known for general p; every argument below uses only the per-prime form of Theorem 3.1. We record this consolidation:

Corollary 3.3. *Condition (I) is automatic at every prime factor of W_p and is therefore non-discriminating: the two-sided $(N-1)/(N+1)$ test reduces to Condition (II) alone, which carries all of the arithmetic content of the sufficiency question.*

3.2. **The two branches.** By Lemma 2.4, every prime factor of W_p is $\equiv 1$ or 3 (mod 8); by Corollary 2.5 these are the only residue classes the case analysis need consider. We call $r \equiv 1 \pmod 8$ the *split branch* (since $(2/r) = 1$ splits $\mathbb{Z}[\sqrt{2}]/(r)$ as $\mathbb{F}_r \times \mathbb{F}_r$), and $r \equiv 3 \pmod 8$ the *inert branch* (since $(2/r) = -1$ makes $\mathbb{Z}[\sqrt{2}]/(r) = \mathbb{F}_{r^2}$).

Sections 4 and 5 treat the two branches separately. Section 6 gives cross-case reductions that act uniformly on any composite W_p regardless of its factor structure.

4. The split branch: $r \equiv 1 \pmod 8$

4.1. **Descent in the split algebra.** For $r \equiv 1 \pmod 8$ with $r \mid W_p$: $(2/r) = 1$, so $\mathbb{Z}[\sqrt{2}]/(r) \cong \mathbb{F}_r \times \mathbb{F}_r$ via $\sqrt{2} \mapsto (s, -s)$ where $s^2 = 2$ in $\mathbb{F}_r$. The element ω_3 maps to $(\alpha_0, \alpha_0^{-1})$ with $\alpha_0 = 3 + 2s$.

Descent. Condition (II) at r becomes $\alpha_0^{(N+1)/2} = -1$ in $\mathbb{F}_r^*$. Therefore $\mathrm{ord}(\alpha_0) \mid N+1 = 4W_{p-2}$ and $v_2(\mathrm{ord}(\alpha_0)) = v_2(N+1) = 2$. Write $\mathrm{ord}(\alpha_0) = 4d$ with d odd and $d \mid W_{p-2}$.

Theorem 4.1 (Order of $\sqrt{2}$). *Let $r \equiv 1 \pmod 8$, $r \mid W_p$. Then $\mathrm{ord}_{\mathbb{F}_r^*}(s) = 4p$.*

Proof. $s^{2p} = 2^p \equiv -1 \pmod r$, so $\mathrm{ord}(s) \nmid 2p$; $s^{4p} = 1$, so $\mathrm{ord}(s) \mid 4p$. For prime p the only divisor not dividing $2p$ is 4 or $4p$; $\mathrm{ord}(s) = 4$ gives $4 \equiv 1 \pmod r$, impossible for $r \geq 5$. □

Proposition 4.2 (Self-supporting exponents). *Let $p \geq 5$ be prime and let W_{p-2} be prime. If W_p satisfies Condition (II), then W_p is prime.*

Proof. Suppose W_p is composite and let $r < W_p$ be a prime factor; Condition (II) restricts along $\mathbb{Z}[\sqrt{2}]/(W_p) \to \mathbb{Z}[\sqrt{2}]/(r)$, so it holds at r. If $r \equiv 1 \pmod 8$, the descent above gives $\mathrm{ord}(\alpha_0) = 4d$ with d odd and $d \mid W_{p-2}$; primality of W_{p-2} leaves $d \in \{1, W_{p-2}\}$. For $d = 1$: $\alpha_0^2 = -1$ forces $r \mid \mathrm{N}(\omega_3^2 + 1) = \mathrm{N}(18 + 12\sqrt{2}) = 36$, impossible for $r \equiv 1 \pmod 8$. For $d = W_{p-2}$: $\mathrm{ord}(\alpha_0) = 4W_{p-2} = W_p + 1$ divides $r - 1$, so $r > W_p$ — impossible. If $r \equiv 3 \pmod 8$, the inert descent (Section 5) gives $\mathrm{ord}(\omega_3) = 4g$ at r with g odd and $g \mid W_{p-2}$, and the Pell-sequence exclusion (Theorem 5.10) gives $g > 43$; so $g = W_{p-2}$ and $W_p + 1 = 4W_{p-2} = \mathrm{ord}(\omega_3) \leq r + 1$, i.e. $r \geq W_p$ — again impossible for a proper factor. □

Corollary 4.3 (The certified ladder). *Condition (II) is an unconditional primality test for W_p at every prime p such that $p - 2$ is also prime and W_{p-2} carries a primality certificate: presently*

$$p \in \{5,\ 7,\ 13,\ 19,\ 103,\ 193,\ 349,\ 3541\},$$

the complete list arising from the certified Wagstaff primes. Three further rungs await only a primality certificate for the corresponding W_{p-2}, currently a strong probable prime: $p = 83{,}341$, $117{,}241$, $127{,}033$.

Remark 4.4 (BLS specialization; the ladder is self-supporting). Proposition 4.2 is the fully-factored-$(N+1)$ specialization of Brillhart–Lehmer–Selfridge [4], via $W_p + 1 = 4W_{p-2}$. What the present machinery adds is that Condition (II) *alone* completes it: the $d = 1$ branch dies on the norm $\mathrm{N}(\omega_3^2 + 1) = 36$ and the inert branch on the Pell-sequence exclusion — no auxiliary base, no extra congruence. Under a twin-prime-type heuristic the ladder $\{p : p,\ p-2,\ W_{p-2} \text{ all prime}\}$ is infinite, and each new Wagstaff primality certificate at an exponent q with $q+2$ prime adds a rung (enumeration: `scripts/cp410_twin_ladder.py`). In the self-supporting reading, sufficiency of Condition (II) — Conjecture 1.1 — would propagate primality *up* the ladder without any factoring at all.

4.2. **Conjecture R3 and its conditional reduction.**

Remark 4.5. Since $\alpha_0 = (1+s)^2$: $\mathrm{ord}(\alpha_0) = \mathrm{ord}(1+s)/\gcd(\mathrm{ord}(1+s), 2)$. The key divisibility is $p \mid \mathrm{ord}(1+s)$.

Conjecture 4.6 (Pell-rank divisibility, R3)**.** For every prime $r \equiv 1 \pmod 8$ with $r \mid W_p$: $p \mid \mathrm{ord}_{\mathbb{F}_r^*}(1+s)$, where $s = \sqrt{2} \bmod r$.

Theorem 4.7 (Conditional on R3)**.** *Let W_p be composite with a prime factor $r \equiv 1 \pmod 8$. If Conjecture R3 holds at r, then r cannot satisfy Condition (II).*

Proof. By R3: $p \mid \mathrm{ord}(1+s)$; since $\alpha_0 = (1+s)^2$, $p \mid \mathrm{ord}(\alpha_0) = 4d$, so $p \mid d \mid W_{p-2}$, contradicting Lemma 2.10. □

Theorem 4.8 (Split branch eliminates under R3)**.** *If Conjecture R3 holds at every split factor of every composite W_p, then every composite W_p passing Condition (II) has all factors inert ($\equiv 3 \pmod 8$).*

Proof. Immediate from Theorem 4.7 and Corollary 2.5. □

4.3. **Proved cases of Conjecture R3.**

Proposition 4.9 (Algebraic identity)**.** *In $\mathbb{Z}[\sqrt{2}]$: $\Phi_{2p}(\omega_3) = U_p \cdot (1+\sqrt{2})^{p-1}$. Consequently $\mathrm{N}(\Phi_{2p}(\omega_3)) = U_p^2$.*

Proof. Since p is an odd prime, $\Phi_{2p}(x) = (x^p+1)/(x+1)$; hence

$$\Phi_{2p}(\omega_3) \;=\; \frac{\omega_3^p + 1}{\omega_3 + 1}.$$

Using $\omega_3 = \alpha^2$ with $\alpha = 1 + \sqrt{2}$ and the Pell formula $\alpha^p = V_p/2 + U_p\sqrt{2}$: $\omega_3^p = \alpha^{2p} = (V_p/2 + U_p\sqrt{2})^2 = (V_p^2/4 + 2U_p^2) + V_pU_p\sqrt{2}$. Using $V_p^2 - 8U_p^2 = -4$ (so $V_p^2/4 = 2U_p^2 - 1$), the rational part simplifies:

$$\omega_3^p = (4U_p^2 - 1) + V_pU_p\sqrt{2},$$
$$\omega_3^p + 1 = 4U_p^2 + V_pU_p\sqrt{2} = U_p(4U_p + V_p\sqrt{2}).$$

Also $\omega_3 + 1 = 4 + 2\sqrt{2} = 2\alpha \cdot \alpha^{-1} \cdot (2+\sqrt{2}) = 2(2+\sqrt{2})$. Dividing:

$$\Phi_{2p}(\omega_3) \;=\; \frac{U_p(4U_p + V_p\sqrt{2})}{2(2+\sqrt{2})} \;=\; U_p \cdot f_p, \qquad f_p := \frac{4U_p + V_p\sqrt{2}}{4+2\sqrt{2}}.$$

The Pell recurrences $U_{n+1} = 2U_n + U_{n-1}$, $V_{n+1} = 2V_n + V_{n-1}$ give $f_{n+1} = 2f_n + f_{n-1}$, and by direct computation $f_0 = 2\sqrt{2}/(4+2\sqrt{2}) = \sqrt{2}-1 = \alpha^{-1}$, $f_1 = (4+2\sqrt{2})/(4+2\sqrt{2}) = 1 = \alpha^0$. The same recurrence is satisfied by α^{n-1} (since $\alpha^2 = 2\alpha+1$), so $f_n = \alpha^{n-1}$ by induction. Hence $\Phi_{2p}(\omega_3) = U_p\alpha^{p-1} = U_p(1+\sqrt{2})^{p-1}$. Norm: $\mathrm{N}((1+\sqrt{2})^{p-1}) = (-1)^{p-1} = 1$ since p is odd; so $\mathrm{N}(\Phi_{2p}(\omega_3)) = U_p^2$. $\square$

Theorem 4.10 (Case A)**.** *If $r \equiv 1 \pmod 8$, $r \mid W_p$, and $r \mid U_p$, then $\mathrm{ord}(\alpha) = 4p$ in the split algebra.*

Proof. Reducing Proposition 4.9 modulo r: $r \mid U_p$ gives $\Phi_{2p}(\omega_3) \equiv 0 \pmod r$ (since $\Phi_{2p}(\omega_3) = U_p \cdot (1+\sqrt{2})^{p-1}$ and $r \nmid (1+\sqrt{2})^{p-1}$ in the split algebra, as $1+\sqrt{2}$ is a unit). So $\omega_3 \bmod r$ is a primitive $2p$-th root of unity in $\mathbb{F}_r^*$: $\mathrm{ord}(\omega_3) = 2p$.

Now $\omega_3 = \alpha^2$, so $\mathrm{ord}(\alpha) \in \{2p, 4p\}$ (the two options whose square has order $2p$). We claim $\mathrm{ord}(\alpha) = 4p$. Since $\mathrm{ord}(\omega_3) = 2p$, we have $\omega_3^p = -1$ (the unique element of order 2 in the cyclic group $\langle\omega_3\rangle$), so $\alpha^{2p} = \omega_3^p = -1$. Hence $\mathrm{ord}(\alpha) \nmid 2p$; combined with $\alpha^{4p} = 1$, this gives $\mathrm{ord}(\alpha) = 4p$ (ruling out $\mathrm{ord}(\alpha) = 4$ since $\alpha^4 = \omega_3^2$ and ω_3 has order $2p \geq 10$). $\square$

Theorem 4.11 (Case B)**.** *If $r \equiv 1 \pmod 8$, $r \mid W_p$, and $r \mid V_p$, then $\mathrm{ord}(\alpha) = 2p$ in the split algebra.*

Proof. With the split embedding $\sqrt{2} \mapsto (s, -s)$ and $r \mid V_p$, the identity $\alpha^p = V_p/2 + U_p\sqrt{2}$ gives

$$\alpha^p \;\equiv\; (U_p s,\; -U_p s) \quad \text{in } \mathbb{F}_r \times \mathbb{F}_r.$$

Squaring coordinatewise: $\alpha^{2p} \equiv (2U_p^2,\; 2U_p^2)$. From the Pell identity $V_p^2 - 8U_p^2 = -4$ with $r \mid V_p$: $8U_p^2 \equiv 4 \pmod r$, so $2U_p^2 \equiv 1 \pmod r$. Hence $\alpha^{2p} \equiv (1,1) = 1$.

Meanwhile $\alpha^p \equiv (U_p s, -U_p s)$ has coordinates of opposite signs, so $\alpha^p \not\equiv (1,1)$. Thus $\mathrm{ord}(\alpha) \mid 2p$ and $\mathrm{ord}(\alpha) \nmid p$, forcing $\mathrm{ord}(\alpha) \in \{2, 2p\}$. If $\mathrm{ord}(\alpha) = 2$: $\omega_3 = \alpha^2 \equiv (1,1)$, i.e., $3+2s \equiv 1$ and $3-2s \equiv 1$ in $\mathbb{F}_r$; adding: $6 \equiv 2$, i.e., $4 \equiv 0 \pmod r$, contradicting $r \geq 17$. Therefore $\mathrm{ord}(\alpha) = 2p$. $\square$

Theorem 4.12 (Case $4U_p^2 \equiv 1$)**.** *If $r \equiv 1 \pmod 8$, $r \mid W_p$, and $4U_p^2 \equiv 1 \pmod r$, then $\mathrm{ord}(\alpha) = 8p$ in the split algebra.*

Proof. With $4U_p^2 \equiv 1 \pmod r$: compute α^{2p} using the identity $\alpha^p = V_p/2 + U_p\sqrt{2}$. Squaring:

$$\alpha^{2p} \;=\; V_p^2/4 + V_pU_p\sqrt{2} + 2U_p^2 \;=\; (V_p^2/4 + 2U_p^2) + V_pU_p\sqrt{2}.$$

From $V_p^2 - 8U_p^2 = -4$: $V_p^2/4 + 2U_p^2 = V_p^2/4 + (V_p^2+4)/4 = V_p^2/2 + 1$. Substituting: $\alpha^{2p} = (V_p^2/2+1) + V_pU_p\sqrt{2} \equiv V_p^2/2 + 1 + V_pU_ps$ in the split algebra.

Now use $4U_p^2 \equiv 1$: $U_p^2 \equiv 1/4$; and $V_p^2 \equiv 8U_p^2 - 4 \equiv 2 - 4 = -2$; so $V_p^2/2 + 1 \equiv -1 + 1 = 0$. Also V_pU_p has square $V_p^2U_p^2 \equiv (-2)(1/4) = -1/2$. Hence $\alpha^{2p} \equiv V_pU_ps$, with $(\alpha^{2p})^2 = (V_pU_p)^2 \cdot s^2 = (-1/2)\cdot 2 = -1$. So $\alpha^{4p} \equiv -1 \pmod{r}$.

Therefore $\mathrm{ord}(\alpha) \mid 8p$ but $\mathrm{ord}(\alpha) \nmid 4p$; the divisors of $8p$ not dividing $4p$ are 8 and $8p$. If $\mathrm{ord}(\alpha) = 8$: $\alpha^8 = (1+\sqrt{2})^8 = 577 + 408\sqrt{2} \equiv 1 \pmod{r}$ would force $r \mid \gcd(576, 408) = 24$, contradicting $r \geq 17$. Hence $\mathrm{ord}(\alpha) = 8p$. $\square$

In each case $p \mid \mathrm{ord}(\alpha)$, confirming Conjecture R3 (via Proposition 4.14 below). Among the 185 known split factors, these three cases cover 29% (1 + 6 + 46 factors respectively); see §B.2.

4.4. **Pell rank and equivalence with Conjecture R3.**

Definition 4.13 (Pell rank of apparition)**.** For a prime r with $(2/r) = 1$, the Pell rank $\rho(r)$ is the least $n \geq 1$ with $r \mid U_n$. We say r is a primitive divisor of U_n if $\rho(r) = n$. Bilu–Hanrot–Voutier [3] gives primitive divisors for all sufficiently large Lucas–Lehmer indices; the small indices used in the finite checks below are handled directly.

Proposition 4.14 (Rank equivalence)**.** *Let $r \equiv 1 \pmod{8}$, $r \mid W_p$. Suppose Condition (II) holds at r in a composite W_p; so $\mathrm{ord}(\alpha_0) = 4d$ with d odd. Then:*

(a) $\rho(r) = 4d$ and r is a primitive divisor of U_{4d}.

(b) Conjecture R3 at r holds iff $p \mid \rho(r)$.

Proof. Set $\beta = 1 + s$, so $\beta^2 = \alpha_0$. We first establish $\mathrm{ord}(\beta) = 8d$. Since $\beta^2 = \alpha_0$,

$$\mathrm{ord}(\alpha_0) = \frac{\mathrm{ord}(\beta)}{\gcd(\mathrm{ord}(\beta), 2)}.$$

An odd $\mathrm{ord}(\beta)$ would force $\mathrm{ord}(\alpha_0)$ odd, contradicting $\mathrm{ord}(\alpha_0) = 4d$, which has $v_2 = 2$. Hence $\mathrm{ord}(\beta) = 2\,\mathrm{ord}(\alpha_0) = 8d$.

(a) From $\mathrm{ord}(\beta) = 8d$: $\beta^{4d} = -1$ (the unique order-2 element in $\langle\beta\rangle$, since $4d = \mathrm{ord}(\beta)/2$). Now $\beta^{2n} = \alpha_0^n = (3+2s)^n$; using the identification of Pell polynomials with Chebyshev, one obtains

$$\beta^{2n} \;=\; (4U_n^2 + (-1)^n) + V_nU_ns \qquad \text{in } \mathbb{F}_r.$$

(Direct verification: for $n = 1$, $\beta^2 = 1 + 2s + s^2 = 3 + 2s = 4U_1^2 - 1 + V_1U_1s = 4 - 1 + 2s = 3 + 2s$ ✓. Induction on n then propagates the formula, using the recurrences $U_{n+1} = 2U_n + U_{n-1}$, $V_{n+1} = 2V_n + V_{n-1}$ together with the Pell cross relation

$$V_n \;=\; 2\,(U_n + U_{n-1}),$$

which the inductive step uses. This relation is obtained by comparing the $\sqrt{2}$-coefficients on the two sides of $\alpha^n = \alpha \cdot \alpha^{n-1}$: from $\alpha^n = V_n/2 + U_n\sqrt{2}$

and $(1+\sqrt{2})(V_{n-1}/2+U_{n-1}\sqrt{2})$ one reads off $U_n = V_{n-1}/2+U_{n-1}$, i.e. $V_{n-1} = 2(U_n - U_{n-1})$; shifting the index and using $U_{n+1} = 2U_n + U_{n-1}$ gives $V_n = 2(U_{n+1} - U_n) = 2(U_n + U_{n-1})$. It is this identity that converts the cross term in $\beta^{2(n+1)}$ into the required $V_{n+1}U_{n+1}$ coefficient.)

So $\beta^{2n} = (-1)^n$ iff the constant term satisfies $4U_n^2 + (-1)^n \equiv (-1)^n \pmod r$ and the s-coefficient V_nU_n vanishes mod r. The constant-term condition reads $4U_n^2 \equiv 0 \pmod r$, hence $r \mid U_n$ (as r is odd); the s-coefficient condition $V_nU_n \equiv 0$ is then automatic.

For $n = 4d$: $\beta^{8d} = 1 = (-1)^{4d}$, so $r \mid U_{4d}$. For any integer $1 \le n < 4d$: if n is even, $\beta^{2n} = 1$ requires $8d \mid 2n$, so $4d \mid n$, contradicting $n < 4d$. If n is odd, $\beta^{2n} = -1$ requires $\beta^{2n} = \beta^{4d}$, i.e., $8d \mid 2n - 4d$, i.e., $2d \mid n$, impossible for odd n. So $\rho(r) = 4d$, and r is a primitive divisor of U_{4d}.

(b) Conjecture R3 at r says $p \mid \mathrm{ord}(1+s) = \mathrm{ord}(\beta) = 8d$. Since p is odd, $p \mid 8d$ iff $p \mid d$, iff $p \mid 4d = \rho(r)$. □

Definition 4.15 (Compatible triple)**.** A compatible triple (d, r, p) is: an odd integer $d \ge 1$; a prime $r \equiv 1 \pmod 8$ that is a primitive divisor of U_{4d}; a prime $p \ge 5$ with $\mathrm{ord}_r(2) = 2p$ and $d \mid W_{p-2}$.

By Proposition 4.14, if Condition (II) passes at some split factor of a composite W_p, a compatible triple exists.

Remark 4.16 (No phantom compatible triples). Every compatible triple has W_p composite: $r \mid W_p$ with $r \equiv 1 \pmod 8$, while $W_p \equiv 3 \pmod 8$ (Lemma 2.4), so $r \ne W_p$ and r is a *proper* divisor. Hence the nonexistence of compatible triples coincides with its restriction to composite W_p — the only form the reduction of §7.6 invokes; no separate "W_p prime" case arises on the split branch. This is in contrast to the inert branch, where $r \equiv 3 \equiv W_p \pmod 8$ permits $r = W_p$ and hence *phantom* danger triples (Definition 5.15, Remark 5.17).

4.5. **Quartic residuosity of 2.**

Theorem 4.17 (Quartic residuosity at primitive Pell divisors)**.** *Let $n \ge 1$ be odd. If $r \equiv 1 \pmod 8$ is a prime primitive divisor of U_{4n}, then $2^{(r-1)/4} \equiv 1 \pmod r$, i.e., 2 is a fourth-power residue modulo r.*

Proof. In $\mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$ we have the identity

$$(*) \qquad \sqrt{2} \;=\; (1-\zeta_8)^2 \cdot \zeta_8^3 \cdot (1+\sqrt{2}),$$

verified as follows: $(1-\zeta_8)^2 = 1 - 2\zeta_8 + \zeta_8^2 = 1 - 2\zeta_8 + i$; multiplying by ζ_8^3 and using $\zeta_8^2 = i$, a direct multiply-out gives $(1-\zeta_8)^2 \cdot \zeta_8^3 = 2 - \sqrt{2}$, then $(2-\sqrt{2})(1+\sqrt{2}) = 2 + 2\sqrt{2} - \sqrt{2} - 2 = \sqrt{2}$. $(*)$ is confirmed.

Since $r \equiv 1 \pmod 8$, r splits completely in $\mathbb{Z}[\zeta_8]$; fix a prime ideal $\mathfrak{P} \mid r$. Reduction mod $\mathfrak{P}$ embeds $\zeta_8 \mapsto \omega$ (a primitive 8-th root of unity in $\mathbb{F}_r^\times$), $\sqrt{2} \mapsto s$, and $1+\sqrt{2} \mapsto \alpha$, giving

$$(**) \qquad s \;\equiv\; (1-\omega)^2 \cdot \omega^3 \cdot \alpha \qquad \text{in } \mathbb{F}_r.$$

Raise $(**)$ to $(r-1)/2$. We first record, *unconditionally*, that a prime $r \equiv 1 \pmod 8$ with $\rho(r) = 4n$ (i.e. a primitive divisor of U_{4n}) and n odd has $\mathrm{ord}(\alpha) = 8n$ and $\alpha^{4n} = -1$. Indeed, in the split algebra $\alpha^m = V_m/2 + U_m s$, so $\alpha^m \in \{\pm 1\}$ forces the s-coefficient to vanish, $r \mid U_m$, i.e. $\rho(r) \mid m$; the converse holds when m is even, the only case used below. At $m = 4n$ the identity $V_{4n}^2 - 8U_{4n}^2 = 4$ with $r \mid U_{4n}$ gives $V_{4n}/2 \equiv \pm 1 \pmod r$, so $\alpha^{4n} \in \{\pm 1\}$. If $\alpha^{4n} = +1$, then $e := \mathrm{ord}(\alpha)$ divides $4n$; but $\alpha^e = 1$ forces $r \mid U_e$, hence $\rho(r) = 4n \mid e$, so $e = 4n$, whence α^{2n} has order 2, $\alpha^{2n} = -1$, and $r \mid U_{2n}$ — contradicting $\rho(r) = 4n \nmid 2n$. Therefore $\alpha^{4n} = -1$ and $\mathrm{ord}(\alpha) = 8n$. (This is the converse direction of the rank–order correspondence; Proposition 4.14(a), which assumes Condition (II) at r, is not invoked here, and Theorem 4.17 makes no Condition-(II) hypothesis.) Writing $r - 1 = 8nM$ for a positive integer M:

- $[(1-\omega)^2]^{(r-1)/2} \equiv 1 \pmod{\mathfrak{P}}$: $(1-\omega)^{r-1} = 1$ by Fermat's little theorem in $\mathbb{F}_r$.
- $[\omega^3]^{(r-1)/2} = \omega^{3(r-1)/2}$: since $\mathrm{ord}(\omega) = 8$ and $(r-1)/2 = 4nM$, $\omega^{3\cdot 4nM} = \omega^{12nM}$. Reducing mod 8: $12nM \equiv 4nM \pmod 8$. Since n is odd, $4nM \equiv 4M \pmod 8$, so $\omega^{4nM} = \omega^{4M} = (\omega^4)^M = (-1)^M$.
- $\alpha^{(r-1)/2} = \alpha^{4nM} = (\alpha^{4n})^M = (-1)^M$.

Multiplying the three factors gives $s^{(r-1)/2} = 1\cdot(-1)^M\cdot(-1)^M = (-1)^{2M} = 1$.

Hence $2^{(r-1)/2} = s^{r-1} = (s^{(r-1)/2})^2 = 1$ trivially, but the sharper conclusion $s^{(r-1)/2} = 1$ means $s \in (\mathbb{F}_r^\times)^2$, so $\sqrt{s} \in \mathbb{F}_r$. Now $s^2 = 2$, so $(\sqrt{s})^4 = 2$, and $\sqrt{s}$ has order dividing $r-1$: writing $\sqrt{s} = t$ with $t^4 = 2$: $2^{(r-1)/4} = t^{r-1} = 1$. □

Remark. Theorem 4.17 is recorded for context. It is cited once, in the obstruction analysis of §A.1, but is not used in the reduction chain itself: the split branch is closed through the Pell-rank reformulation (Proposition 4.14) and the nonexistence of compatible triples (Theorem 4.18 and Conjecture 4.20). Further split-local character refinements are relegated to the supplementary material.

4.6. **No Compatible Triples.**

Theorem 4.18 (NCT for $d \le 99$)**.** *For each odd d with $1 \le d \le 99$, no compatible triple (d, r, p) exists.*

Proof. Call d *admissible* if every prime factor q of d satisfies $v_2(\mathrm{ord}_q(2)) = 1$; otherwise $q \mid W_{p-2}$ is impossible (if $\mathrm{ord}_q(2)$ is odd, no half-order m_q exists; if $4 \mid \mathrm{ord}_q(2)$, then m_q is even but $p-2$ is odd, contradicting $m_q \mid p-2$). The parity obstruction eliminates $\sim$80% of odd $d \le 500$; within $[1, 99]$ it leaves 14 values — the 12 in $[1, 81]$ together with $\{83, 99\}$. The seven excluded values $\{85, 87, 89, 91, 93, 95, 97\}$ in $(81, 99]$ are parity-blocked: each has a prime factor q with $q \notin A$ (e.g., $85 = 5 \cdot 17$ with $5 \notin A$; $91 = 7 \cdot 13$ with $7 \notin A$; etc.).

Cases $d = 1, 3$. $U_4 = 12$ and $U_{12} = 13{,}860$ have no prime factor $\equiv 1 \pmod 8$, so no split r exists.

Cases $d \in \{9, 11, 19, 27, 33, 43, 57, 59, 67, 81\}$. U_{4d} is completely factored (largest at $d = 81$, 124 digits); each of the 18 primitive divisors $r \equiv 1 \pmod 8$ with $\rho(r) = 4d$ is checked and found to have $\mathrm{ord}_r(2)$ either odd or $\mathrm{ord}_r(2)/2$ composite, so no Wagstaff prime p can correspond.

Cases $d \in \{83, 99\}$. Rather than factor the full U_{4d} (sizes $\approx \alpha^{4d}$, prohibitive at $d = 99$ for general-purpose ECM), we use the cyclotomic decomposition

$$V_{2d} = \prod_{\substack{m \mid d \\ m \text{ odd}}} \Psi_{4m}, \qquad U_n = \prod_{e \mid n} \Psi_e \quad \text{(Möbius-inverted)},$$

where Ψ_n is the n-th *primitive part* of the Pell U-sequence. The decomposition is derived from $U_{2n} = V_n U_n$ (specialise $n = 2d$, so $V_{2d} = U_{4d}/U_{2d}$; the surviving primitive parts are exactly Ψ_e with $e \mid 4d$, $e \nmid 2d$, i.e., $e = 4m$ with $m \mid d$ odd). Each piece Ψ_{4m} has size $\approx \alpha^{2\varphi(m)}$, far smaller than V_{2d} itself. Every primitive split divisor of V_{2d} — and hence every potential primitive Wagstaff factor at this d — lies in some Ψ_{4m}.

For $d = 83$ (prime, $\mathrm{ord}_{83}(2) = 82 = 2 \cdot 41$, so $83 \in A$): $\Psi_{4 \cdot 83}$ has 63 decimal digits and five prime factors; four are $\equiv 3 \pmod 8$ (inert, not relevant to the split-branch search), and the single split factor

$$r = 7{,}370{,}827{,}889{,}154{,}320{,}981{,}521 \quad (\equiv 1 \pmod 8)$$

satisfies

$$\mathrm{ord}_r(2) = 1{,}842{,}706{,}972{,}288{,}580{,}245{,}380,$$
$$\mathrm{ord}_r(2)/2 = 921{,}353{,}486{,}144{,}290{,}122{,}690.$$

The half-order is even, hence composite. Not Wagstaff.

For $d = 99 = 9 \cdot 11$ (with $\mathrm{ord}_{11}(2) = 10 = 2 \cdot 5$, $11 \in A$, and $3 \in A$): $\Psi_{4 \cdot 99}$ has 46 decimal digits, a balanced semiprime of two split factors. One factor satisfies $\mathrm{ord}_r(2)$ odd; the other has $\mathrm{ord}_r(2)/2$ divisible by 5^2, hence composite. Neither is Wagstaff.

No primitive split divisor at $d \in \{83, 99\}$ yields a compatible triple. Combined with the cases above, this closes NCT for $d \le 99$ unconditionally. $\square$

(Detailed per-d obstruction table, factorizations of U_{4d} for $d \le 81$, and factorizations of Ψ_{4m} for odd $m \le 99$, are archived in the supplementary Zenodo deposit; each asserted prime factor used in the finite exhaustion is independently certified by APR-CL in PARI/GP, with BPSW and deterministic Miller–Rabin used only as preliminary screens.)

Proposition 4.19 (q-th power necessity)**.** *In any compatible triple* (d, r, p), *for every odd prime* $q \mid d$*: 2 must be a* q*-th power residue modulo* r.

Proof. Suppose not. Since r is a primitive divisor of U_{4d} with $r \equiv 1 \pmod 8$, its rank of apparition $\rho(r) = 4d$ divides $r - 1$; hence $q \mid d \mid r - 1$, and the failure of 2 to be a q-th-power residue modulo r forces $q \mid \mathrm{ord}_r(2) = 2p$. So $q = p$, whence $p \mid d \mid W_{p-2}$, contradicting Lemma 2.10. $\square$

Conjecture 4.20 (NCT for general d)**.** For every odd $d \geq 1$, no compatible triple (d, r, p) exists.

Remark 4.21. A natural strengthening of Proposition 4.19 — that $v_2(\mathrm{ord}_r(2)) \geq 2$ whenever 2 is a q-th power modulo a primitive divisor r of U_{4q} — is *false*. For $q = 163$, the primitive divisor $r = 12{,}036{,}824{,}328{,}615{,}957{,}881$ of U_{652} satisfies $2^{(r-1)/163} \equiv 1 \pmod r$ yet $v_2(\mathrm{ord}_r(2)) = 1$. Conjecture 4.20 is unaffected: $\mathrm{ord}_r(2)/2$ is composite (§B.3). The take-away: the split-branch gap is structural, and must be attacked directly via compatible triples.

4.7. **The Mixed Pair Conjecture.** The main reduction (Theorem 7.2) invokes NCT only inside a global Condition-(II) pass, and a global pass always supplies a second, inert, locally passing factor (Lemma 2.4: the inert multiplicity is odd, hence at least one inert factor exists). The split-branch residual can therefore be weakened from the single-coincidence statement NCT to the following pair-type statement.

Conjecture 4.22 (Mixed Pair Conjecture, MPC)**.** No prime $p \geq 5$ admits simultaneously

(i) a compatible triple (d_s, r_s, p): an odd $d_s \geq 1$, a prime $r_s \equiv 1 \pmod 8$ that is a primitive divisor of U_{4d_s}, with $\mathrm{ord}_{r_s}(2) = 2p$ and $d_s \mid W_{p-2}$ (Definition 4.15);

(ii) an inert prime $r_{\mathrm{in}} \equiv 3 \pmod 8$ with $\mathrm{ord}_{r_{\mathrm{in}}}(2) = 2p$ and $d_{r_{\mathrm{in}}} \mid W_{p-2}$, where $d_{r_{\mathrm{in}}} = \mathrm{ord}_{r_{\mathrm{in}}}(\omega_3)/4$.

Remark 4.23 (Reading the two data). The divisibilities $r_s \mid W_p$ and $r_{\mathrm{in}} \mid W_p$ are automatic: $\mathrm{ord}_r(2) = 2p$ forces $r \mid 2^p + 1$ with $r \neq 3$, hence $r \mid W_p$. The two data are not symmetric, and datum (ii) depends on the conjunction. *Given* (i), W_p is composite — $r_s \equiv 1 \pmod 8$ is a proper divisor of $W_p \equiv 3 \pmod 8$ (Remark 4.16) — and *then* the inert datum (ii) automatically satisfies $d_{r_{\mathrm{in}}} > 43$ and admissibility (every prime factor of $d_{r_{\mathrm{in}}}$ in A), by the Pell-sequence exclusion (Theorem 5.10) and the parity obstruction (Theorem 5.3); so (ii) is then exactly an inert local Condition-(II) pass at a composite W_p. Standalone, datum (ii) has *phantom* instances at prime W_p: for example $(p, r, d) = (5, 11, 3)$ satisfies (ii) with $W_5 = 11$ prime. The conjunction with (i) excludes them.

Proposition 4.24 (NCT-general implies MPC)**.** *Conjecture 4.20 implies Conjecture 4.22.*

Proof. A violation of MPC contains, in datum (i), a compatible triple. □

No converse implication is known, so MPC is *formally weaker as a hypothesis* than NCT-general; this is the only comparison we assert. (Establishing that MPC is properly weaker would require exhibiting a compatible triple without an inert co-passer — a configuration that would itself falsify NCT-general; and if NCT-general is true, the material implication MPC $\Rightarrow$ NCT-general holds trivially.) MPC forbids a coincidence of two pinned local passes at one p — the same shape as PPPC — so the two principal residuals of the sharpened reduction (§7.7) are structurally uniform: both are pair-type statements. MPC inherits every unconditionally proved NCT range: a violation requires $d_s > 99$ with $d_s \notin \{107, 121, 129, 139, 171, 177\}$ (Theorem 4.18, §B.3); on the inert side, the order-alignment in datum (ii) is independently rare (§7.3: 0 of 369,876 tested candidates satisfy it). A joint heuristic for MPC is recorded in §7.4. Proving MPC meets the same structural obstruction as NCT — Observation 8.1 applies unchanged to its split datum — so MPC is offered as a weaker residual, not as a more tractable theorem target.

For the cutoff-split form of the reduction (Theorem 7.4) write $\mathbf{MPC}^{>}$ for the restriction of MPC in which datum (ii) is additionally required to satisfy $r_{\text{in}} > 10^{12}$. The complementary leg $r_{\text{in}} \le 10^{12}$ is absorbed by the Platinum enumeration restricted to composite W_p and by the witness-exponent statement X1 (Conjecture 6.29); see §7.7.

5. The inert branch: $r \equiv 3 \pmod 8$

5.1. **The gap structure.** For $r \equiv 3 \pmod 8$ with $r \mid W_p$ composite: $(2/r) = -1$, so $\mathbb{Z}[\sqrt{2}]/(r) \cong \mathbb{F}_{r^2}$. The element ω_3 has norm $+1$, so lies in the norm-1 subgroup μ_{r+1}; by Lemma 2.8, $v_2(\mathrm{ord}(\omega_3)) = 2$, so $\mathrm{ord}(\omega_3) = 4g$ for g odd. Combined with $\mathrm{ord}(\omega_3) \mid r + 1$ and $v_2(r + 1) = 2$: $g \mid (r + 1)/4$.

Descent. Condition (II) at r becomes $\omega_3^{(N+1)/2} = -1$, i.e., $\omega_3^{2W_{p-2}} = -1$. With $\mathrm{ord}(\omega_3) = 4g$: this holds iff $g \mid W_{p-2}$.

Notation. For inert $r \mid W_p$:

$$G_r \;=\; \gcd\!\left(\frac{r+1}{4},\; W_{p-2}\right).$$

If Condition (II) passes, then $g \mid G_r$.

Remark 5.1 (Why the norm argument doesn't suffice). Condition (II) already implies $\omega_3^N \equiv \omega_3^{-1} \pmod N$, so quadratic Frobenius strengthenings [10] add nothing. The norm $\mathrm{N}(\alpha) = -1$ constrains only the 2-part of orders; the odd part g is not constrained by the norm argument alone.

5.2. **Parity and valuation obstructions.**

Definition 5.2 (Class III primes)**.** An odd prime q is Class III if $v_2(\mathrm{ord}_q(2)) = 1$. Write $A = \{q : v_2(\mathrm{ord}_q(2)) = 1\}$.

Class-III primes are exactly the odd primes that can divide $W_n = (2^n + 1)/3$ for odd $n \ge 3$: $q \mid W_n$ requires $\mathrm{ord}_q(2) = 2m$ with m odd. Every $q \equiv 3 \pmod 8$ lies in A; $q \equiv 5$ or $7 \pmod 8$ do not.

Theorem 5.3 (Parity obstruction for inert factors)**.** *Let $r \equiv 3 \pmod 8$, $r \mid W_p$ composite, $g = \mathrm{ord}_r(\omega_3)/4$. If g has a prime factor $q \notin A$, then $g \nmid W_{p-2}$ and Condition (II) fails at r.*

Proof. $q \notin A$ and $q \mid g$: then $q \nmid W_n$ for any odd n, so $q \nmid W_{p-2}$. $\square$

Theorem 5.4 (Generalized LTE for Class III primes)**.** *Let $q \in A$ with $\mathrm{ord}_q(2) = 2m$, m odd. For odd n with $m \mid n$: $v_q(2^n + 1) = v_q(2^m + 1) + v_q(n/m)$. For $q \geq 5$: $v_q(W_n) = v_q(2^m + 1) + v_q(n/m)$; for $q = 3$: $v_3(W_n) = v_3(n)$.*

Proof. Apply the lifting-the-exponent lemma to $(2^m)^\ell + 1$ with $q \mid 2^m + 1$. □

Computationally, $v_q(2^m + 1) = 1$ for all 194 Class III primes $q < 5000$, where $m = \mathrm{ord}_q(2)/2$; no "Wieferich-at-minus-1" behaviour is observed.

Valuation obstruction. If $v_q(g) > v_q(W_{p-2})$ for some $q \in A$ with $q \mid g$, then $g \nmid W_{p-2}$ and Condition (II) fails.

Among the 869 inert factors with $r < 2 \times 10^5$ tested in §B.4, the parity and valuation obstructions account for the bulk of the exclusions; together with the Pell-sequence exclusion (Theorem 5.10) they block 845 of the 869 (Corollary 5.11).

5.3. **Pell-sequence exclusion.**

Theorem 5.5 (Pell-sequence divisibility)**.** *Let $r \equiv 3 \pmod 8$, $r \mid W_p$, and $\mathrm{ord}_r(\omega_3) = 4d$. Then*

$$r \;\Big|\; \gcd\!\left(\frac{V_{4d} + 2}{2},\ U_{4d}\right).$$

Proof. $\mathrm{ord}_r(\omega_3) = 4d$ gives $\omega_3^{2d} = -1$, hence $\alpha^{4d} = -1$. Expanding $\alpha^{4d} = V_{4d}/2 + U_{4d}\sqrt{2}$ and comparing with $-1 = -1 + 0 \cdot \sqrt{2}$: $r \mid U_{4d}$ and $r \mid (V_{4d}/2 + 1) = (V_{4d} + 2)/2$. □

Definition 5.6 (Pell exclusion number)**.** For odd $d \geq 1$: $N_d = \gcd((V_{4d} + 2)/2,\ U_{4d})$.

Proposition 5.7 (Closed form)**.** *For odd $d \geq 1$: $N_d = V_{2d} = V_d^2 + 2$.*

Proof. Three Pell identities: $V_{4d} = V_{2d}^2 - 2$, so $V_{4d} + 2 = V_{2d}^2$; $U_{4d} = U_{2d}V_{2d}$; $(V_{2d}/2)^2 - 2U_{2d}^2 = 1$, so $\gcd(V_{2d}/2, U_{2d}) = 1$. Since $V_{2d} = 2k$ even, $N_d = \gcd(2k^2, 2kU_{2d}) = 2k = V_{2d}$. Finally $V_{2d} = V_d^2 - 2(-1)^d = V_d^2 + 2$ for odd d. □

Lemma 5.8 (Inert primitivity)**.** *Let $r \equiv 3 \pmod 8$ be prime and suppose $\mathrm{ord}_r(\omega_3) = 4d$. Then $r \mid V_{2d}$ and $r \nmid V_{2e}$ for every integer $0 < e < d$. Thus r is primitive for the d-indexed Pell–Lucas family $\{V_{2e}\}_{e \geq 1}$ at index d.*

Proof. Write $V_{2e} = \omega_3^e + \omega_3^{-e}$. Since $\mathrm{ord}_r(\omega_3) = 4d$, the element ω_3^d has order 4, so $(\omega_3^d)^2 = -1$ and $\omega_3^d + \omega_3^{-d} = 0$; hence $r \mid V_{2d}$.

Conversely, if $r \mid V_{2e}$, then $\omega_3^e + \omega_3^{-e} = 0$ in $\mathbb{F}_{r^2}$, so $\omega_3^{2e} = -1$. Also $\omega_3^{2d} = -1$. Therefore $\omega_3^{2(e-d)} = 1$, and $4d \mid 2(e - d)$, equivalently $e \equiv d \pmod{2d}$. No integer e with $0 < e < d$ satisfies this congruence. □

Remark 5.9. Since $N_d = V_d^2 + 2 = \mathrm{N}_{\mathbb{Z}[\sqrt{-2}]}(V_d + \sqrt{-2})$, every prime $r \equiv 3 \pmod 8$ dividing N_d splits in $\mathbb{Z}[\sqrt{-2}]$ (since $(-2/r) = 1$). This Zsigmondy-type realization is the structural basis of attack angle N2 (Stewart–Tijdeman bounds on $\omega(N_d)$); see §A.4.

Theorem 5.10 (Pell-sequence exclusion for $d \leq 43$)**.** *For each odd d in the admissible set $\{1, 3, 9, 11, 19, 27, 33, 43\}$ (the $d \leq 43$ with all prime factors in A): no prime $r \equiv 3 \pmod 8$ that divides a composite W_p and passes Condition (II) can have $\mathrm{ord}_r(\omega_3) = 4d$.*

Proof. Suppose, for contradiction, that $r \equiv 3 \pmod 8$ divides a composite W_p, passes Condition (II), and has $\mathrm{ord}_r(\omega_3) = 4d$ for some d in the admissible set. By the descent of §5.1, Condition (II) at r is equivalent to $d \mid W_{p-2}$; and by Theorem 5.5, $r \mid N_d = V_d^2 + 2$ (Proposition 5.7). We exhaust the admissible d:

$d = 1$: $N_1 = V_1^2 + 2 = 2^2 + 2 = 6 = 2 \cdot 3$. The only odd prime factor is 3, and $3 \mid W_p$ is impossible by Lemma 2.3. No valid r.

$d = 3$: $N_3 = V_3^2 + 2 = 14^2 + 2 = 198 = 2 \cdot 3^2 \cdot 11$. The odd prime factors $\equiv 3 \pmod 8$ are 3 (excluded) and 11. For $r = 11$: $\mathrm{ord}_{11}(2) = 10 = 2 \cdot 5$, so the unique prime p with $\mathrm{ord}_r(2) = 2p$ is $p = 5$. But $W_5 = 11$ is itself prime, so $r = 11$ does not divide any *composite* W_p. No valid r.

$d \in \{9, 11, 19, 27, 33, 43\}$: We use the complete factorisations of $N_d = V_d^2 + 2$ (computed with SymPy `factorint`, results in the supplementary archive; the largest factorisation is N_{43}, a 33-digit integer whose prime decomposition has six prime factors, all under 20 digits). For each prime factor r of N_d with $r \equiv 3 \pmod 8$ and $r > 11$, we compute $\mathrm{ord}_r(2)$ and check:

(1) If $v_2(\mathrm{ord}_r(2)) \neq 1$, then $r \nmid W_p$ for any prime p (by Lemma 2.4); r is excluded.
(2) If $\mathrm{ord}_r(2) = 2m$ with m composite or m not prime, then no prime p satisfies $\mathrm{ord}_r(2) = 2p$; r is excluded.
(3) Otherwise, $\mathrm{ord}_r(2) = 2p$ for a unique prime p, and Condition (II) at r requires $d \mid W_{p-2}$. One survivor must be set aside first: $r = 43$, with $\mathrm{ord}_{43}(2) = 14$ and hence $p = 7$, divides only $W_7 = 43$, which is itself prime — so $r = 43$ divides no *composite* W_p (the exclusion already used for $r = 11$ at $d = 3$). For every other case-3 survivor the divisibility $d \mid W_{p-2}$ fails — either the arithmetic progression of Theorem 6.7 ($p \equiv a_d \pmod{2m_d}$) is incompatible with $p = \mathrm{ord}_r(2)/2$, or $v_3(d) > v_3(p-2)$ (Theorem 5.4, valuation excess for the prime 3), or an analogous q-adic mismatch occurs for some $q \mid d$ with $q > 3$ — so $d \nmid W_{p-2}$, contradicting Condition (II).

Condition 1 or 2 alone disposes of the overwhelming majority of the primitive prime factors of N_d for $d \in \{9, 11, 19, 27, 33, 43\}$; condition 3 is needed only for a handful of survivors. The complete per-r verification table is archived with the paper (supplementary data). No admissible $d \leq 43$ yields a valid r.

For d not in the admissible set (i.e. $d \leq 43$ with some prime factor $q \notin A$): Theorem 5.3 gives $g \nmid W_{p-2}$, so $\mathrm{ord}_r(\omega_3)/4 \neq d$ in any Condition-(II)-passing scenario, regardless of the value of N_d. □

Corollary 5.11 (Inert factors with small G_r)**.** *If $r \equiv 3 \pmod 8$ divides a composite W_p and $G_r \leq 43$, then Condition (II) fails at r.*

Proof. Condition (II) gives $g \le G_r \le 43$, g odd. The prime factors of g are all in A (else Theorem 5.3 applies); so g is in the admissible set of Theorem 5.10, which excludes it. $\square$

5.4. **Lower bound on g.**

Lemma 5.12 (Class-III division)**.** *For $q \in A$ with $q > 3$ and $m_q = \mathrm{ord}_q(2)/2$: $q \mid W_{p-2}$ iff $m_q \mid p-2$.*

Proof. Let $n = p-2$, so n is odd. If $q \mid W_n$, then $2^n \equiv -1 \pmod{q}$; since $2^{m_q} \equiv -1 \pmod{q}$ and $\mathrm{ord}_q(2) = 2m_q$, this is equivalent to $n \equiv m_q \pmod{2m_q}$, in particular $m_q \mid n$. Conversely, if $m_q \mid n$, then n/m_q is odd (both m_q and n are odd), so $2^n = (2^{m_q})^{n/m_q} \equiv -1 \pmod{q}$, whence $q \mid 3W_n$ and, as $q > 3$, $q \mid W_n$. $\square$

Lemma 5.13 (Lower bound)**.** *For inert $r \mid W_p$ composite: $g > \log(2p)/\log(3+2\sqrt{2}) \approx 0.393 \cdot \log_2 p$.*

Proof. From $\mathrm{ord}_r(\omega_3) = 4g$: $\omega_3^{2g} = -1 \pmod{r}$, hence $V_{2g} \equiv 0 \pmod{r}$. Since $r \ge 2p+1$: $2p+1 \le V_{2g} < (3+2\sqrt{2})^g + 1$. $\square$

Product bound. For a given p:

$$G_{\max}(p) \;=\; 3^{v_3(p-2)} \cdot \prod_{\substack{q>3,\ q\in A,\\ m_q \mid (p-2)}} q^{v_q(W_{p-2})}$$

is an upper bound on any compatible g. The product ranges over all Class III primes, not only over primes $q \equiv 3 \pmod{8}$: Class III primes with $q \equiv 1 \pmod{8}$ can divide W_{p-2} and can occur in $(r+1)/4$ for inert r. When $G_{\max}(p) < 3$, no inert factor can pass Condition (II).

5.5. **Density limits.** Corollary 5.11 handles 845 of 869 known inert factors unconditionally (§B.4). A natural subclass of the residual configurations has density zero:

Proposition 5.14 (Density zero for pure Class III inert factors)**.** *The set of primes $r \equiv 3 \pmod{8}$ for which every odd prime factor of $(r+1)/4$ belongs to Class III has density zero within the primes $r \equiv 3 \pmod{8}$.*

Proof sketch. Let $S = \{q : v_2(\mathrm{ord}_q(2)) \ge 2\}$. S contains all $q \equiv 5 \pmod{8}$, so $\sum_{q\in S} 1/q = \infty$. The hypothesis forces $q \nmid (r+1)$ for every $q \in S$, equivalently $r \not\equiv -1 \pmod{q}$. For each finite $S' \subset S$, the primes satisfying these conditions have density at most $\prod_{q\in S'}(q-2)/(q-1)$ in the set of primes $\equiv 3 \pmod{8}$; since $\sum 1/(q-1) = \infty$ this product has infimum zero. $\square$

The full residual set is slightly larger than this subclass (it requires only g's prime factors to be Class III, not all of $(r+1)/4$'s), and Proposition 5.14 is a density-zero (not pointwise) statement; a naive GRH/Chebotarev argument in the style of Hooley [12] and Heath-Brown [11] fails to produce a pointwise closure. See §A.2 for the structural barrier.

5.6. **Multi-factor reduction.**

Definition 5.15 (Danger triple)**.** A danger triple (d, r, p) consists of: an odd integer $d > 43$ with every prime factor in A; a prime $r \equiv 3 \pmod 8$ that is primitive for the d-indexed Pell–Lucas family at V_{2d} and has $\mathrm{ord}_r(\omega_3) = 4d$ in $\mathbb{F}_{r^2}$; and the unique prime p with $\mathrm{ord}_r(2) = 2p$, satisfying $d \mid W_{p-2}$. A danger triple is phantom if W_p is prime; otherwise real.

By Theorem 5.5 and Lemma 5.8, any inert factor r of a composite W_p with $d > 43$ gives rise to a danger triple.

Theorem 5.16 (Multi-factor reduction)**.** *Let W_p be composite with every prime factor $r \equiv 3 \pmod 8$. If Condition (II) holds at every factor, then W_p has $t \geq 3$ prime factors counted with multiplicity, and each factor r_i has $d_i = \mathrm{ord}_{r_i}(\omega_3)/4 > 43$ with every prime factor of d_i in A.*

Proof. Three-or-more factors. Let t be the number of prime factors of W_p counted with multiplicity. Each prime factor $r_j \equiv 3 \pmod 8$, so the product $\prod r_j \equiv 3^t \pmod 8$, the product running over the t prime factors with multiplicity. Combined with $W_p \equiv 3 \pmod 8$ (Lemma 2.4): $3^t \equiv 3 \pmod 8$, forcing t odd. Since W_p is composite, $t \geq 3$.

Per-factor $d_i > 43$. Each factor r_i passes Condition (II), so $d_{r_i} \mid W_{p-2}$ (§5.1); hence every prime factor of d_{r_i} lies in A by Theorem 5.3 (the parity obstruction). Pell-sequence exclusion (Theorem 5.10) rules out $d_{r_i} \leq 43$ for an inert factor of a composite W_p. Therefore $d_{r_i} > 43$. □

Remark 5.17 (Known danger triples, $d \leq 1000$)*.* A primitive-divisor survey of the admissible $d \leq 1000$ has identified five danger triples:

$$(d, p) \in \{(57, 11),\ (57, 85{,}684{,}865{,}933),$$
$$(67, 24{,}822{,}855{,}876{,}938{,}710{,}967),\ (331, 17),\ (683, 13)\}.$$

Three are *phantom* (with W_p prime): $(57, 11)$ since $W_{11} = 683$, $(331, 17)$ since $W_{17} = 43{,}691$, $(683, 13)$ since $W_{13} = 2731$; the two real triples are disposed of at secondary factors (§B.4). The survey is complete — full factorisation of the Pell–Lucas number V_{2d} — only for the d whose V_{2d} was factored completely (notably $d \in \{57, 67\}$, via V_{114} and V_{134}); for the remaining admissible d the factorisation is partial (e.g. V_{1366} at $d = 683$ retains composite half-cofactors totalling more than five hundred digits), so the catalog was recorded as a lower bound; Proposition 5.18 below makes it exhaustive for every admissible $d \leq 400$, and it remains a lower bound only on $(400, 1000]$. The catalog itself is *not* used in the reduction: the cross-case argument of Section 6 routes through the Platinum Lemma — an r-indexed enumeration (Theorem 6.6) — and not through any d-indexed catalog; the census enters only through Corollary 5.19 and the diagonal closures.

Proposition 5.18 (Complete danger census through $d = 400$)**.** *For the admissible odd d with $43 < d \leq 400$ — the 34 values*

57, 59, 67, 81, 83, 99, 107, 121, 129, 131, 139, 163, 171, 177, 179, 201, 209,
211, 227, 243, 249, 251, 281, 283, 297, 307, 321, 331, 347, 361, 363, 379, 387, 393

— the danger triples (d, r, p) of Definition 5.15 are exactly the four rows of Remark 5.17 with $d \le 400$: the phantoms $(57, 683, 11)$ and $(331, 43{,}691, 17)$, and the real triples

$$(57,\ 171{,}369{,}731{,}867,\ 85{,}684{,}865{,}933), \qquad (67,\ 148{,}937{,}135{,}261{,}632{,}265{,}803,\ 24{,}822{,}855{,}876{,}938{,}710{,}9$$

In particular the $d \le 400$ part of that catalog is exhaustive, and no two danger triples with the same $d \le 400$ share the pinned exponent.

Verification. A danger triple at d has r primitive at V_{2d} (Definition 5.15, Lemma 5.8); non-admissible d carry none, by the Class III requirement of the definition. For each of the 34 admissible rungs the complete factorization of V_{2d} was assembled — from the fixed-d certificate bundle of §B.3, whose pinned Ψ_{4d} factorizations cover the 28 rungs in $(100, 400]$; from the diagonal certificates of Proposition A.9; and from the public factor database [19] with local factoring for the remaining rungs — and re-verified by the exact product identity, with multiplicities, against the recomputed V_{2d}; the 226 distinct prime factors occurring across the 34 rungs are each certified prime by APR-CL. For each of the 340 (rung, prime) pairs the danger prerequisites are then decided exactly as in Proposition A.9 — residue $r \equiv 3 \pmod 8$; primitivity $\mathrm{ord}_r(\omega_3) = 4d$, by order descent in $\mathbb{F}_{r^2}$; $v_2(\mathrm{ord}_r(2)) = 1$, by the square chain; primality of the pinned exponent $p = \mathrm{ord}_r(2)/2$ — together with condition (v), $d \mid W_{p-2}$, by the exact criterion $2^{p-2} \equiv -1 \pmod{3d}$. Exactly the four listed rows survive. Every verdict was reproduced by an independent re-implementation, and the survivor rungs' factorizations were additionally re-derived from scratch. Scripts (§B.5): `scripts/cp412_danger_census_d400.py` and `scripts/cp412b_t1_aprcl_sweep.py`; the per-rung certificate is archived with the supplementary data. □

Corollary 5.19 (Inert order-parameter floor $d_r > 400$). *Let W_p be composite and suppose Condition (II) holds at every prime factor. Then every inert prime factor $r \equiv 3 \pmod 8$ of W_p has $d_r = \mathrm{ord}_r(\omega_3)/4 > 400$. This subsumes the per-factor conclusion $d_r > 43$ of Theorems 5.10 and 5.16.*

Proof. Suppose $d_r \le 400$ for some inert factor r. Condition (II) at r gives $d_r \mid W_{p-2}$ with d_r odd (descent of §5.1); every prime factor of d_r is Class III (Theorem 5.3); $d_r > 43$ (Theorem 5.10, applicable as W_p is composite); r is primitive at V_{2d_r} (Lemma 5.8); and $\mathrm{ord}_r(2) = 2p$ (Theorem 6.1). So (d_r, r, p) is a real danger triple with $43 < d_r \le 400$, and Proposition 5.18 forces $p = 85{,}684{,}865{,}933$ or $p = 24{,}822{,}855{,}876{,}938{,}710{,}967$. At either exponent a secondary prime factor of W_p fails Condition (II) (§B.4): at the first, the inert $r_2 = 7{,}675{,}646{,}348{,}774{,}307{,}083$, by the parity obstruction (Proposition 6.30); at the second, the split $r_2 = 5{,}474{,}333{,}343{,}676{,}555{,}561{,}818{,}313$, where $\mathrm{ord}_{r_2}(\omega_3) = 4d_2$ with $d_2 = 3^3 \cdot 1021 \cdot p$ divisible by p itself, so $d_2 \nmid W_{p-2}$ ($p \nmid W_{p-2}$: $2^{p-2} \equiv 2^{-1} \not\equiv -1 \pmod p$ for $p > 3$) and Condition (II) fails at r_2. Either way Condition (II) fails at a prime factor of W_p, contradicting the hypothesis. □

Remark 5.20 (Two complementary slices). The census is the d-indexed complement of the r-indexed Platinum enumeration (Theorem 6.6): Platinum enumerates every locally passing inert factor with $r \le 10^{12}$ and unrestricted d_r; the census covers every $d_r \le 400$ with no bound on r. Between them, a locally passing inert factor of a composite W_p has $d_r > 400$ or is one of the two census rows (both at exponents closed by secondary factors), and has $r > 10^{12}$ or is one of the three Platinum rows (two closed; the third is X1). In the mixed-pair notation of §4.7, datum (ii) of any MPC-violating pair therefore carries $d_{r_{\mathrm{in}}} > 400$; on the all-inert branch, every member of $T_p^{>}$ at an undischarged exponent carries $d_r > 400$. The reduction chain of Theorem 7.4 is unchanged; the census enters as a floor, not as a new hypothesis.

6. Cross-case reductions from pinning

This section records the cross-case reductions that act uniformly across all candidate composite W_p, beyond the branch-by-branch analysis of Sections 4 and 5. None depends on Conjecture R3 or on computational input at a single factor; each is a structural consequence of $r \mid W_p$ with Condition (II). The end product (Theorem 6.15) reduces the all-inert branch of Conjecture 1.1 to PPPC together with the mild Wieferich–Wagstaff exclusion $W = 0$, using the Platinum enumeration (Theorem 6.6) as a stated computational input.

6.1. **Order-Pinning Lemma.**

Theorem 6.1 (Order-Pinning). *Let r be a prime factor of W_p with $r > 3$. Then* $\mathrm{ord}_r(2) = 2p$.

Proof. $r \mid W_p$ gives $2^p \equiv -1 \pmod r$, so $\mathrm{ord}_r(2) \mid 2p$ and $\mathrm{ord}_r(2) \nmid p$. Divisors of $2p$ not dividing p: $\{2, 2p\}$. If $\mathrm{ord}_r(2) = 2$ then $r \mid 3$, contradicting $r > 3$. □

Order-Pinning is the single most load-bearing structural identity in the paper. It converts the Wagstaff condition "$r \mid W_p$" into the sharp equation $\mathrm{ord}_r(2) = 2p$, which means every prime factor of W_p lies in the *exactly one* arithmetic progression $r \equiv 1 \pmod{2p}$ and shares a single prime p.

6.2. **Multi-Factor Pinning.**

Theorem 6.2 (Multi-Factor Pinning). *Let W_p be composite. All prime factors $r > 3$ of W_p share the common value* $\mathrm{ord}_r(2)/2 = p$. *Consequently, two composite Wagstaff numbers W_{p_1}, W_{p_2} with $p_1 \ne p_2$ share no prime factor $r > 3$.*

Proof. Immediate from Theorem 6.1. □

Multi-Factor Pinning is a *global* constraint relating all prime factors of one W_p to a common "pinned" parameter p; the same principle blocks any prime from being a factor of two different W_p.

6.3. **The finite danger catalog.**

Definition 6.3. A danger triple at d is a triple (r, p, d) with r an inert prime factor of W_p, $d = \mathrm{ord}_r(\omega_3)/4$, and (r, p, d) surviving every constraint of Section 5 (parity, Pell exclusion, Condition (II) pass at r); cf. Definition 5.15.

The primitive-divisor survey of §5.6 identifies the known danger triples (Remark 5.17).

Remark 6.4 (Phantom exclusion of the known catalog). The five known danger triples (Remark 5.17) have pairwise distinct pinned exponents, $p \in \{11,\ 85\,684\,865\,933,\ 24\,822\,855\,876\,938\,710\,967,\ 17,\ 13\}$. Since all prime factors of one W_p share a single pinned exponent (Theorem 6.2), no composite W_p carries two factors drawn from this list.

Remark 6.5 (The catalog is not load-bearing). Were the $d \leq 1000$ danger catalog known to be exhaustive, Remark 6.4 together with Theorem 5.16 would force any composite all-inert W_p passing Condition (II) to have at least two factors with $d_i > 1000$. The catalog is *not* known to be exhaustive (Remark 5.17: V_{2d} is only partially factored for most admissible d), so the reduction does not take this route. The cross-case bound of §6.7 instead invokes the Platinum Lemma (Theorem 6.6) — an r-indexed enumeration requiring no d-indexed completeness.

6.4. **Platinum Lemma.**

Theorem 6.6 (Platinum Lemma)**.** *For every prime p, at most one inert prime factor $r \leq 10^{12}$ of W_p passes Condition (II) at r.*

Verification. Every prime $r \equiv 3 \pmod 8$ with $r \leq 10^{12}$ is enumerated (§B.4); by Order-Pinning each has a unique pinned exponent $p = \mathrm{ord}_r(2)/2 < 5 \times 10^{11}$. Pell-sequence exclusion (Theorem 5.10) disposes of $d_r \leq 43$; for $d_r > 43$, Condition (II) at r — equivalently $d_r \mid W_{p-2}$ — is tested by direct evaluation of $\omega_3^{2W_{p-2}} \pmod r$. The enumeration exhibits exactly three inert factors $r \leq 10^{12}$ passing Condition (II), and these lie at three *distinct* exponents p (the local-pass witnesses of §B.4). Hence every prime p has at most one passing inert factor $r \leq 10^{12}$. Full scripts and certificates are archived. $\square$

The Platinum Lemma is the only computational input in the cross-case reductions; every downstream consequence including Theorem 6.15 is deterministic once the enumeration is accepted.

6.5. **Exact-AP characterization.**

Theorem 6.7 (Exact AP characterization)**.** *Let $q > 3$ be prime. If $q \notin A$, then $q \nmid W_{p-2}$ for every prime $p \geq 5$. If $q \in A$, set $m_q = \mathrm{ord}_q(2)/2$ (odd); then for every prime $p \geq 5$:*

$$q \mid W_{p-2} \iff p \equiv m_q + 2 \pmod{2m_q}.$$

In particular, for each $q \in A$ with $q > 3$ the set of admissible p forms a single AP of period $2m_q$; for $q \notin A$ the divisibility $q \mid W_{p-2}$ is impossible. The prime $q = 3 \in A$ is the lone exception to the period-$2m_q$ form: by Theorem 5.4, $3 \mid W_{p-2}$ iff $3 \mid p-2$.

Proof. Since $q > 3$, $\gcd(q, 3) = 1$, so $q \mid W_{p-2} = (2^{p-2}+1)/3$ iff $q \mid 2^{p-2}+1$ iff $2^{p-2} \equiv -1 \pmod q$. The latter forces $\mathrm{ord}_q(2) = 2m_q$ with $m_q \mid p-2$ and $m_q \nmid (p-2)/2$, i.e., $v_2(\mathrm{ord}_q(2)) = 1$; 2^{m_q} has order 2 so equals -1. The condition $2^{p-2} \equiv -1 \equiv 2^{m_q} \pmod q$ is equivalent to $p - 2 \equiv m_q \pmod{2m_q}$. □

Remark 6.8 (Prime powers and the valuation-adjusted modulus). For composite d, the prime-level periods in Theorem 6.7 must be lifted with valuations. If $q > 3$, $q \in A$, $m_q = \mathrm{ord}_q(2)/2$, and $a_q = v_q(2^{m_q} + 1)$, then for $e \geq 1$ and odd n,

$$q^e \mid W_n \iff n \equiv m_q q^{\max(0,e-a_q)} \pmod{2m_q q^{\max(0,e-a_q)}}.$$

For $q = 3$, Theorem 5.4 gives $v_3(W_n) = v_3(n)$, so $3^e \mid W_n$ iff $3^e \mid n$. Thus if

$$d = 3^{e_3} \prod_{q>3} q^{e_q}$$

is admissible, the exact condition $d \mid W_{p-2}$ is a single compatible system of congruences modulo the valuation-adjusted modulus

$$L_d = \mathrm{lcm}\Big(3^{e_3},\ 2m_q q^{\max(0,e_q-a_q)}\ :\ q > 3,\ q^{e_q} \| d\Big),$$

with the corresponding residue class determined by the displayed congruences. The shorthand $M_d = \mathrm{lcm}(m_q : q \mid d)$ is useful only for squarefree or heuristic prime-level discussions; it is not the exact modulus for general d.

Proposition 6.9 (Small-k valuation obstruction)**.** *Let $r = 2pk + 1$ be an inert local-pass candidate, with $p \geq 5$ prime, $r \equiv 3 \pmod 8$, $d = d_r = \mathrm{ord}_r(\omega_3)/4$, and*

$$d \mid \frac{r+1}{4} = \frac{pk+1}{2}, \qquad d \mid W_{p-2}.$$

Let $q^e \| d$.

If $q = 3$, then

$$3^e \mid 2k + 1.$$

If $q > 3$, then $q \in A$; with $m_q = \mathrm{ord}_q(2)/2$, $a_q = v_q(2^{m_q} + 1)$, and $b = \max(0, e - a_q)$, one has

$$q^b \mid 2k + 1.$$

In particular, for $k = 1$ one has $v_3(d) \leq 1$ and $e \leq a_q$ for every prime power $q^e \| d$ with $q > 3$.

Proof. Because q is odd and $q^e \mid d \mid (pk + 1)/2$, we have

$$pk \equiv -1 \pmod{q^e}.$$

In particular $q \nmid k$.

For $q = 3$, the condition $3^e \mid W_{p-2}$ is equivalent by Theorem 5.4 to $p \equiv 2 \pmod{3^e}$. Combining with $pk \equiv -1 \pmod{3^e}$ gives $2k \equiv -1 \pmod{3^e}$, hence $3^e \mid 2k + 1$.

For $q > 3$, the divisibility $q^e \mid W_{p-2}$ forces $q \in A$ by Theorem 6.7. The valuation-adjusted congruence of Remark 6.8 gives

$$p - 2 \equiv m_q q^b \pmod{2m_q q^b}, \qquad b = \max(0, e - a_q).$$

Modulo q^b this is $p \equiv 2 \pmod{q^b}$ (with nothing to show when $b = 0$). Reducing $pk \equiv -1 \pmod{q^e}$ modulo q^b and substituting $p \equiv 2$ yields $2k \equiv -1 \pmod{q^b}$.

If $k = 1$, the displayed divisibilities become $3^e \mid 3$ and $q^b \mid 3$ for every $q > 3$, so $e \leq 1$ for $q = 3$ and $b = 0$ for $q > 3$, equivalently $e \leq a_q$. □

6.6. **Pair separation.** Proposition 6.9 constrains each inert candidate separately. The same-p pinning also couples two candidates at one exponent; until now the only pair-level constraint was the immediate consequence $2p \mid \gcd(r_1 - 1, r_2 - 1)$ of Order-Pinning (Remark 6.20). The next theorem is the first pair constraint beyond it. The mechanism is elementary: everything two candidates *share* must divide their k-gap.

Theorem 6.10 (Pair separation)**.** *Let $p \geq 5$ be prime and let $r_i = 2pk_i + 1$ ($i = 1, 2$) be distinct primes with $r_i \equiv 3 \pmod 8$ and $k_1 < k_2$. Let d_1, d_2 be odd positive integers with $d_i \mid (r_i + 1)/4$. Then:*

(1) $k_1 \equiv k_2 \equiv p \pmod 4$*;*
(2) $\gcd(pk_1 + 1,\, pk_2 + 1) \mid k_2 - k_1$*;*
(3) $4 \gcd(d_1, d_2) \mid k_2 - k_1$*; equivalently,* $8p \gcd(d_1, d_2) \mid r_2 - r_1$*;*
(4) *if moreover (d, r_1, p) and (d, r_2, p) are danger triples with the same parameter $d = d_1 = d_2$ (Definition 5.15; here $d_i = d_{r_i} = \mathrm{ord}_{r_i}(\omega_3)/4$, an odd divisor of $(r_i + 1)/4$ by §5.1), then $4d \mid k_2 - k_1$ with $d \geq 57$; hence $k_2 - k_1 \geq 228$ and $r_2 - r_1 \geq 456\,p$.*

Items (1)–(3) use neither Condition (II) nor the primality of p: they hold verbatim for any odd $p \geq 3$ and any odd $d_i \mid (r_i + 1)/4$. Item (4) is where the Condition-(II) context enters, through $d > 43$ (Pell-sequence exclusion) and admissibility of d (every prime factor in A, forced by $d \mid W_{p-2}$ and Theorem 6.7).

Proof. (1) $r_i \equiv 3 \pmod 8$ means $2pk_i \equiv 2 \pmod 8$, i.e. $pk_i \equiv 1 \pmod 4$. Since p is odd, $p^2 \equiv 1 \pmod 4$, so multiplying by p gives $k_i \equiv p \pmod 4$ for both i; in particular $4 \mid k_2 - k_1$.

(2) The integer identity $k_2(pk_1 + 1) - k_1(pk_2 + 1) = k_2 - k_1$ shows that every common divisor of $pk_1 + 1$ and $pk_2 + 1$ divides $k_2 - k_1$.

(3) Set $g = \gcd(d_1, d_2)$, an odd integer. First, $\gcd(g, p) = 1$: a common prime divisor q of g and p would divide $d_1 \mid (r_1 + 1)/4 = (pk_1 + 1)/2$, hence $pk_1 + 1$, and also pk_1 (as $q \mid p$), forcing $q \mid 1$. Next, g divides both $(pk_i + 1)/2$, hence their difference $p(k_2 - k_1)/2$; by (1) we have $4 \mid k_2 - k_1$, so $(k_2 - k_1)/2$ is an (even) integer, and coprimality to p gives $g \mid (k_2 - k_1)/2$, i.e. $2g \mid k_2 - k_1$.

Combining $2g \mid k_2 - k_1$ with $4 \mid k_2 - k_1$ and g odd yields $4g \mid k_2 - k_1$. Multiplying by $2p$: $8pg \mid 2p(k_2 - k_1) = r_2 - r_1$.

(4) Here $d_1 = d_2 = d$, so (3) gives $4d \mid k_2 - k_1$, and $k_2 > k_1$ gives $k_2 - k_1 \geq 4d$. In a danger triple, $d > 43$ and every prime factor of d lies in A; the least odd integer > 43 with all prime factors in A is $57 = 3 \cdot 19$: among $45, \dots, 55$ the odd values $45 = 3^2 \cdot 5$, $49 = 7^2$, $51 = 3 \cdot 17$, $55 = 5 \cdot 11$ contain the non-Class-III primes $5, 7, 17$, while $47 \notin A$ ($\mathrm{ord}_{47}(2) = 23$, odd) and $53 \notin A$ ($v_2(\mathrm{ord}_{53}(2)) = 2$). Hence $d \geq 57$, $k_2 - k_1 \geq 228$, and $r_2 - r_1 = 2p(k_2 - k_1) \geq 456\,p$. $\square$

The next proposition transfers valuations *across* the pair. It strictly strengthens Proposition 6.9 in two ways: part (a) lets the factor with the higher q-valuation constrain the other factor's k, and part (b) routes a constraint through the half-order of one prime divisor of d acting on another.

Proposition 6.11 (Pair-coupled valuation transfer). *In the setting of Theorem 6.10, assume additionally that $d_i \mid W_{p-2}$ for both i (condition (v) at both candidates). Then every prime $q > 3$ dividing $d_1 d_2$ lies in A; write $m_q = \mathrm{ord}_q(2)/2$ and $a_q = v_q(2^{m_q} + 1)$ as in Remark 6.8. For every such q:*

*(a) (*cross-factor channel*) with $e_i = v_q(d_i)$, $E = \max(e_1, e_2)$ and $b = \max(0,\, E - a_q)$:*

$$q^{\min(e_i, b)} \,\Big|\, 2k_i + 1 \qquad (i = 1, 2);$$

*(b) (*half-order channel*) for every prime ℓ with $v = v_\ell(m_q) \geq 1$ and $e = v_\ell(d_i) \geq 1$ for some i:*

$$\ell^{\min(v, e)} \,\Big|\, 2k_i + 1.$$

Proof. Both d_i divide W_{p-2}, hence so does $\mathrm{lcm}(d_1, d_2)$; in particular every prime $q > 3$ dividing $d_1 d_2$ divides W_{p-2} and so lies in A (Theorem 6.7).

(a) $q^E \mid \mathrm{lcm}(d_1, d_2) \mid W_{p-2}$, so the valuation-adjusted congruence of Remark 6.8 gives $p - 2 \equiv m_q q^b \pmod{2 m_q q^b}$; since $q \nmid m_q$, reducing modulo q^b gives $p \equiv 2 \pmod{q^b}$. For each i, $q^{e_i} \mid d_i \mid (pk_i + 1)/2$ gives $pk_i \equiv -1 \pmod{q^{e_i}}$. Reducing both congruences modulo $q^{\min(e_i, b)}$ and substituting $p \equiv 2$ yields $2k_i \equiv -1$.

(b) $q \mid W_{p-2}$ gives $m_q \mid p - 2$ (Lemma 5.12), hence $p \equiv 2 \pmod{\ell^v}$. Since d_i is odd, ℓ is odd, and $\ell^e \mid d_i \mid (pk_i + 1)/2$ gives $pk_i \equiv -1 \pmod{\ell^e}$. Reduce modulo $\ell^{\min(v,e)}$ and substitute $p \equiv 2$. $\square$

Remark 6.12 (Strict strengthenings of the small-k obstruction). Proposition 6.9 is the single-candidate shadow of Proposition 6.11: its exponent $\max(0, e_i - a_q)$ never exceeds the $\min(e_i, b)$ of part (a), and part (a) is strictly stronger whenever the *other* factor carries the higher valuation ($e_i < E$). Part (b) needs no pair at all — its proof uses only $q \mid W_{p-2}$ — so it already strengthens Proposition 6.9 for a single candidate (r, p, k, d) with $q \mid d$. Example: $d = 737 = 11 \cdot 67$ is admissible with $a_{11}, a_{67} \geq 1$, so Proposition 6.9

imposes no constraint at $k = 1$; but $m_{67} = 33 = 3 \cdot 11$, and part (b) with $q = 67$, $\ell = 11$ forces $11 \mid 2k+1$, killing $k = 1$ (and every $k \not\equiv 5 \pmod{11}$).

Remark 6.13 (No off-diagonal kill at the congruence level). Two honest limits, both witnessed constructively. (i) The natural strengthening of Proposition 6.9 to $\mathrm{rad}(d) \mid 2k+1$ is *false*: the inert prime $r = 43 = 2 \cdot 7 \cdot 3 + 1$ has $\mathrm{ord}_{43}(\omega_3) = 44$, giving $(r, p, k, d) = (43, 7, 3, 11)$ with $11 \mid W_5$, yet $11 \nmid 2k+1 = 7$. The reason is structural: the progression condition on p (modulus $2m_q q^b$) and the divisor congruence $pk \equiv -1 \pmod{q}$ live on coprime moduli unless a channel of Proposition 6.11 happens to link them. (ii) No combination of the constraints of this subsection rules out bounded-k *off-diagonal* pairs: at $(k_1, k_2) = (1, 5)$, $(d_1, d_2) = (57, 59)$, every prime $p \equiv 166{,}781 \pmod{1{,}170{,}324}$ (the displayed residue is itself prime) satisfies simultaneously every congruence imposed by items (1)–(3), by Proposition 6.9, by Proposition 6.11, and by condition (v) for both d_i (Theorem 6.7: $3 \mid p-2$, $p - 2 \equiv 9 \pmod{18}$, $p - 2 \equiv 29 \pmod{58}$). These congruences force only the modulus $2^2 \cdot 3^2 \cdot 29 = 1{,}044$; the finer modulus $1{,}170{,}324 = 1{,}044 \cdot 19 \cdot 59$ of the displayed witness carries the realizability factors that fix the prescribed ω_3-orders ($d_1 = 57 = 3 \cdot 19$, $d_2 = 59$), not additional forced congruences. What the witness class does not supply is *realizability*: a prime p in the class for which $2p+1$ and $10p+1$ are both prime, both inert, with the prescribed ω_3-orders. Separating congruence-consistency from realizability at a fixed divisor is exactly the analytic wall of §A.3: the separation theorem yields repulsion for free, while attraction — what PPPC must exclude — is untouched at this level.

Remark 6.14 (The $r+1$ side of pair gcds). For any pair as in Theorem 6.10, $v_2(r_i + 1) = 2$ and $(r_i + 1)/4 = (pk_i + 1)/2$ is odd, so

$$\gcd(r_1 + 1,\, r_2 + 1) \;=\; 2\gcd(pk_1 + 1,\, pk_2 + 1),$$

which is 4 times an odd integer and divides $2(k_2 - k_1)$ by item (2). This explains a regularity of the candidate-pair scans (§7.3): while the $r - 1$ side carries the forced large gcd $2p \mid \gcd(r_1 - 1, r_2 - 1)$, the observed values of $\gcd(r_1 + 1, r_2 + 1)$ on real candidate pairs are tiny — 4 and 12 in the inert pairs examined — being capped by twice the k-gap rather than scaled by p.

6.7. **Second-Moment Reduction.** Write E_3 for the cardinality of the set of primes $p \geq 5$ for which W_p is composite, every factor is inert, and Condition (II) holds at every factor — the set of p contributing to the all-inert branch of Conjecture 1.1. For each p put

$$T_p^{>} \;=\; \{\, r : r \mid W_p,\ r \equiv 3 \pmod{8},\ r > 10^{12},\ d_r > 43,\ d_r \mid W_{p-2} \,\},$$

the set of residual inert danger factors at p — those not excluded by Pell-sequence exclusion or the Platinum Lemma.

Theorem 6.15 (Second-Moment Reduction)**.** *Deterministically, given the Platinum enumeration of Theorem 6.6,*

$$E_3 \;\leq\; \Pi + W,$$

where

$$\Pi \;:=\; \sum_{p\ge 5}\binom{|T_p^{>}|}{2},$$

and W is the number of primes $p \ge 5$ for which W_p is composite, every prime factor of W_p is inert and passes Condition (II), and W_p is not squarefree. The inequality $E_3 \le \Pi + W$ is a deterministic consequence of the structural lemmas once the finite Platinum enumeration is accepted. The enumeration enters only through the cutoff $r > 10^{12}$ in the definition of $T_p^{>}$; Remark 6.16 records a computational-input-free form of the bound, with a triple-count residual in place of Π.

Proof. Step 1 (split by squarefreeness). The set of p counted by E_3 splits according to whether W_p is squarefree: $E_3 = E_3^{\mathrm{sf}} + W$, where E_3^{sf} counts the p with W_p squarefree (all factors inert, all passing Condition (II)) and W counts those with W_p not squarefree — exactly the set in the statement.

Step 2 (the squarefree part). Let $p \in E_3^{\mathrm{sf}}$. By Theorem 5.16, W_p has $t \ge 3$ inert prime factors counted *with multiplicity*; since W_p is squarefree, these are ≥ 3 *distinct* inert primes $r_1, r_2, r_3, \ldots$, each $\equiv 3 \pmod 8$, each passing Condition (II) — so $d_{r_i} \mid W_{p-2}$ — and (Theorem 5.16) each with every prime factor of d_{r_i} in A. Pell-sequence exclusion (Theorem 5.10) rules out $d_{r_i} \le 43$ for an inert factor of a composite W_p, so $d_{r_i} > 43$ for every i. By the Platinum Lemma (Theorem 6.6), at most one of these factors satisfies $r_i \le 10^{12}$; hence at least two satisfy $r_i > 10^{12}$. Each such factor then has $r > 10^{12}$, $d_r > 43$, and $d_r \mid W_{p-2}$, so it lies in $T_p^{>}$. Therefore $|T_p^{>}| \ge 2$ for every $p \in E_3^{\mathrm{sf}}$.

Step 3 (integer inequality and summation). For every integer $X \ge 0$: $\mathbf{1}_{X\ge 2} \le \binom{X}{2}$ (verified directly for $X \le 4$, a fortiori for $X \ge 3$). Hence

$$E_3^{\mathrm{sf}} \;\le\; \#\{p : |T_p^{>}| \ge 2\} \;\le\; \sum_{p}\binom{|T_p^{>}|}{2} \;=\; \Pi.$$

Combining with Step 1, $E_3 = E_3^{\mathrm{sf}} + W \le \Pi + W$. □

Remark 6.16 (Computational-input-free form of the bound). The Platinum Lemma (Theorem 6.6) enters the proof above only at Step 2, to force $r_i > 10^{12}$ for at least two factors; dropping it costs nothing essential. Write T_p' for the set of inert prime factors r of W_p with $d_r > 43$ and $d_r \mid W_{p-2}$, *without* the lower cutoff $r > 10^{12}$. For every p counted by E_3 with W_p squarefree (the set E_3^{sf} of Step 1), Theorem 5.16 together with Pell-sequence exclusion (Theorem 5.10) supplies at least three distinct inert factors with $d_r > 43$, all lying in T_p', so $|T_p'| \ge 3$. The integer inequality $\mathbf{1}_{X\ge 3} \le \binom{X}{3}$ then yields, with no computational input whatever,

$$E_3 \;\le\; \Pi_3' + W, \qquad \Pi_3' \;:=\; \sum_{p\ge 5}\binom{|T_p'|}{3}.$$

The residual $\Pi_3' = 0$ is a different cutoff-free triple-count statement: no prime p admits three distinct inert factors r with $d_r > 43$ and $d_r \mid W_{p-2}$, regardless of size. It is *not* implied by the above-cutoff PPPC alone, because PPPC only controls factors with $r > 10^{12}$. Conversely, $\Pi_3' = 0$ is not the same as PPPC. Thus the computational-input-free form replaces PPPC by a separate triple residual. The Platinum Lemma is used in the main statement to convert the deterministic three-factor information into the sharper above-cutoff pair-count Π, whose vanishing is the clean free-standing conjecture PPPC.

Remark 6.17 (The residual W is the Wieferich–Wagstaff branch)*.* W counts the composite, all-inert, Condition-(II)-passing W_p that fail to be squarefree. A prime r with $v_r(W_p) \geq 2$ satisfies $r^2 \mid 3W_p = 2^p + 1$; since $\mathrm{ord}_r(2) = 2p$ (Order-Pinning, Theorem 6.1), the congruence $2^p \equiv -1 \pmod{r^2}$ forces $\mathrm{ord}_{r^2}(2) = 2p$ as well, hence $\mathrm{ord}_{r^2}(2) = \mathrm{ord}_r(2)$, which is exactly the condition $2^{r-1} \equiv 1 \pmod{r^2}$ — i.e. r is a base-2 *Wieferich prime.* This is the **(E3) Wieferich–Wagstaff condition** of §A.2. Passing Condition (II) at the repeated factor in fact requires *more* than that r be base-2 Wieferich, and the extra requirement is half-automatic. Condition (II) concerns $\omega_3 = \alpha^2$, and the inert factor already gives $\alpha^{N+1} \equiv -1 \pmod r$; write the modulo-$r^2$ defect as

$$\alpha^{N+1} \;=\; -1 + r\,(u + v\sqrt{2}) \quad \text{in } \mathbb{Z}[\sqrt{2}]/r^2, \qquad (u,v) \in (\mathbb{Z}/r)^2,$$

so the factor passes Condition (II) modulo r^2 iff $(u,v) = (0,0)$. The *rational* part vanishes unconditionally: since $N(\alpha) = -1$ and $N+1 = 4W_{p-2}$ is even, $N(\alpha^{N+1}) = (-1)^{N+1} = 1$, while $N(-1 + r(u + v\sqrt{2})) \equiv 1 - 2ru \pmod{r^2}$, forcing $u \equiv 0 \pmod r$. The residual requirement is thus the single congruence $v \equiv 0 \pmod r$ — equivalently $\mathrm{ord}_{r^2}(\omega_3) = \mathrm{ord}_r(\omega_3)$, the order not growing (the alternative $\mathrm{ord}_{r^2}(\omega_3) = 4d_r\,r$ would force $4d_r r \mid 4W_{p-2}$, hence $r \mid W_{p-2}$, impossible as $\gcd(W_p, W_{p-2}) = 1$) — a base-$(1+\sqrt{2})$, or *Pell–Wall–Sun–Sun-type*, condition independent of the rational base-2 one. A repeated factor of a Condition-(II)-passing W_p therefore carries two Wieferich-type conditions at once, of joint heuristic density r^{-2} rather than r^{-1}; this makes $W = 0$ still more strongly expected, though no unconditional proof follows — each is a Wieferich-type non-existence statement, open in general. The only base-2 Wieferich primes below 2^{64} are 1093 and 3511, congruent to 5 and 7 (mod 8) respectively, so no base-2 Wieferich prime $\equiv 3 \pmod 8$ occurs below 2^{64} [16]; consequently $W = 0$ whenever every prime factor of W_p lies below 2^{64}, and conjecturally $W = 0$ for all p. Thus W is a third residual — the Wieferich–Wagstaff branch — alongside Π (PPPC) and the split-branch residual; see §A.2.

Corollary 6.18 (Residual pair-count bound)**.** *If $\Pi = 0$ and $W = 0$ — equivalently, if no composite W_p admits two distinct residual inert danger factors $r_1, r_2 > 10^{12}$ at the same p, and no composite all-inert Condition-(II)-passing W_p has a repeated prime factor — then $E_3 = 0$, and Conjecture 1.1* *holds in the all-inert branch. Any finite bounds on* Π *and* W *yield a finite bound on* E_3*.*

Remark 6.19 (No independence or heuristic hypothesis). Every step in Theorem 6.15 is deterministic. The proof uses only the squarefree split, the Pell-sequence exclusion, the Platinum enumeration (Theorem 6.6), and the integer inequality $\mathbf{1}_{X\geq 2} \leq \binom{X}{2}$. This is in contrast with heuristic third-moment counts in earlier formulations, which are inputs to conjectural density estimates on Π, not to Theorem 6.15 itself.

Remark 6.20 (Structure of $T_p^{>}$ and shared pinning). By Multi-Factor Pinning every element of $T_p^{>}$ satisfies $\mathrm{ord}_r(2) = 2p$ for the *same* p, so a pair $\{r_1, r_2\} \subset T_p^{>}$ is much more tightly constrained than a pair of independent danger primes. The Exact-AP characterization (Theorem 6.7 and Remark 6.8) sharpens this further: both factors share the pinned p and each $d_{r_i} \mid W_{p-2}$ is a rigid valuation-adjusted congruence condition.

6.8. **The W-residual trichotomy.** The residual W of Theorem 6.15 can be decomposed further. The engine is an elementary valuation identity sharpening the order argument of Remark 6.17.

Lemma 6.21 (Valuation transfer)**.** *For every prime* $p \geq 5$ *and every prime factor* $r > 3$ *of* W_p*:*

$$v_r(W_p) \;=\; v_r\big(2^{\,r-1} - 1\big)\,.$$

Proof. Since $r > 3$, $v_r(W_p) = v_r(2^p + 1)$. Order-Pinning (Theorem 6.1) gives $\mathrm{ord}_r(2) = 2p$, so $r \mid 2^p + 1$ and $r \nmid 2^p - 1$; as $2^{2p} - 1 = (2^p - 1)(2^p + 1)$, this yields $v_r(2^{2p} - 1) = v_r(2^p + 1)$. By Lemma 2.3, $r \equiv 1 \pmod{2p}$; write $r - 1 = 2pk$ with $1 \leq k < r$. The lifting-the-exponent lemma applied to $x^k - 1$ with $x = 2^{2p} \equiv 1 \pmod r$ gives

$$v_r\big(2^{\,r-1} - 1\big) = v_r\Big((2^{2p})^k - 1\Big) = v_r(2^{2p} - 1) + v_r(k) = v_r(2^{2p} - 1),$$

since $0 < k < r$ forces $v_r(k) = 0$. Combining the two displays: $v_r(2^{\,r-1} - 1) = v_r(2^p + 1) = v_r(W_p)$. $\square$

Definition 6.22 (Super-Wieferich prime)**.** A prime r is *base-*2 *super-Wieferich* if $r^3 \mid 2^{\,r-1} - 1$, i.e. $v_r(2^{\,r-1} - 1) \geq 3$. No base-2 super-Wieferich prime is known in *any* congruence class: the only base-2 Wieferich primes below 2^{64}, namely 1093 and 3511, both have $v_r(2^{\,r-1} - 1) = 2$ exactly [16].

Corollary 6.23 (Multiplicity forces Wieferich behaviour)**.** *Let* $r > 3$ *be a prime factor of* W_p*. If* $v_r(W_p) \geq 2$*, then* r *is a base-*2 *Wieferich prime; if* $v_r(W_p) \geq 3$*, then* r *is base-*2 *super-Wieferich.*

We call the three exponents

$$p \;\in\; \{\,1{,}251{,}488{,}009,\quad 10{,}916{,}765{,}939,\quad 85{,}684{,}865{,}933\,\}$$

at which the Platinum enumeration records a local-pass inert factor $r \leq 10^{12}$ (Theorem 6.6, §B.4) the *witness exponents.*

Theorem 6.24 (W-residual trichotomy)**.** *With E_3, $T_p^{>}$ and Π as in §6.7, and given the Platinum enumeration (Theorem 6.6), define*

$$\begin{aligned} W_1 &= \#\{\, p \in E_3 : \ W_p = r^t \text{ is a prime power} \,\}, \\ W_2 &= \#\{\, p \in E_3 : \ W_p \text{ has exactly two distinct prime factors,} \\ &\qquad\qquad\qquad \text{the smaller of them} \le 10^{12} \,\}. \end{aligned}$$

Then:

(a) $E_3 \le \Pi + W_1 + W_2$.

(b) Every p counted by W_1 has $W_p = r^t$ with t odd, $t \ge 3$, and r a base-2 super-*Wieferich prime $\equiv 3 \pmod 8$; in particular $r > 2^{64}$, and $(2^p+1)/3 = y^q$ is solvable with integers $y > 1$, $q \ge 3$ — a Nagell–Ljunggren instance at $x = -2$, excluded unconditionally by Lemma 6.26 below.*

(c) Every p counted by W_2 is a witness exponent, and $W_p = r_{\text{wit}} \cdot s^b$ where $r_{\text{wit}} \le 10^{12}$ is the Platinum witness factor at p, $b \ge 2$ is even, and $s > 2^{64}$ is a base-2 Wieferich prime $\equiv 3 \pmod 8$. In particular $W_2 \le 3$, the number of witness exponents.

All three buckets are defined by per-factor local *passes — the defining property of E_3 — and no global-pass assumption enters the statement or the proof.*

Proof. Partition $E_3 = \#\{p \in E_3 : |T_p^{>}| \ge 2\} + \#\{p \in E_3 : |T_p^{>}| \le 1\}$. The first part is at most Π by the integer inequality $\mathbf{1}_{X \ge 2} \le \binom{X}{2}$, exactly as in Step 3 of Theorem 6.15.

Let $p \in E_3$ with $|T_p^{>}| \le 1$. By the definition of E_3, W_p is composite, every prime factor is inert, and Condition (II) holds at every factor; hence each factor r has $d_r \mid W_{p-2}$ (§5.1) and $d_r > 43$ (Theorem 5.10, applicable since W_p is composite and Condition (II) holds at r). A factor $r \le 10^{12}$ then satisfies $p \le (r-1)/2 < 5 \times 10^{11}$ (Lemma 2.3), so (p, r) is a row of the Platinum enumeration that passes Condition (II) locally: by Theorem 6.6 and §B.4, at most one factor $\le 10^{12}$ exists at p, and if one exists then p is a witness exponent. A factor $r > 10^{12}$ lies in $T_p^{>}$.

Write D for the set of *distinct* prime factors of W_p. If W_p were squarefree, Theorem 5.16 would give $|D| \ge 3$ distinct factors, of which at most one is $\le 10^{12}$, so $|T_p^{>}| \ge 2$ — contradiction. So W_p is not squarefree. If $|D| \ge 3$: again at least two factors exceed 10^{12}, so $|T_p^{>}| \ge 2$ — contradiction; hence $|D| \le 2$. If $|D| = 2$ with both factors $> 10^{12}$: $|T_p^{>}| = 2$ — contradiction. Two cases remain.

Case $|D| = 1$. Then $W_p = r^t$, and t, the total multiplicity, is odd by Lemma 2.4; since W_p is composite, $t \ge 3$. Lemma 6.21 gives $v_r(2^{r-1} - 1) = v_r(W_p) = t \ge 3$, so r is base-2 super-Wieferich (Corollary 6.23); and $r \equiv 3 \pmod 8$ since all factors are inert. Below 2^{64} the only base-2 Wieferich primes are $1093 \equiv 5$ and $3511 \equiv 7 \pmod 8$, with $v_r(2^{r-1} - 1) = 2$ exactly, so $r > 2^{64}$. The equation $W_p = r^t$ exhibits $(2^p + 1)/3 = y^q$ with $y = r$, $q = t \ge 3$. These p are counted by W_1.

Case $|D| = 2$ *with the smaller factor* $\leq 10^{12}$. The smaller factor passes locally, so (as shown above) it is the Platinum witness factor r_{wit} at p and p is a witness exponent. Write $W_p = r_{\mathrm{wit}}^a s^b$ with $a, b \geq 1$; here $s > 10^{12}$, since at most one locally passing factor is $\leq 10^{12}$ (as shown above) and both factors pass. The total multiplicity $a + b$ is odd (Lemma 2.4) and ≥ 2, hence $a + b \geq 3$. If $a \geq 2$, Corollary 6.23 would make $r_{\mathrm{wit}} \leq 10^{12} < 2^{64}$ a base-2 Wieferich prime $\equiv 3 \pmod 8$ — none exists below 2^{64}. So $a = 1$ and $b = (a + b) - 1 \geq 2$ is even; s is repeated, hence a base-2 Wieferich prime $\equiv 3 \pmod 8$, hence $s > 2^{64}$. These p are counted by W_2, and W_2 is at most the number of witness exponents.

Every $p \in E_3$ with $|T_p^{>}| \leq 1$ therefore falls in the W_1 or W_2 bucket, proving (a)–(c). □

Remark 6.25 (Pell–Wall–Sun–Sun condition for global passers). The buckets of Theorem 6.24 are local-pass-defined, and Condition (II) modulo r^2 does *not* follow from local passes at the individual factors. If, however, W_p moreover passes Condition (II) *globally* — the only case Conjecture 1.1 needs — then the congruence holds modulo r^2 for any repeated factor r, and the analysis of Remark 6.17 applies: the repeated factor additionally satisfies the Pell–Wall–Sun–Sun-type condition $\mathrm{ord}_{r^2}(\omega_3) = \mathrm{ord}_r(\omega_3)$, a second Wieferich-type event on top of the base-2 one. The Wieferich and super-Wieferich conclusions of Theorem 6.24 themselves require no global hypothesis. For *squarefree* W_p the situation is rigid in the opposite direction: by the Chinese Remainder isomorphism $\mathbb{Z}[\sqrt{2}]/(W_p) \cong \prod_i \mathbb{Z}[\sqrt{2}]/(r_i)$, the global Condition (II) is *exactly* the conjunction of the local passes — the same element $\omega_3^{2W_p - 2}$ must equal -1 in every component, and Lemma 2.8 makes the (-1)-landing condition automatic and simultaneous at every inert factor — so a global pass adds nothing beyond the conjunction of local passes, and the local/global gap is confined to repeated factors.

The prime-power bucket can in fact be emptied unconditionally: the equation behind it is a Nagell–Ljunggren instance at a small base, and the negative-base case of that equation is closed in the literature.

Lemma 6.26 (W_p is never a perfect power). *For every odd prime* $p \geq 5$ *there are no integers* $y > 1$, $q \geq 2$ *with* $W_p = y^q$. *In particular* $W_1 = 0$, *unconditionally.*

Proof. Suppose $W_p = y^q$ with integers $y > 1$, $q \geq 2$; we may take q prime. For $p \geq 3$ one has $2^p + 1 \equiv 1 \pmod 8$, so $W_p \equiv 3^{-1} \equiv 3 \pmod 8$; since squares are $\equiv 0, 1, 4 \pmod 8$, this rules out $q = 2$, and $q \geq 3$ is odd. For odd p,

$$W_p = \frac{2^p + 1}{3} = \frac{(-2)^p - 1}{(-2) - 1},$$

so $W_p = y^q$ is an instance of the Nagell–Ljunggren equation $(x^n - 1)/(x - 1) = y^q$ with $x = -2$, $n = p > 2$. By the consolidated statement of Bennett and

Levin [2, Proposition 1] — which incorporates the negative-x theorem of Bugeaud and Mignotte [7] — every solution with $|x|, |y|, q > 1$ and $n > 2$, other than the four known solutions

$$(x, y, n, q) \in \{(3, 11, 5, 2), (7, 20, 4, 2), (18, 7, 3, 3), (-19, 7, 3, 3)\},$$

satisfies $|x| \geq 10^4$. Since $|-2| = 2 < 10^4$ and $x = -2$ appears in none of the four, no solution with $x = -2$ exists. (Two independent walls corroborate the instance: the same proposition also gives a prime divisor of x congruent to $1 \bmod q$, and the only prime divisor of -2 is 2, which excludes $x = -2$ without the 10^4 computation; and Bugeaud, Cao and Mignotte [6] had already solved $(2^n + 1)/3 = y^q$ directly for exponents greater than 1.) □

Corollary 6.27. *The trichotomy of Theorem 6.24(a) reads $E_3 \leq \Pi + W_2$: the prime-power bucket is empty for every p.*

Theorem 6.24(b) retains independent interest as a cross-check: any hypothetical solution of $W_p = y^q$ would also force a base-2 super-Wieferich prime $\equiv 3 \pmod 8$ above 2^{64}, so the bucket was doubly walled.

Remark 6.28 (Bucket-level weakening). With $W_1 = 0$ settled unconditionally (Lemma 6.26), the bucket-level weakening concerns only W_2. Every p counted by W_2 has W_p non-squarefree, all-inert and locally passing everywhere, hence is counted by W; so

$$W = 0 \implies W_2 = 0,$$

and the residual $W_2 = 0$ is weaker *as a hypothesis* than the blanket $W = 0$ of Theorem 7.2. (W_1 needs no hypothesis at all, by Lemma 6.26.) The weakening is claimed at the bucket level only.

6.8.1. *The witness-exponent statement X1.* The W_2 bucket, and the small-partner leg of the mixed-pair reduction (§7.7), are pinned to the three witness exponents. Two of them are discharged by explicit failing factors; the third is the content of a single-integer statement.

Conjecture 6.29 (X1, at-a-factor form)**.** Condition (II) fails at some prime factor of $W_{10916765939}$.

$W_{10916765939}$ *is* composite: the Platinum witness $r = 152{,}834{,}723{,}147 = 14p + 1$ divides it and passes Condition (II) locally (§B.4). A directed scan of the progression $r = 2pk + 1$ at this exponent, complete for $k \leq 10^{12}$, finds no second prime factor: the witness factor itself, at $k = 7$, is the only hit. The at-a-factor form is deliberate: a single failing factor removes the exponent from every local-pass-defined set — in particular from E_3, hence from the W_2 bucket — and a fortiori excludes a global pass. The global form ("W_p fails Condition (II)") would *not* by itself empty the non-squarefree W_2 bucket, since local passes at every factor do not imply a global pass when W_p is not squarefree. The at-a-factor form is also exactly what the discharges below establish.

Proposition 6.30 (Discharged witness exponents)**.** *Condition (II) fails at an explicit prime factor of W_p for two of the three witness exponents:*

(a) $p = 85{,}684{,}865{,}933$*: the prime*

$$r_2 = 7{,}675{,}646{,}348{,}774{,}307{,}083 \equiv 3 \pmod 8$$

divides W_p with $\operatorname{ord}_{r_2}(2) = 2p$ *and*

$$d_{r_2} = \operatorname{ord}_{r_2}(\omega_3)/4 = 3 \cdot 487 \cdot 1{,}313{,}423{,}399{,}858{,}711;$$

the prime factors 487 *and* 1,313,423,399,858,711 *of d_{r_2} are not Class III, so Condition (II) fails at r_2 by the parity obstruction (Theorem 5.3). This is the secondary factor of the $d = 57$ witness already recorded in §B.4.*

(b) $p = 1{,}251{,}488{,}009$*: the prime*

$$r = 33{,}829{,}735{,}778{,}964{,}491 = 2pk + 1, \quad k = 13{,}515{,}805, \quad r \equiv 3 \pmod 8,$$

divides W_p with $\operatorname{ord}_r(2) = 2p$ *and*

$$d_r = \operatorname{ord}_r(\omega_3)/4 = 3 \cdot 41 \cdot 68{,}759{,}625{,}567{,}001;$$

the prime factors 41 *and* 68,759,625,567,001 *of d_r are not Class III, so Condition (II) fails at r by the parity obstruction.*

Consequently neither exponent lies in E_3, and neither W_p passes Condition (II).

Verification. Computational, in the at-a-factor logic above: a factor failing Condition (II) excludes the exponent from every local-pass-defined set and from any global pass. For each of the two factors, primality is certified by APR-CL, the divisibilities $2^p \equiv -1 \pmod r$, the pinned orders $\operatorname{ord}_r(2) = 2p$, the orders $\operatorname{ord}_r(\omega_3) = 4d_r$, the factorizations of d_r, and the non-Class-III property of the listed prime factors ($\operatorname{ord}_{487}(2) = 243$ odd; $\operatorname{ord}_{41}(2) = 20$ with $v_2 = 2$; analogously for the two large factors) are verified by the archived scripts (§B.5), with an independent end-to-end recomputation in `scripts/cp352_verify_discharges_recompute.py` that also re-tests Condition (II) at each factor by direct congruence evaluation, bypassing the descent. □

Corollary 6.31 ($W_2 \le 1$; $W_2 = 0$ under X1)**.** *Given the Platinum enumeration and Proposition 6.30, only the witness exponent* $p = 10{,}916{,}765{,}939$ *can contribute to W_2; hence $W_2 \le 1$. If X1 holds (Conjecture 6.29), then $W_2 = 0$.*

Proof. By Theorem 6.24(c) every p counted by W_2 is a witness exponent and lies in E_3; Proposition 6.30 removes two of the three, and X1 removes the third. □

7. The Pell Primitive Pair Conjecture

7.1. **Statement.** The residual of the reduction chain above is the Pell Primitive Pair Conjecture, already stated informally in §1.3. For completeness:

Conjecture 7.1 (Pell Primitive Pair Conjecture, PPPC)**.** For every prime $p \geq 5$: $|T_p^{>}| \leq 1$, where $T_p^{>}$ is the set of primes r with

$$r \equiv 3 \pmod 8, \quad \mathrm{ord}_r(2) = 2p, \quad r > 10^{12}, \quad d_r > 43, \quad d_r \mid W_{p-2}.$$

Equivalently, no two distinct primes r_1, r_2 of this form exist for the same p.

7.2. **Equivalent forms.** PPPC admits several equivalent formulations, each useful in different contexts. The diagonal case $d_1 = d_2$ is part of the statement: two distinct primes in $T_p^{>}$ may have the same order parameter d_r.

(a) Pair-count form. $\Pi := \sum_p \binom{|T_p^{>}|}{2} = 0$.

(b) Primitive-divisor form. No prime p admits two distinct primitive divisors r_1, r_2 (of V_{2d_1} and V_{2d_2}, with $d_1, d_2 > 43$ not necessarily distinct) such that

$$r_i \equiv 3 \pmod 8, \qquad \mathrm{ord}_{r_i}(2) = 2p, \qquad r_i > 10^{12}, \qquad d_i \mid W_{p-2}.$$

Equivalently, both the off-diagonal coincidences $d_1 < d_2$ and the diagonal multiplicities at a fixed d vanish.

(c) AP-coincidence form. Let S_d be the set of primes p satisfying the valuation-adjusted congruence condition equivalent to $d \mid W_{p-2}$ (prime moduli are described in Theorem 6.7; prime powers are governed by the LTE refinement below it). Then PPPC asserts that the pinned-order map attached to the primitive inert divisors of V_{2d} has no common value across S_{d_1} and S_{d_2} for $d_1 < d_2$, and is injective on the relevant primitive-divisor set when $d_1 = d_2$.

(d) Number-field form. Let $\mathcal{P}_d^{\mathrm{prime}} \subset \mathrm{Spec}\,\mathbb{Z}$ be the set of primes r that are primitive divisors of V_{2d} with $r \equiv 3 \pmod 8$, $r > 10^{12}$, $p_r := \mathrm{ord}_r(2)/2$ prime, and $d \mid W_{p_r-2}$. The maps

$$\mathcal{P}_d^{\mathrm{prime}} \longrightarrow \mathbb{Z}, \qquad r \longmapsto p_r$$

have pairwise disjoint images across distinct admissible $d > 43$, and each map is injective on each fixed d.

Both coordinates of these equivalent forms now carry unconditional constraints. Any two members of $T_p^{>}$ at one p are separated, by the Pair Separation Theorem (Theorem 6.10), at scale $8p\gcd(d_{r_1}, d_{r_2})$ in r — at scale $3208\,p$ on the diagonal, where the census floor $d > 400$ applies; and the diagonal multiplicities of form (c)/(d) vanish entirely for every admissible $d \leq 400$, for all p and all k, by the certified closures of Propositions A.9 and 5.18 — in particular, no diagonal pair with both $k_i = (r_i - 1)/2p \leq 1604$ exists (Corollary A.10). The residue-level structure of the diagonal sub-case, and the question that remains open there, are developed in Appendix A.2 and §A.4.

7.3. **Empirical evidence. Small-factor enumeration.** The Platinum enumeration of §B.4 records every inert factor $r \le 10^{12}$ in the stated range and tests Condition (II). It verifies that, for each p, at most one such *small* inert factor passes locally. This is not a verification of PPPC, whose set $T_p^{>}$ is defined by the opposite cutoff $r > 10^{12}$. Its role in Theorem 6.15 is sharper and more limited: in a squarefree all-inert counterexample with at least three locally passing inert factors, at most one can be small, so at least two must lie in $T_p^{>}$.

Extended p-indexed enumeration. Beyond the Platinum range $r \le 10^{12}$, a p-indexed scan covered $p^* \in [5 \cdot 10^{11}, 5 \cdot 10^{11} + 10^9]$ (37,123,274 primes). For each p^* and each $k = 1, 2, \dots, k_{\max}$, the candidate $r = 2p^*k + 1$ was tested for primality, residue $r \equiv 3 \pmod 8$, and order $\mathrm{ord}_r(2) = 2p^*$; the pre-condition-(v) window

$$T_{p^*}^{\le k_{\max},\,\mathrm{pre}} := \{\, r = 2p^*k + 1 : 1 \le k \le k_{\max},\ r \text{ prime}, \\ r \equiv 3 \pmod 8,\ \mathrm{ord}_r(2) = 2p^* \,\}.$$

contains the candidates before the PPPC condition $d_r \mid W_{p^*-2}$ is applied. Intersecting it with $r > 10^{12}$, $d_r > 43$, and condition (v) gives the portion of $T_{p^*}^{>}$ visible in the window; the inclusion is therefore from $T_{p^*}^{>}$ into this pre-window, not conversely. Distribution at $k_{\max} = 500$ on the 37M-prime sample:

$\lvert T_{p^*}^{\le 500,\mathrm{pre}} \rvert$	count	fraction
0	33,052,256	89.03%
1	3,886,080	10.47%
2	180,116	0.485%
3	4,750	0.0128%
4	72	0.000194%

Sample mean $\lambda = 0.1148$, sample variance 0.1121, ratio Var/mean $= 0.977$ (compared to Poisson's 1.0); χ^2 against Poisson(λ) is approximately 10^4, indicating significantly sub-Poisson behaviour. Extending to $k_{\max} = 5000$ on a 1M-prime subsample multiplies the pair fraction by $1.75\times$ and the triple fraction by $2.5\times$, consistent with a non-degenerate but slowly-growing tail.

Selection-bias note: condition (v) filter. Of the 184,938 candidate pairs (r_1, r_2) within the $k_{\max} = 500$ window satisfying conditions (i)–(iv) of PPPC ($r_i \equiv 3 \pmod 8$, $r_i > 10^{12}$, $\mathrm{ord}_{r_i}(2) = 2p^*$, primality), independent verification computed condition (v) — $d_{r_i} \mid W_{p^*-2}$ — for each individual r_i via $\mathbb{F}_{r_i^2} = \mathbb{F}_{r_i}[\sqrt{2}]$ arithmetic: 0 **of the** 369,876 **tested** r_i **satisfy (v).** The conditional density of (v) given (i)–(iv) is thus $\ll 10^{-5}$ empirically, consistent with the heuristic $E \approx 10^{-3}$ of §7.4 and with $\Pi = 0$. In every tested case the failure is of condition (v) itself — the order-alignment $d_{r_i} \mid W_{p^*-2}$, equivalently $2^{p^*-2} \equiv -1 \pmod{3d_{r_i}}$ — with conditions (i)–(iv) holding by construction of the candidate list. A deep-k complement to this record was computed at 30,000 exponents p^* from 5×10^{11} with the window extended two thousandfold to $k \le 10^6$: 6,559 candidates satisfy (i)–(iv), and again *zero* satisfy condition (v); 623 of these exponents carry two or more (i)–(iv) candidates, and every such pair satisfies the unconditional separation constraints of Theorem 6.10 (script `scripts/cp358_falsification_prewindow.py`).

Methodological remark. The d-indexed enumeration (§B.4 Platinum, §5.6 primitive-divisor survey, §B.3 sieve) and the p-indexed enumeration above probe complementary slices of the PPPC search space: d-indexed scans capture pairs where both factors have small order parameter, while p-indexed scans capture the arithmetic progression $r \equiv 1 \pmod{2p^*}$ before condition (v) is applied. The $d \le 1000$ catalog of the primitive-divisor survey covers a small fraction of the p-indexed pre-window at $p^* = 5 \times 10^{11}$ — most of the r's appearing in the $T_{p^*}^{\le 500,\mathrm{pre}}$ distribution above have d_r well above 1000. Absence of pairs in the $d \le 1000$ catalog is rigorous within scope but is not directly comparable with the p-indexed pre-window density; both lines of empirical evidence are needed.

7.4. **Heuristic support. Heuristic count.** Modelling each primitive inert divisor of V_{2d} as a random coincidence with $\mathrm{ord}_r(2)/2 = p$ among its admissible residues, the expected total number of single danger triples (d, r, p) with $d > D$ is

$$E(D) \;\approx\; \sum_{d>D,\ d \text{ admissible}} \frac{(\ln\ln|V_{2d}|)^2}{2\ln|V_{2d}| \cdot \varphi(L_d)},$$

where L_d is the valuation-adjusted modulus of Remark 6.8 and $|V_{2d}| \sim (1+\sqrt{2})^{2d}$. For admissible prime $d = q$ with the generic valuation $a_q = 1$, $L_d = 2m_q$ and typically $\varphi(L_d)$ is of order q, giving a rapidly convergent prime-level model. The displayed expression carries an implicit *inert-fraction* factor of $\approx 1/2$ — only the primitive divisors lying in the inert class $r \equiv 3 \pmod 8$ are relevant, and the displayed sum itself evaluates to ≈ 0.0076 at $D = 1000$ before that factor is applied. Numerical evaluation at $D = 1000$ in the squarefree prime-level model, with the inert fraction included, gives $E \approx 0.003$, consistent with zero observed.

A formalized pair-count model for PPPC (heuristic only). The pair heuristic can be made an explicit expected-count computation. The model has three assumptions, stated here once and numbered; *none is a theorem*, and nothing in this paragraph proves anything about PPPC.

M1 (*Candidate production.*) At a prime exponent p, the candidates satisfying conditions (i)–(iv) of Conjecture 7.1 are the primes $r = 2pk + 1 > 10^{12}$ with $k \equiv p \pmod 4$ (the class forced by $r \equiv 3 \pmod 8$, as in Theorem 6.10(1)); the model takes them to arrive independently with intensity

$$\Pr\big[\text{(i)–(iv) at } k\big] \;\approx\; \frac{C_{\mathrm{cal}}}{2k\,\ln(2pk)},$$

the factor $1/(2k) = p/(r-1)$ being the order-alignment probability ($\mathrm{ord}_r(2) = 2p$) and C_{cal} a single calibration constant absorbing the AP-primality and mod-8 corrections. Calibrating on the measured window mean of §7.3, $\lambda(500) = 0.1148$ at $p^* \approx 5 \times 10^{11}$, gives $C_{\mathrm{cal}} = 3.69$; the model then predicts $\lambda(10^6) = 0.208$ against the measured $6559/30000 = 0.219$ (ratio 0.95). Integrated over the full factor spectrum this intensity gives $\mu_p \approx \frac{1}{2}(\ln\ln W_p - \ln\ln \max(2p, 10^{12}))$ expected above-cutoff candidates per exponent ($\mu_{p^*} \approx 11.6$), the normal-order count of inert factors.

M2 (*Condition-(v) density.*) Given (i)–(iv), the alignment $d_r \mid W_{p-2}$ holds with probability δ, anchored on the paper's two measured records in three readings:

$$\delta_{\mathrm{ceil}} = \frac{3}{376{,}435} = 8.0 \times 10^{-6}, \qquad \delta_{\mathrm{cen}} = \frac{3}{15{,}587{,}021} = 1.9 \times 10^{-7},$$

$$\delta_{\mathrm{best}}(r) = \delta_{\mathrm{cen}} \cdot \min\bigl(1,\ 10^{12}/r\bigr).$$

The ceiling is the 95% rule-of-three bound from the above-cutoff zero-pass record ($369{,}876 + 6{,}559$ tested candidates, 0 passes, §7.3); the central value is the only nonzero measured density — the Platinum enumeration's 3 passes among the 15,587,021 rows surviving the $G_r \le 43$ exclusion (§B.4), at $r \le 10^{12}$; the best estimate decays the measured density like $1/r$ beyond the measurement scale (the alignment mass of divisors $d_r \mid (r+1)/4$ against the fixed integer W_{p-2}), the same structural-decay device as the MPC heuristic's best column.

M3 (*Independence.*) Candidate production and the (v)-events at distinct candidates are independent, so the pass count $X_p = |T_p^{>}|$ is Poisson with mean $\nu_p = \sum_k \Pr[\text{(i)–(iv)}]\,\delta$. The measured variance ratio 0.977 (sub-Poisson, §7.3) and the unconditional repulsions of this paper — Theorem 6.10, and the diagonal closures of Proposition A.9 modulo their stated certificates, neither credited by the model — both push the true pair count below the model's.

Per-exponent violation probability. Under M1–M3, $\Pr[|T_p^{>}| \ge 2] \le \mathbb{E}\binom{X_p}{2} = \nu_p^2/2$. At the Platinum frontier $p^* = 5 \times 10^{11}$: $\nu_{p^*} = 9.3 \times 10^{-5}$ (ceiling) / 2.2×10^{-6} (central) / 7.8×10^{-9} (best), giving per-exponent pair probabilities

$$\frac{\nu_{p^*}^2}{2} \;=\; 4.3 \times 10^{-9} \,/\, 2.5 \times 10^{-12} \,/\, 3.1 \times 10^{-17} \qquad \text{(ceiling / central / best)}.$$

For self-consistency: the central reading predicts 0.07 expected (v)-passes among the 369,876 tested candidates (observed: 0), and the ceiling reading predicts 3 (it *is* the 95% bound built from that record).

Summed expectation. Under the flat readings the sum over p of ν_p — let alone of $\nu_p^2/2$ — diverges: with a constant δ and $\mu_p \sim \frac{1}{2}\ln\ln W_p$, single passes recur infinitely often. This is the model face of Remark 7.10, which expects the single-pass first moment to be divergent (the enumeration already exhibits three witness exponents), and it is why the flat readings are summed only through the d-indexed form of the count, where each order parameter d contributes the convergent single-triple summand $E_1(d) = \frac{1}{2}(\ln\ln V_{2d})^2/(2\ln V_{2d}\,\varphi(L_d))$ of the display above and a partner factor $\min(1,\, \omega_{\mathrm{in}}(\bar{p}_d)\,\delta)$, $\omega_{\mathrm{in}}(x) = \frac{1}{2}(\ln x - \ln\ln x)$, at the pinned scale $\ln\bar{p}_d = \ln V_{2d} - \ln 4$ (the same partner device as the MPC heuristic). The best reading needs no such cap: the $1/r$ decay makes the p-indexed sum $\sum_p \nu_p^2/2$ converge outright, dominated within a decade of the anchor scale p^* — exactly where the calibration is measured. The results:

$$\mathbb{E}[\Pi] \;\approx\; 1.1\times10^{-3}\ \text{(ceiling)},\qquad \sim 10^{-4}\ \text{(central)},\qquad 6.0\times10^{-7}\ \text{(best)}.$$

The three-orders spread between ceiling and best is genuine model uncertainty — it is the cost of extrapolating a condition-(v) density measured at $r \le 10^{12}$ to factors of unbounded size — and is reported bracketed rather than resolved, exactly as for the MPC joint heuristic (whose bracket, in the same framework, is 8.7×10^{-4} / 4.4×10^{-4} / 9.8×10^{-10}). The best-estimate reading reproduces the $O(10^{-6})$ pair figure quoted in earlier versions of this section. Constants and calibration gates: `scripts/cp395_pppc_heuristic_constants.py` (the script first reproduces the $E(1000)$ figures of the display above, then the window means, before computing the bracket). These figures are heuristics only; they prove nothing about PPPC, which remains an open conjecture, and Conjecture 1.1 is expected to hold for the structural reason described below, not because of these numbers. The rigorous proof of $\Pi = 0$ is open.

Joint heuristic for MPC (heuristic only). The Mixed Pair Conjecture (Conjecture 4.22) admits an analogous joint count: a violation requires a split compatible triple *and* an inert local pass at the same pinned p. Summing the joint expectation over admissible $d_s > 199$, with the then-open trio $\{131, 163, 179\}$ carried as separate line items (all three have since been closed, §B.3; their retained line items contribute negligibly to the totals), gives an expected violation count E_{MPC} with ceiling 8.7×10^{-4}, central value 4.4×10^{-4}, and best estimate 9.8×10^{-10}, against an NCT-general *single-event* heuristic of 1.27×10^{-2} (ceiling) and 2.4×10^{-5} (structural) computed in the same framework. The joint heuristic is thus 1–5 orders of magnitude below the corresponding NCT-general single-event heuristic (model-dependent), approaching PPPC's pair scale $O(10^{-6})$ only in the structural-decay reading. The ceiling/best-estimate spread of six orders is genuine model uncertainty — it extrapolates the empirically measured condition-(v) density from $r \sim 10^{12}$ to factors of size e^{200} and beyond — and is reported bracketed rather than resolved. These figures are heuristics only; they prove nothing about MPC, which remains an open conjecture.

A structural ingredient of the split branch deserves separate mention: in any compatible triple, the unconditionally proved band $8pd \mid r_s - 1$ (Proposition A.3) forces the index of 2 in $\mathbb{F}_{r_s}^{\times}$ — namely $(r_s-1)/\operatorname{ord}_{r_s}(2) = (r_s-1)/2p$ — to absorb the entire d-part of $r_s - 1$, the order of 2 being coprime to d.

In the index model this is an extra alignment of heuristic cost $\sim 1/(3d)$ per split factor, a suppression with no inert analogue (the inert order datum lives on the $r+1$ side, where no corresponding index constraint arises).

The conjecture is structural, not probabilistic. The rapid convergence above is an artefact of the pinning constraints and would not survive their removal. The Chua congruence is a strong Lucas test with parameter $Q = \mathrm{N}(\omega_3) = +1$; for a Lucas or Frobenius test with $|Q| = 1$ the standard pseudoprime heuristic predicts *infinitely many* composite numbers passing the single congruence, absent further structure (cf. [10]). What forces Π to be small is not a generic scarcity of pseudoprimes but the Wagstaff-specific Order-Pinning (Theorem 6.1) and Multi-Factor Pinning (Theorem 6.2): every prime factor of one W_p shares a single pinned exponent p *exactly*, and each divisibility $d_r \mid W_{p-2}$ is the rigid valuation-adjusted congruence condition of Theorem 6.7 and Remark 6.8. Conjecture 1.1 is therefore expected to hold for a structural reason rather than a probabilistic one — which is also why a purely analytic count treating the factors as independent random primes cannot reach it, the precise content of §A.3.

7.5. **Connection to classical problems.** PPPC is structurally related to several classical problems on Pell–Lucas sequences:

- **Primitive-divisor structure of V_{2d}.** Bilu–Hanrot–Voutier [3] guarantees a primitive divisor for V_n for all $n \geq 30$. Stewart–Tijdeman [18] gives $\omega(V_n) \ll n/\log n$. A sharpening of the Stewart bound at small $n \cdot \log\log n$ scales could contribute to bounding components of the pair count Π (attack angle N2).
- **Prime divisors of $a^2 + 2b^2$.** By Remark 5.9, any $r \in T_p^{>}$ divides $N_d = V_d^2 + 2$, so r is a prime of form $a^2 + 2b^2$. Iwaniec's half-dimensional sieve [13] bounds the count of such primes in arithmetic progressions (attack angle N3).
- **Effective Chebotarev density.** The Artin-like question "for a fixed $\alpha \in \mathbb{Z}[\sqrt{2}]^\times$, how often does a primitive divisor of V_n fall in a prescribed AP?" is related to Chebotarev on the tower $\mathbb{Q}(\alpha^{1/m})$; effective forms (Lagarias–Odlyzko; under GRH, Hooley–Heath-Brown [12, 11]; see the survey [15]) could supply inputs relevant to AP components of Π, but do not by themselves control the fixed-divisor count.
- **Bombieri–Friedlander–Iwaniec-type levels of distribution.** An analogue of the Bombieri–Friedlander–Iwaniec level-of-distribution technology for the non-standard Lucas sequence $\{V_{2d} : d \leq X\}$ would be relevant to components of the pair count (attack angle N4). No such direct application is used here.

Section A.2 elaborates these connections.

7.6. **Main reduction theorem.**

Theorem 7.2 (Main reduction)**.** *Assume PPPC (Conjecture 7.1), NCT for general d (Conjecture 4.20), and $W = 0$ — equivalently, no composite W_p whose prime factors are all inert and all pass Condition (II) has a repeated prime factor. Then Conjecture 1.1 holds. The implication uses, as a stated computational input, the Platinum enumeration of Theorem 6.6.*

Proof. Let W_p be composite. By Corollary 3.3, if W_p passes Condition (II) then it passes at every prime factor. By Corollary 2.5, every factor is $\equiv 1$ or $3 \pmod 8$.

If some factor is split ($\equiv 1 \pmod 8$), then Condition (II) at that factor and Proposition 4.14 produce a compatible triple. NCT for general d (Conjecture 4.20) rules this out.

Otherwise every factor is inert ($\equiv 3 \pmod 8$). Then W_p contributes to E_3. By Theorem 6.15, $E_3 \le \Pi + W$. PPPC asserts $|T_p^{>}| \le 1$ for every p, so $\Pi = 0$; the third hypothesis is $W = 0$. Hence $E_3 \le \Pi + W = 0 + 0 = 0$. □

The third hypothesis $W = 0$ is the **(E3) Wieferich–Wagstaff exclusion**: by Remark 6.17, a composite all-inert Condition-(II)-passing W_p with a repeated prime factor r would force r to be a base-2 Wieferich prime $\equiv 3 \pmod 8$. The exhaustive search below 2^{64} finds only $1093 \equiv 5 \pmod 8$ and $3511 \equiv 7 \pmod 8$ [16]; so $W = 0$ holds *unconditionally* for every W_p all of whose prime factors lie below 2^{64}, and conjecturally $W = 0$ for all p. It is therefore by far the mildest of the three residual inputs to Theorem 7.2.

7.7. **The mixed-pair form of the main reduction.** The proof of Theorem 7.2 uses Condition (II) at the split factor only, discarding the fact that every *other* factor of a global passer also passes. Retaining that information weakens the split-branch hypothesis from NCT-general to the Mixed Pair Conjecture of §4.7.

Theorem 7.3 (Mixed-pair form of the main reduction)**.** *Assume PPPC (Conjecture 7.1), the Mixed Pair Conjecture (Conjecture 4.22), and $W = 0$. Then Conjecture 1.1 holds. The implication uses, as a stated computational input, the Platinum enumeration of Theorem 6.6.*

Proof. Let W_p be composite and suppose it passes Condition (II). The congruence $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$ restricts along the ring surjection

$$\mathbb{Z}[\sqrt{2}]/(W_p) \longrightarrow \mathbb{Z}[\sqrt{2}]/(r)$$

for every prime $r \mid W_p$, so Condition (II) holds at every prime factor. By Corollary 2.5, every factor is $\equiv 1$ or $3 \pmod 8$.

Case A: some factor $r_s \equiv 1 \pmod 8$. Condition (II) at r_s and Proposition 4.14 produce a compatible triple (d_s, r_s, p) — datum (i) of MPC. By Lemma 2.4, the total multiplicity of inert factors of W_p is odd, hence at least one inert prime factor $r_{\text{in}} \equiv 3 \pmod 8$ exists. Condition (II) at r_{in} and the descent of §5.1 give $\operatorname{ord}_{r_{\text{in}}}(\omega_3) = 4\, d_{r_{\text{in}}}$ with $d_{r_{\text{in}}}$ odd and $d_{r_{\text{in}}} \mid W_{p-2}$, while

Order-Pinning (Theorem 6.1) gives $\operatorname{ord}_{r_{\text{in}}}(2) = 2p$ — datum (ii). The pair of data violates MPC.

Case B: all factors inert. Exactly as in Theorem 7.2: W_p contributes to E_3; Theorem 6.15 gives $E_3 \le \Pi + W$; PPPC gives $\Pi = 0$, and $W = 0$ by hypothesis — contradiction. □

Since NCT-general implies MPC (Proposition 4.24), Theorem 7.2 is an immediate corollary of Theorem 7.3; the hypotheses of Theorem 7.3 are formally weaker. Combining the mixed-pair form with the W-residual trichotomy of §6.8 gives the sharpest chain of this paper with PPPC kept intact.

Theorem 7.4 (Sharpened reduction chain)**.** *Assume*

- *PPPC (Conjecture 7.1);*
- *MPC$^{>}$ (Conjecture 4.22 with datum (ii) restricted to $r_{\text{in}} > 10^{12}$, §4.7);*
- *X1 (Conjecture 6.29).*

Then Conjecture 1.1 holds. The implication uses, as stated computational inputs, the Platinum enumeration (Theorem 6.6) and the discharge verifications of Proposition 6.30. Moreover, the conjunction {PPPC, MPC$^{>}$, X1} is implied by the conjunction {PPPC, NCT-general, $W = 0$} of Theorem 7.2.

Proof. Let W_p be composite and pass Condition (II); as in Theorem 7.3, it passes at every prime factor.

Case A: some factor $r_s \equiv 1 \pmod 8$. As in Theorem 7.3, data (i) and (ii) of MPC hold at p, with inert co-passer r_{in}. If $r_{\text{in}} > 10^{12}$, MPC$^{>}$ is violated. If $r_{\text{in}} \le 10^{12}$: W_p is composite and r_{in} passes Condition (II) locally with $d_{r_{\text{in}}} > 43$ (Theorem 5.10, as W_p is composite), and $p \le (r_{\text{in}} - 1)/2 < 5 \times 10^{11}$, so (p, r_{in}) is a locally passing row of the Platinum enumeration (which ranges over inert factors of *composite* W_p), and p is one of the three witness exponents (§6.8). The compositeness restriction matters here: phantom local passes at *prime* W_p, such as $r = W_p \in \{683, 2731, 43691\}$ at $p \in \{11, 13, 17\}$, are not rows of the enumeration and are excluded. For the witness exponents 85,684,865,933 and 1,251,488,009, Proposition 6.30 exhibits a prime factor at which Condition (II) fails, contradicting the pass at every factor; for $p =$ 10,916,765,939, X1 supplies the failing factor. Case A is closed.

Case B: all factors inert. Then p contributes to E_3. By Theorem 6.24, $E_3 \le \Pi + W_1 + W_2$; PPPC gives $\Pi = 0$; $W_1 = 0$ by Lemma 6.26; and $W_2 = 0$ by Corollary 6.31 under X1. Hence $E_3 = 0$ — contradiction.

Conjunction implication. Assume PPPC, NCT-general and $W = 0$. NCT-general implies MPC (Proposition 4.24), hence MPC$^{>}$; $W = 0$ implies $W_2 = 0$ (Remark 6.28). For X1: suppose Condition (II) held at every prime factor of the composite $W_{10916765939}$. A split factor would, with Proposition 4.14, yield a compatible triple against NCT-general; so every factor would be inert and $p =$ 10,916,765,939 would lie in E_3, while $E_3 \le \Pi + W = 0$ under PPPC and $W = 0$ — contradiction. Hence some factor fails, which is X1.

The implication is claimed at the level of the conjunctions; no comparison between individual hypotheses across the two lists is asserted. ☐

Remark 7.5. Theorem 7.4 is the sharpest conditional statement of this paper with PPPC kept intact: PPPC is unchanged, and each of the other inputs is qualitatively lighter than its counterpart in Theorem 7.2. $\mathrm{MPC}^{>}$ is a pair-type statement of the same shape as PPPC (§4.7); and X1 concerns one explicit integer and is discharged by exhibiting any single failing factor of $W_{10916765939}$. The statement $W_1 = 0$, a hypothesis in earlier versions of this chain, is now unconditional (Lemma 6.26), so the chain reads: Conjecture 1.1 follows from PPPC + $\mathrm{MPC}^{>}$ + X1. None of the three inputs is proved here; the chain remains a conditional reduction.

7.8. **The triple form: eroding the pair count.** The pair count Π enters Theorem 7.4 through the W-residual trichotomy (Theorem 6.24), whose proof splits E_3 at $|T_p^{>}| \geq 2$. Splitting at $|T_p^{>}| \geq 3$ instead pushes the same structural lemmas one step further: the pair count is replaced by a *triple* count, and the overflow is absorbed by the prime-power bucket W_1 — which is empty by Lemma 6.26 — one further corner bucket, and an explicit witness-exponent term. The role of PPPC in the chain is thereby eroded from a pair-counting to a triple-counting statement, with the same two computational inputs as Theorem 7.4 and no new ones. For the triple count, put

$$\Pi_3 \;:=\; \sum_{p \geq 5} \binom{|T_p^{>}|}{3},$$

with $T_p^{>}$ exactly as in §6.7: the index set, the cutoff $r > 10^{12}$ and the defining conditions are identical to those of Π, and only the pair $\binom{\cdot}{2}$ is replaced by the triple $\binom{\cdot}{3}$.

Conjecture 7.6 (PPPC3, above-cutoff triple form)**.** $\Pi_3 = 0$; equivalently, $|T_p^{>}| \leq 2$ for every prime $p \geq 5$.

PPPC implies PPPC3: $\Pi = 0$ forces $|T_p^{>}| \leq 1$ for every p, hence $\Pi_3 = 0$. The converse implication is not asserted, so PPPC3 is formally weaker as a hypothesis. PPPC3 must also be distinguished from the computational-input-free triple residual $\Pi_3' = 0$ of Remark 6.16: Π_3' counts triples in the cutoff-free sets T_p' and is *not* implied by the above-cutoff PPPC, whereas Π_3 retains the cutoff $r > 10^{12}$ of $T_p^{>}$ and *is* implied by PPPC. The two triple residuals are distinct statements with different roles: $\Pi_3' = 0$ removes the computational input; PPPC3 weakens the conjectural input while keeping the Platinum cutoff.

The second new object is a corner bucket. With E_3 as in §6.7, define

$$\begin{aligned} W_2' \;=\; \#\{\, p \in E_3 : W_p &\text{ has exactly two distinct prime factors,} \\ &\text{both factors} > 10^{12},\ \text{and } W_p \text{ is not squarefree} \,\}, \end{aligned}$$

the *Wieferich-paired corner*: by Theorem 7.8(c) below, each such p carries a base-2 Wieferich prime $\equiv 3 \pmod 8$ above 2^{64} *and* a second large locally passing factor at the same p — a doubly-conditioned configuration. The non-squarefree clause is in fact redundant: for $p \in E_3$ with exactly two distinct prime factors, every factor is inert, so the total multiplicity of factors is odd (Lemma 2.4), whereas a squarefree W_p with two distinct factors would have total multiplicity 2; the clause is kept for robustness of the definition.

Conjecture 7.7 (Wieferich-paired corner exclusion)**.** $W_2' = 0$.

Conjecture 7.7 is implied by either of two hypotheses already in play: by PPPC, since each p counted by W_2' has $|T_p^>| = 2$ exactly and so contributes $\binom{2}{2} = 1$ to Π (Theorem 7.8(c) below); and by $W = 0$, since the W_2'-set is a subset of the W-set of Theorem 6.15 (non-squarefree, all-inert, locally passing at every factor).

Theorem 7.8 (Triple trichotomy)**.** *With E_3, $T_p^>$ and Π as in §6.7, W_1 as in Theorem 6.24, and Π_3, W_2' as above, write X_{E_3} for the number of witness exponents (§6.8) that lie in E_3. Deterministically, given the Platinum enumeration (Theorem 6.6):*

(a) $E_3 \le \Pi_3 + W_1 + W_2' + X_{E_3}$.

(b) Given also the discharge verifications of Proposition 6.30: $X_{E_3} \le 1$, only the witness exponent $p =$ 10,916,765,939 being able to lie in E_3; and, still given the discharges, X1 (Conjecture 6.29) implies $X_{E_3} = 0$.

(c) Every p counted by W_2' carries a repeated prime factor $s > 2^{64}$ that is a base-2 Wieferich prime $\equiv 3 \pmod 8$, together with a second distinct locally passing prime factor $r > 10^{12}$ at the same p; and $|T_p^>| = 2$ exactly. Consequently PPPC implies $W_2' = 0$ — each p counted by W_2' contributes $\binom{2}{2} = 1$ to Π — and $W = 0$ implies $W_2' = 0$ — the W_2'-set is a subset of the W-set of Theorem 6.15. If W_p moreover passes Condition (II) globally*, the repeated factor additionally satisfies the Pell–Wall–Sun–Sun-type condition of Remark 6.17, exactly as in Remark 6.25; this global-passer statement is a remark on the bucket, not part of its definition.*

As in Theorem 6.24, all buckets are defined by per-factor local *passes, and no global-pass assumption enters the statement or the proof.*

Proof. (a) Split E_3 according to $|T_p^>| \ge 3$ or $|T_p^>| \le 2$. For every integer $X \ge 0$, $\mathbf{1}_{X\ge 3} \le \binom{X}{3}$ (the indicator vanishes for $X \le 2$; $\binom{3}{3} = 1$; $\binom{X}{3} \ge 4$ for $X \ge 4$), so, summing over p as in Step 3 of Theorem 6.15,

$$\#\{\, p \in E_3 : |T_p^>| \ge 3 \,\} \;\le\; \Pi_3.$$

Now let $p \in E_3$ with $|T_p^>| \le 2$. By the definition of E_3, W_p is composite, every prime factor is inert, and Condition (II) holds at every factor; hence every prime factor r has $d_r \mid W_{p-2}$ (§5.1) and $d_r > 43$ (Theorem 5.10, applicable since W_p is composite and Condition (II) holds at r). Therefore every *distinct* prime factor $r > 10^{12}$ of W_p lies in $T_p^>$; since $|T_p^>| \le 2$, W_p has at most two distinct prime factors exceeding 10^{12}.

Case 1: some prime factor $r \le 10^{12}$. Then r passes locally with $d_r > 43$, and $p \le (r-1)/2 < 5 \times 10^{11}$ (Lemma 2.3), so (p, r) is a locally passing row of the Platinum enumeration — which ranges over inert factors of *composite* W_p, so the phantom local passes at prime W_p are excluded, exactly as in Case A of Theorem 7.4. Hence p is a witness exponent (§6.8); since $p \in E_3$, it is counted by X_{E_3}.

Case 2: no prime factor $\le 10^{12}$. Then every distinct prime factor of W_p exceeds 10^{12}, so by the paragraph above W_p has at most two distinct prime factors. If W_p were squarefree, the number of distinct factors would equal the total multiplicity, which is ≥ 3 by Theorem 5.16 — contradiction; so W_p is not squarefree. If $W_p = r^t$ has a single distinct factor, then t, the total multiplicity, is odd (Lemma 2.4) and ≥ 2 (W_p is not squarefree), hence $t \ge 3$; p is counted by W_1. If W_p has exactly two distinct prime factors — both $> 10^{12}$, with W_p not squarefree — then p is counted by W_2'.

Every $p \in E_3$ with $|T_p^{>}| \le 2$ is therefore counted by $X_{E_3} + W_1 + W_2'$, and (a) follows.

(b) A witness exponent lying in E_3 passes Condition (II) at *every* prime factor of its W_p. For $p \in \{85{,}684{,}865{,}933,\ 1{,}251{,}488{,}009\}$, Proposition 6.30 exhibits an explicit prime factor at which Condition (II) fails, so neither exponent lies in E_3. Only the witness exponent 10,916,765,939 can therefore be counted by X_{E_3}; hence $X_{E_3} \le 1$. X1 asserts a failing factor of $W_{10916765939}$, which likewise removes that exponent from E_3; hence X1 implies $X_{E_3} = 0$.

(c) Let p be counted by W_2', with $W_p = r^a s^b$, $r \ne s$ both $> 10^{12}$. The total multiplicity $a + b$ is odd (Lemma 2.4) and ≥ 2 (W_p is composite), hence ≥ 3, so at least one of the exponents is ≥ 2. Any repeated factor is a base-2 Wieferich prime (Corollary 6.23) and is $\equiv 3 \pmod 8$ (all factors are inert), hence exceeds 2^{64} — below 2^{64} the only base-2 Wieferich primes are $1093 \equiv 5$ and $3511 \equiv 7 \pmod 8$. Both r and s pass locally, hence each has order parameter > 43 dividing W_{p-2} (as in (a)), and both exceed 10^{12}, so both lie in $T_p^{>}$; since $T_p^{>}$ consists of distinct prime factors of W_p, of which there are exactly two, $|T_p^{>}| = 2$ exactly. Then $\binom{|T_p^{>}|}{2} = 1$, so p contributes 1 to Π, and PPPC ($\Pi = 0$) forces $W_2' = 0$. The W_2'-set consists of non-squarefree all-inert local passers, hence is a subset of the W-set of Theorem 6.15, and $W = 0$ forces $W_2' = 0$. The global-passer Pell–Wall–Sun–Sun statement is Remark 6.17 applied to the repeated factor, exactly as in Remark 6.25: the congruence modulo s^2 is guaranteed only under a global pass and does not follow from local passes at the individual factors. ☐

Corollary 7.9 (Triple-sharpened chain)**.** *Assume*

- *$MPC^{>}$ (Conjecture 4.22 with datum (ii) restricted to $r_{\text{in}} > 10^{12}$, §4.7);*
- *PPPC3 (Conjecture 7.6);*
- *$W_2' = 0$ (Conjecture 7.7);*
- *X1 (Conjecture 6.29).*

Then Conjecture 1.1 holds. The implication uses, as stated computational inputs, the Platinum enumeration (Theorem 6.6) and the discharge verifications of Proposition 6.30 — the same two inputs as Theorem 7.4, with no new computational input. Moreover, the conjunction {*MPC*$^{>}$*, PPPC3,* $W_2' = 0$*, X1*} *is implied by the conjunction* {*PPPC, MPC*$^{>}$*, X1*} *of Theorem 7.4, hence also by the conjunction* {*PPPC, NCT-general,* $W = 0$} *of Theorem 7.2.*

Proof. Let W_p be composite and pass Condition (II); as in Theorem 7.3, the congruence restricts along the ring surjection $\mathbb{Z}[\sqrt{2}]/(W_p) \to \mathbb{Z}[\sqrt{2}]/(r)$ for every prime $r \mid W_p$, so Condition (II) holds at every prime factor; and every factor is $\equiv 1$ or $3 \pmod 8$ (Corollary 2.5), so the two cases below are exhaustive.

Case A: some factor $r_s \equiv 1 \pmod 8$*.* Verbatim as in Case A of Theorem 7.4: that argument uses MPC$^{>}$, the Platinum enumeration, the discharges of Proposition 6.30 and X1 only — no use of Π or PPPC is made in it — so it closes Case A under the present hypotheses unchanged.

Case B: all factors inert. Then p contributes to E_3. Theorem 7.8(a) gives $E_3 \le \Pi_3 + W_1 + W_2' + X_{E_3}$; PPPC3 gives $\Pi_3 = 0$; $W_1 = 0$ by Lemma 6.26; $W_2' = 0$ by hypothesis; and $X_{E_3} = 0$ by Theorem 7.8(b) under X1. Hence $E_3 = 0$ — contradiction.

Conjunction implication. PPPC implies PPPC3 ($\Pi = 0$ forces $|T_p^{>}| \le 1$ for every p, hence $\Pi_3 = 0$), and PPPC implies $W_2' = 0$ (Theorem 7.8(c)); the remaining hypotheses MPC$^{>}$ and X1 are common to the two conjunctions. The final implication then follows from the conjunction implication of Theorem 7.4. As there, the implications are claimed at the level of the conjunctions. ☐

Remark 7.10 (Heuristic and empirical status of the triple form). Heuristically, the erosion replaces a squared coincidence by a cubed one. A violation of PPPC requires two, and a violation of PPPC3 three, simultaneous local passes at a single exponent: in the model of §7.4 the expected pair and triple counts are governed by the second and the third power of the same small per-exponent single-pass rate — whose *first* moment, the expected number of single passes over all p, is expected to be divergent (single passes recur: the enumeration already exhibits three witness exponents) — so no first-moment count can close either form. These figures are heuristic only; they prove nothing about PPPC3 or Conjecture 7.7. Empirically, the two separate records of Table 1 — the Platinum enumeration (every inert factor $r \le 10^{12}$ in its range; exactly three locally passing rows, at pairwise distinct exponents) and the above-cutoff p-window scan of §7.3 (0 of 369,876 tested candidates satisfy condition (v)) — give $|T_p^{>}| \le 1$ wherever computed, so Π, Π_3 and W_2' all vanish as far as computed. Structurally, PPPC3 meets the obstruction of Observation 8.1 unchanged, including its analytic counting face: the objects counted are the same primitive divisors with the same pinned order data, and only the multiplicity of the required coincidence at one exponent is raised from two to three. The erosion shrinks the footprint of the residual, not the wall.

Remark 7.11 (The triple form is terminal for the count axis). The pair-to-triple step does not iterate usefully. First, a structural identity: $3 \nmid W_p$ for prime $p \geq 5$ (lifting the exponent gives $v_3(2^p + 1) = 1 + v_3(p) = 1$, exactly cancelled by the division by 3), so Lemma 6.21 applies at *every* prime factor of every composite W_p, and

$$W_p \text{ squarefree} \iff \text{every prime factor of } W_p \text{ is base-2 non-Wieferich.}$$

Repeating the proof of Theorem 7.8 one level up yields, with the same two computational inputs,

$$E_3 \;\leq\; \Pi_4 + W_1 + W_2' + W_3' + W_3^{\mathrm{sf}} + X_{E_3}, \qquad \Pi_4 := \sum_{p \geq 5} \binom{|T_p^{>}|}{4},$$

where W_3' counts the $\omega = 3$, all-factors-large, non-squarefree exponents — Wieferich-tagged exactly as W_2' — and W_3^{sf} counts the $\omega = 3$, all-factors-large, *squarefree* ones; there is no $\omega = 4$ corner (a squarefree small-plus-three-large configuration has even total multiplicity, excluded by Lemma 2.4). The display is stated for orientation, not as a numbered result, because it gains nothing: every rarity tag used in this paper is inapplicable to W_3^{sf} by definition — the parity dividend is spent ($\Omega = 3$ is odd), the multiplicity tag is absent (squarefree means precisely all-non-Wieferich, by the identity above), and the smallness tag is absent (no factor lies in the Platinum range) — so the hypothesis $W_3^{\mathrm{sf}} = 0$ is the bare triple coincidence again, carrying the heuristic mass of Π_3 itself. The same stratum recurs at every level $N+1 \geq 4$. Two further closures are worth recording. The global-pass route cannot tag the stratum either: by Remark 6.25, on squarefree W_p a global pass is exactly the conjunction of local passes. And the configuration is realized in the function-field calibration of §A.3: in the Carlitz analogue ($q = 3, 5$) there exist moduli with three and with four distinct large primitive prime factors, all of multiplicity one — so no technique transferable to $\mathbb{F}_q[t]$ can empty the stratum (the analogue carries no condition-(v) linkage, matching the coarsening under which the pair statement is false there; orders beyond four were not searched). Continuing the ladder would therefore require a genuinely new archimedean rarity input for squarefree large-triple configurations — precisely the missing input named in Observation 8.1.

Remark 7.12 (Cross-factor cofactor congruence). At an inert prime factor r of a composite W_p, with $d_r = \mathrm{ord}_r(\omega_3)/4$ as in §5.1, Condition (II) at r is *equivalent* to a congruence on the complementary cofactor:

$$\text{Condition (II) holds at } r \iff \frac{W_p}{r} \equiv 1 \pmod{4d_r}.$$

Forward: the pass gives $d_r \mid W_{p-2}$, hence $4d_r \mid 4W_{p-2} = W_p{+}1$ (Lemma 2.9), while the gap structure gives $4d_r \mid r + 1$; thus $W_p \equiv r \equiv -1 \pmod{4d_r}$, r is invertible with $r^{-1} \equiv -1$, and the cofactor is $\equiv (-1)(-1) = 1$. Backward:

$W_p/r \equiv 1$ and $r \equiv -1$ force $W_p \equiv -1 \pmod{4d_r}$, i.e. $d_r \mid W_{p-2}$, which is the pass. The congruence holds modulo $8d_r$ for free ($W_p \equiv r \equiv 3 \pmod 8$, Lemma 2.4). At an all-inert passer with $\omega = 3$ this becomes a closure system: for each i, the *other two* factors are mutual inverses, $r_j r_l \equiv 1 \pmod{4d_{r_i}}$. The congruence is verified at all three Platinum witness factors and at the $d = 67$ danger factor of Remark 5.17. Like the pair-separation constraints of Theorem 6.10, these are perimeter, not a tag: the system has nonempty congruence-consistent classes (it transfers to the function-field calibration verbatim), and for $\omega \geq 3$ its use as an exclusion is circular without new input.

8. Open problems

The full sufficiency (Conjecture 1.1) remains open. The unconditional gap consists of two problems — NCT for general d (§A.1) and PPPC (§A.2) — together with a third, far milder residual, the Wieferich–Wagstaff exclusion $W = 0$ of Theorem 6.15. The third is not a free-standing open problem of the same weight: by Remark 6.17 a failure of $W = 0$ would exhibit a base-2 Wieferich prime $\equiv 3 \pmod 8$, so $W = 0$ holds unconditionally for every W_p with all prime factors below 2^{64} [16] and is universally expected; it is recorded as component (R) of the structural obstruction in §A.2 and is not discussed separately below. The sharpened chain of Theorem 7.4 replaces these targets by the formally weaker MPC$^{>}$ (§4.7) and the single-exponent statement X1 (Conjecture 6.29) — the former third hypothesis $W_1 = 0$ is now unconditional (Lemma 6.26); but *proving* MPC meets the same structural obstruction as NCT — Observation 8.1 applies unchanged to its split datum — so the discussion below is phrased for the two principal residuals. The triple form (Corollary 7.9) erodes the inert target once more — PPPC is replaced by PPPC3 together with the corner exclusion $W_2' = 0$ (Conjectures 7.6 and 7.7) — but PPPC3 meets the identical obstruction (Remark 7.10), so the phrasing below is likewise unchanged.

The two principal problems are *technically* independent — they live on different branches ($r \equiv 1$ versus $r \equiv 3 \pmod 8$), in different algebras ($\mathbb{F}_r$ versus $\mathbb{F}_{r^2}$), and admit different attacks. We record at the outset that they nonetheless meet a single *structural* obstruction, which names precisely what new arithmetic input a closure of either would require. This is a unification of the obstruction, not a reduction of one problem to the other: the split side is delimited by the μ_p-tautology of §A.1 and the finiteness band of Proposition A.3; the inert side is delimited by Proposition A.13 and the refined second-moment identity of Proposition A.5. Neither side implies the other.

Observation 8.1 (The additive-coupling / multiplicative-projection obstruction). Both problems demand forcing a coincidence between two multiplicative orders attached to a single prime factor r of W_p: the order of the *non-unit* $2 = (\sqrt{2})^2$ and the order of a *unit* u of $\mathbb{Z}[\sqrt{2}]^\times$ — $u = \omega_3 = \alpha^2$ (norm $+1$)

on the inert branch, $u = \alpha = 1 + \sqrt{2}$ (norm -1) on the split branch. The obstruction has two levels, not to be conflated.

(1) Global level. In $\mathbb{Z}[\sqrt{2}]^\times$ the cyclic subgroups $\langle 2\rangle$ and $\langle u\rangle$ meet trivially, $\langle 2\rangle \cap \langle u\rangle = \{1\}$, by the valuation $v_{(\sqrt{2})}$ at the ramified prime: $v_{(\sqrt{2})}(2) = 2$ while every unit has $v_{(\sqrt{2})} = 0$, so $2^m = u^n$ forces $m = 0$ and then $n = 0$. Hence there is *no* multiplicative relation between 2 and u; the only relation is *additive* — $\sqrt{2} = \alpha - 1$, whence $2 = (\alpha - 1)^2$ (inert) and $2 = \alpha - \alpha^{-1}$ (split).

(2) Residue level. The datum deciding membership lives in a multiplicative power-residue projection of $\mathbb{F}_r^\times$ or $\mathbb{F}_{r^2}^\times$, and the one available relation — the additive $\sqrt{2} = \alpha - 1$ — has *no image* there: $x \mapsto x^{(r-1)/p}$ does not factor through addition, since $(x + y)^{(r-1)/p}$ is not a function of $x^{(r-1)/p}$ and $y^{(r-1)/p}$ (e.g. in $\mathbb{F}_{73}$ with $p = 3$, the elements 1 and 3 share the projection value $1^{24} = 3^{24} = 1$, yet $(1 + 1)^{24} = 64 \neq 8 = (3 + 1)^{24}$). At this level "multiplicative independence" is *false* and must not be invoked: in the cyclic group $\mathbb{F}_r^\times$ the subgroups $\langle\alpha\rangle$ and $\langle 2\rangle$ meet in a subgroup of order $\gcd(\operatorname{ord}_r \alpha, \operatorname{ord}_r 2)$, generically large, and equal to $\{\pm 1\}$ — never $\{1\}$ — on the split branch where $\gcd(8d, 2p) = 2$. The two branches realise the obstruction differently:

- *Split, $r \equiv 1 \pmod 8$.* Both 2 and α lie in $\mathbb{F}_r$; the decisive datum is $j_\alpha = \log_2(\alpha) \bmod p$, read in the μ_p-coordinate $x \mapsto x^{(r-1)/p}$. The three exact $\mathbb{F}_r$-internal identities (i)–(iii) of §A.1 all descend from $\sqrt{2} = \alpha - 1$ and collapse under $x^{(r-1)/p}$ to $0 \equiv 0 \pmod p$ (the μ_p-tautology), constraining j_α not at all.
- *Inert, $r \equiv 3 \pmod 8$.* Here $2 \in \mathbb{F}_r$ but $\omega_3 \in \mathbb{F}_{r^2}$ on the norm-1 line μ_{r+1}; r must simultaneously divide $2^p + 1$ with $\operatorname{ord}_r(2) = 2p$ (an $r - 1$ datum on the non-unit 2) and be a primitive divisor of the fixed V_{2d_r} with $\operatorname{ord}_r(\omega_3) = 4d_r$, $d_r \mid W_{p-2}$ (an $r + 1$ datum on the unit). These sit on opposite sides of $\gcd(r - 1, r + 1) = 2$; there is no single field element whose μ_p-coordinate decides membership, and no analogue of the tautological trio — the realisation here is the integer order-coincidence itself.

Consequently every coupled method surveyed in §§A.1–A.2 meets the same wall, in three faces, none a claim of mathematical impossibility, only of the reach of presently-known methods:

(R) reciprocity / Stickelberger / class field theory *relocate* the power-residue symbol over the Galois orbit of r (symmetrising it into a norm) but cannot *create* its value;

(C) fixed-conductor Galois / Iwasawa / Chebotarev / abc-radical invariants are *blind* to the order datum, because $\operatorname{ord}_r(2)$ is integer data — "a difference of infinitely many splitting conditions, not the splitting condition of any one finite extension" (Proposition A.13, unconditional);

(A) additive, resultant and elimination identities — and the Bugeaud–Corvaja–Zannier gcd bounds for the two recurrences $2^p + 1$ and V_{2d} — *die*, because the projection does not factor through addition (and, for the gcd bounds, because they constrain *size* in families-in-n, never the *count* at a fixed value).

On the inert (PPPC) side a fourth wall, analytic rather than algebraic, compounds these three. The residual there is a *count* — the pair-sum Π — and the condition carrying its finiteness, "r is a primitive divisor of the *fixed integer* V_{2d}", is not a prime-in-arithmetic-progression condition; sieve and effective-Chebotarev bounds count only the latter, so they do not by themselves bound Π. This is the counting analogue of face (C), and like (C) it is unconditional as a limitation of fixed-Frobenius data; see §A.3.

A method closing *either* problem must supply genuinely new arithmetic content living in the multiplicative power-residue projection itself — an object of odd order p (split) or a determinacy law for the multiset of orders of the divisors of one fixed integer (inert) — coupling 2 and the unit *across* the additive bridge $\sqrt{2} = \alpha - 1$, rather than scoring r against either structure alone. No such input is presently known. The asymmetry is real and is the only difference: PPPC is inert ($\mathbb{F}_{r^2}$, two prime factors per composite W_p available, residual a pair-count Π); NCT is split ($\mathbb{F}_r$, one factor per W_p, so two-prime contradictions are structurally unavailable, and every class-field-theoretic angle yields an equivalence, never a forcing, endpoint "$a \equiv 1 \pmod 8$"). The decoupling integers $\gcd(r-1, r+1) = 2$ (inert) and $\gcd(8d, 2p) = 2$ (split) are the same "$= 2$" obstruction in two guises.

The computational data and certificates underlying the finite ranges quoted above and in Table 1 are collected in Appendix B; the residual-obstruction facts needed to delimit the two open problems are collected in Appendix A. In brief:

Problem 1 (NCT, general d): prove Conjecture 4.20, no compatible triple for any odd $d > 1$ (proved unconditionally for every odd $d \leq 200$ — Theorem 4.18 for $d \leq 99$ and complete fixed-d closures at all nine parity-unblocked values $107 \leq d \leq 179$ — and at all nineteen parity-unblocked values of $(200, 400]$, §B.3). The split realisation of Observation 8.1 confines each d to an effective finite search (Proposition A.3) but leaves a μ_p-coordinate that the known $\mathbb{F}_r$-internal identities do not constrain.

Problem 2 (PPPC): prove $|T_p^{>}| \leq 1$ for every prime p, i.e. $\Pi = 0$ (Conjecture 7.1), which by Theorem 6.15 closes the all-inert branch of Conjecture 1.1. The inert realisation of Observation 8.1, the unconditional Proposition A.13, and the reason prime-in-AP estimates alone do not control Π are in §A.2. The unconditional perimeter of the problem is tightened in this version: two members of $T_p^{>}$ are separated at scale $8p\gcd(d_{r_1}, d_{r_2})$ (Theorem 6.10), and the diagonal part of Π vanishes at every admissible $d \leq 400$ (Propositions A.9 and 5.18), so a violation of PPPC is either off-diagonal — congruence-consistent but blocked at the realizability level, Remark 6.13 —

or diagonal at some admissible $d > 400$, where the residue theory is complete and the remaining content is the anti-aligned corner question of §A.4.

8.1. **Generalization.**

Remark. The structural ingredients used in Sections 4–5 — the v_2 classification of $\mathrm{ord}_r(\omega_3)$ (Lemma 2.8), the Pell exclusion numbers $N_d = V_d^2 + 2$ (Proposition 5.7), and the compatible-triple formulation of Conjecture R3 (Proposition 4.14) — are properties of $\omega_3 \in \mathbb{Z}[\sqrt{2}]$ and the Pell sequences. Wagstaff-specific content enters through Lemma 2.3 ($\mathrm{ord}_r(2) = 2p$) and the identity $W_p + 1 = 4W_{p-2}$; both have analogues in the generalised Chua family $N_b = (b^p + 1)/(b + 1)$. For each base b, the Chebyshev element $\omega_b = b + \sqrt{b^2 - 1}$ plays the role of ω_3, and $N_b + 1$ factors similarly. The extension requires $D = b^2 - 1$ to be non-square, so Pell sequences are replaced by the Lucas sequences of $\mathbb{Z}[\sqrt{D}]$.

Appendix A. Residual problems and obstruction summary

This appendix records only the residual facts needed to make the conditional reduction precise. The longer diagnostic log of closed approaches, together with independent split-local character refinements, is better treated as supplementary material: it is useful for future work but not load-bearing for Theorem 7.2.

Every ingredient of the reduction that carries no conjectural input can be assembled into a single statement: the complete list of unconditional constraints that a counterexample to Conjecture 1.1 must satisfy. The corollary below is proved unconditionally *modulo the named computational inputs* of Remark A.2 — the same certified finite computations already stated as inputs elsewhere in the paper — and assumes none of PPPC, MPC, NCT-general, $W = 0$, or X1.

Corollary A.1 (Shape of a counterexample). *Let $p \geq 5$ be prime, let W_p be composite, and suppose W_p satisfies Condition (II). Then:*

(0) (Common structure.) *Condition (II) holds at every prime factor of W_p; every prime factor is $\equiv 1$ or $3 \pmod 8$, and the total multiplicity of factors $\equiv 3 \pmod 8$ is odd; every prime factor r satisfies $\mathrm{ord}_r(2) = 2p$ exactly, so $r \equiv 1 \pmod{2p}$ and $r = 2pk + 1$ for an integer $k \geq 1$, and no prime divides two composite Wagstaff numbers; W_p is not a perfect power, so W_p has at least two distinct prime factors; and any* repeated *prime factor of W_p is a base-2 Wieferich prime $\equiv 1$ or $3 \pmod 8$ exceeding 2^{64} (base-2* super-*Wieferich if its multiplicity is ≥ 3), which, if inert, additionally satisfies the Pell–Wall–Sun–Sun-type condition $\mathrm{ord}_{r^2}(\omega_3) = \mathrm{ord}_r(\omega_3)$. Moreover exactly one of the configurations (a), (b) occurs.*

(a) (Split branch: some prime factor $r_s \equiv 1 \pmod 8$.) *Then:*

(a1) *a compatible triple* (d, r_s, p) *exists:* d *is odd,* $d \mid W_{p-2}$*, and* r_s *is a primitive divisor of* U_{4d} *(Pell rank* $\rho(r_s) = 4d$*), hence a divisor of the primitive part* Ψ_{4d}*;*
(a2) *every prime factor of* d *is Class III, and* $d > 400$*;*
(a3) $8pd + 1 \le r_s \le (2\sqrt{2})^{2\varphi(d)}$*; in particular* $8pd \mid r_s - 1$*, so* $r_s > 3200\,p$*, and* $p < (2\sqrt{2})^{2\varphi(d)}/(8d)$*;*
(a4) 2 *is a fourth-power residue modulo* r_s*, and a* q*-th power residue modulo* r_s *for every prime* $q \mid d$*;*
(a5) W_p *also has at least one inert prime factor* $r_{\mathrm{in}} \equiv 3 \pmod 8$*, and every inert prime factor satisfies the per-factor constraints (b1)–(b3) below; furthermore, either every inert prime factor exceeds* 10^{12} *— so that* p *violates MPC*$^{\triangleright}$ *— or* $p = 10{,}916{,}765{,}939$ *and the conclusions of (b4) hold verbatim.*

(b) (Inert branch: every prime factor $\equiv 3 \pmod 8$.) *Then* W_p *has* $t \ge 3$ *prime factors counted with multiplicity, of which at least two are distinct, and:*
(b1) *every prime factor* r *has odd order parameter* $d_r = \mathrm{ord}_r(\omega_3)/4$ *with* $d_r \mid \gcd\big((r+1)/4,\, W_{p-2}\big)$*,* $d_r > 400$ *(Corollary 5.19), and* $d_r > \log(2p)/\log(3 + 2\sqrt{2})$*;*
(b2) *every prime factor of every* d_r *is Class III;*
(b3) *every prime factor* r *is a primitive divisor of the Pell–Lucas number* $V_{2d_r} = V_{d_r}^2 + 2$ *in the* d*-indexed family, and its complementary cofactor satisfies* $W_p/r \equiv 1 \pmod{4d_r}$*;*
(b4) *at most one prime factor is* $\le 10^{12}$*; if one exists, then* $p = 10{,}916{,}765{,}939$*, that factor is the Platinum witness* $r = 152{,}834{,}723{,}147 = 14p{+}1$ *(with* $d_r = 38{,}208{,}680{,}787$*), and every other prime factor of* W_p *exceeds* $2p \cdot 10^{12} > 2.1 \times 10^{22}$*;*
(b5) *if* $p \ne 10{,}916{,}765{,}939$*, then every prime factor exceeds* 10^{12}*, and* W_p *has at least two distinct prime factors — at least three if* W_p *is squarefree — all lying in* $T_p^{>}$*; in particular* $|T_p^{>}| \ge 2$*, a violation of PPPC. Any two of them,* $r_1 = 2pk_1 + 1 < r_2 = 2pk_2 + 1$ *with* $d_i = d_{r_i}$*, satisfy:* $k_1 \equiv k_2 \equiv p \pmod 4$*;* $\gcd(pk_1 + 1,\, pk_2{+}1) \mid k_2{-}k_1$*;* $8p\gcd(d_1, d_2) \mid r_2{-}r_1$*; the per-candidate and pair-coupled valuation congruences of Propositions 6.9 and 6.11; and, if* $d_1 = d_2 = d$*, then* $d > 400$*,* $4d \mid k_2 - k_1$*, hence* $k_2 - k_1 \ge 1604$ *and* $r_2 - r_1 \ge 3208\,p$*;*
(b6) *if* W_p *is not squarefree, its repeated factor is a base-*2 *Wieferich prime* $\equiv 3 \pmod 8$ *exceeding* 2^{64} *satisfying the Pell–Wall–Sun–Sun-type condition of item (0), and — if additionally* $p \ne 10{,}916{,}765{,}939$ *—* $W_p \ge (2^{64})^2 \cdot 10^{12}$*, so* $p \ge 173$*.*

Proof. Pure bookkeeping; every item is an existing result applied to the hypothesis.

(0). Condition (II) restricts along the ring surjection $\mathbb{Z}[\sqrt{2}]/(W_p) \to \mathbb{Z}[\sqrt{2}]/(r)$ for every prime $r \mid W_p$, as in the proof of Theorem 7.3; the

residue classes and the parity of the inert multiplicity are Lemma 2.4; the pinned order is Lemma 2.3 (equivalently Theorem 6.1), and the disjointness across exponents is Theorem 6.2. W_p is not a perfect power by Lemma 6.26; with compositeness this gives two distinct factors. A repeated factor r has $v_r(2^{r-1} - 1) = v_r(W_p) \geq 2$ (Lemma 6.21), so r is base-2 Wieferich, super-Wieferich when $v_r(W_p) \geq 3$ (Corollary 6.23); its residue class is 1 or 3 (mod 8) (Lemma 2.4), and the exhaustive search below 2^{64} finds only $1093 \equiv 5$ and $3511 \equiv 7 \pmod 8$ [16], so $r > 2^{64}$. For an inert repeated factor, the global pass gives Condition (II) modulo r^2 and the Pell–Wall–Sun–Sun condition follows as in Remark 6.17 (cf. Remark 6.25).

(a1). Condition (II) at r_s in a composite W_p yields $\mathrm{ord}(\alpha_0) = 4d$ with d odd, $d \mid W_{p-2}$ (descent of §4), and Proposition 4.14(a) gives $\rho(r_s) = 4d$; this is a compatible triple (Definition 4.15), and a primitive divisor of U_{4d} divides Ψ_{4d} (as in Corollary A.4).

(a2). Each prime $q \mid d$ divides W_{p-2}; for $q > 3$, Theorem 6.7 forces $q \in A$, and $3 \in A$. NCT holds unconditionally for every odd $d \leq 400$ (Theorem 4.18 and the twenty-eight complete fixed-d certificates of §B.3), so $d > 400$.

(a3). Proposition A.3; $r_s > 3200\,p$ follows from $8pd \mid r_s - 1$ with $d \geq 401$.

(a4). Theorem 4.17 (its hypotheses — $r_s \equiv 1 \pmod 8$ a primitive divisor of U_{4d}, d odd — hold by (a1)) and Proposition 4.19.

(a5). By Lemma 2.4 the inert multiplicity is odd, hence ≥ 1. Every inert factor passes Condition (II) locally (item (0)), so the descent of §5.1 gives (b1)'s divisibilities; $d_r > 400$ by Corollary 5.19 (applicable since W_p is composite and passes at every factor), the Class III property by Theorem 5.3, the lower bound by Lemma 5.13, and (b3) by Lemma 5.8, Proposition 5.7 (with Theorem 5.5) and Remark 7.12 — exactly as in Remark 4.23. If every inert factor exceeds 10^{12}, the pair of data (i), (ii) with $r_{\mathrm{in}} > 10^{12}$ violates $\mathrm{MPC}^{>}$ (§4.7), as in Case A of Theorem 7.4. Otherwise some inert factor is $\leq 10^{12}$: then $p \leq (r-1)/2 < 5 \times 10^{11}$ (Lemma 2.3), so (p, r) is a locally passing row of the Platinum enumeration (which ranges over inert factors of *composite* W_p), p is one of the three witness exponents (Theorem 6.6, §B.4), and Proposition 6.30 exhibits failing factors at the other two — contradicting item (0) — so $p = 10{,}916{,}765{,}939$, and (b4)'s conclusions follow as below.

(b). $t \geq 3$ with every $d_i > 43$ and Class III is Theorem 5.16; two distinct factors from item (0). Items (b1)–(b3) are per-factor and were justified under (a5). (b4): at most one factor $\leq 10^{12}$ is Theorem 6.6; if one exists, the witness-exponent identification and the elimination of the two discharged exponents are as in (a5), and §B.4 records the witness factor and its d_r at $p = 10{,}916{,}765{,}939$; the directed scan at this exponent, complete for $k \leq 10^{12}$, finds no second prime factor in the progression $r = 2pk+1$ (§6.8.1, §B.4), and every prime factor lies in that progression (item (0)), so every other factor has $k > 10^{12}$, i.e. exceeds $2p \cdot 10^{12} = 2.18 \times 10^{22}$.

(b5). If $p \neq 10{,}916{,}765{,}939$, no factor is $\leq 10^{12}$ (else (b4) would force $p = 10{,}916{,}765{,}939$). Every distinct prime factor then satisfies all five defining conditions of $T_p^{>}$ (§6.7) by (b1) and item (0); there are at least two distinct

factors, and at least three when W_p is squarefree (Theorem 5.16: $t \geq 3$ with multiplicity = distinct count), exactly as in Step 2 of Theorem 6.15. The pair constraints are Theorem 6.10(1)–(3) (its hypotheses hold with $d_i = d_{r_i} \mid (r_i+1)/4$ odd), Propositions 6.9 and 6.11 (condition (v) holds at both candidates). On the diagonal $d_1 = d_2 = d$: the two factors form two same-d danger triples (Definition 5.15), so Theorem 6.10(4) gives $4d \mid k_2 - k_1$, and Propositions A.9 and 5.18 exclude every admissible $d \leq 400$; hence $d \geq 401$ (d odd), $k_2 - k_1 \geq 4d \geq 1604$ and $r_2 - r_1 = 2p(k_2 - k_1) \geq 3208\,p$ (cf. Corollary A.10).

(b6). Item (0) restricted to the all-inert case (every factor $\equiv 3 \pmod 8$); under (b5) the repeated factor exceeds 2^{64} and a second distinct factor exceeds 10^{12}, and the repeated factor has multiplicity ≥ 2, so $W_p \geq (2^{64})^2 \cdot 10^{12}$, giving $2^p \geq 3W_p - 1 > 2^{167}$ and $p \geq 168$, hence $p \geq 173$ (p prime). □

Remark A.2 (Computational inputs of the corollary). Corollary A.1 is unconditional modulo exactly the following certified finite computations, each already a stated input elsewhere in the paper; no conjecture enters.

- *Platinum enumeration* (Theorem 6.6): the 684,965,381-row enumeration of all inert prime factors $r \leq 10^{12}$ ($p < 5 \times 10^{11}$), used in (a5), (b4), (b5).
- *Witness discharges* (Proposition 6.30): the certified failing factors at $p = 85{,}684{,}865{,}933$ and $p = 1{,}251{,}488{,}009$, used in (a5), (b4), (b5).
- *Directed scan at* $p = 10{,}916{,}765{,}939$ (§6.8.1, §B.4): completeness of the second-factor search for $k \leq 10^{12}$, used in (b4).
- *Base-*2 *Wieferich search below* 2^{64} [16]: used in item (0) and (b6).
- *NCT certificates for* $d \leq 400$ (Theorem 4.18 and §B.3): complete primitive-part factorizations with every prime factor APR-CL-certified, used in (a2).
- *Diagonal closures* (Proposition A.9): the complete APR-CL-certified factorizations of V_{114}, V_{118}, V_{134}, V_{162}, V_{166}, V_{198}, V_{214}, V_{242}, used in (b5).

Items (0), (b1)–(b3), (a1), (a3), (a4) and the pair-separation congruences of (b5) carry *no* computational input at all: they are fully algebraic (the perfect-power exclusion rests on the published Nagell–Ljunggren results [2, 7, 6]).

Two readings of Corollary A.1 deserve emphasis. First, the *quotable contrapositive*: any composite Condition-(II) passer either contains a split factor — and then a compatible triple at some admissible $d > 400$, where each further value of d is a finite certificate whose cost is a complete factorization of $\Psi_{4d} \approx (2\sqrt{2})^{2\varphi(d)}$ (the certified range ends, at present, with the 290-digit Ψ_{1516} at $d = 379$, §B.3) — or it is all-inert, and then either $p = 10{,}916{,}765{,}939$ (the single exponent that X1 addresses) or W_p realizes a violation of PPPC outright. Second, the *scale*: the size constraints alone force $W_p \geq 10^{36}$, i.e. $p \geq 127$, in the all-inert branch ($p \geq 173$ in its non-squarefree corner), but size is not the content — the content is a coincidence of a type never yet observed. The 684,965,381-row Platinum enumeration contains exactly three locally passing rows, at three pairwise distinct exponents; the above-cutoff scans of §7.3 tested 376,435 candidates satisfying conditions (i)–(iv) and found *zero* passing condition (v); and the 9,506,750-rectangle corner census of Remark A.11 found no danger event at any composite W_p. A counterexample in the main branch (b5) requires two condition-(v) passes above the cutoff *at the same exponent* — two events of a kind not observed even once, individually, anywhere in the computed record — subject, in addition, to the separation congruences of (b5). The corollary thus quantifies, without any conjectural input, how far outside every computed window a counterexample must live.

A.1. **Problem 1: NCT for general d.** Conjecture 4.20 asks for no compatible triple (d, r, p) at any odd $d > 1$. Theorem 4.18 proves the range $d \le 99$ unconditionally; Appendix B gives complete fixed-d closures for all nine parity-unblocked values of $100 \le d \le 200$ and for every parity-unblocked value of $(200, 400]$, so NCT holds unconditionally for every odd $d \le 400$. What remains for $d > 400$ is not an unbounded search at a fixed d: each d has an explicit finite band.

Proposition A.3 (NCT structural finiteness band)**.** *Let (d, r, p) be a compatible triple (Definition 4.15). Then*

$$8pd + 1 \le r \le (2\sqrt{2})^{2\varphi(d)}.$$

Consequently $p < (2\sqrt{2})^{2\varphi(d)}/(8d)$, and for each fixed odd $d > 1$ the possible pairs (r, p) are finite and explicitly bounded.

Proof. By Proposition 4.14, $\mathrm{ord}_r(\alpha) = 8d$ for $\alpha = 1+\sqrt{2}$, while compatibility gives $\mathrm{ord}_r(2) = 2p$. Since $p \nmid d$ (Lemma 2.10), the two cyclic subgroups of $\mathbb{F}_r^\times$ generated by α and 2 intersect exactly in $\{\pm 1\}$. Their product has order $(8d)(2p)/2 = 8pd$, hence $8pd \mid r - 1$ and $r \ge 8pd + 1$.

For the upper bound, r is a primitive divisor of U_{4d} and hence divides the primitive part

$$\Psi_{4d} = \prod_{\zeta} (\alpha - \zeta\bar{\alpha}),$$

where ζ runs over primitive $4d$-th roots. Each factor has absolute value at most $\alpha + |\bar{\alpha}| = 2\sqrt{2}$, so

$$r \le |\Psi_{4d}| \le (2\sqrt{2})^{\varphi(4d)} = (2\sqrt{2})^{2\varphi(d)}.$$

∎

Corollary A.4 (Fixed-d NCT certificate)**.** *Fix an odd $d > 1$. If a complete factorisation of Ψ_{4d} is known, then NCT at this d is checked by the following finite certificate:*

1. *retain only prime factors $r \mid \Psi_{4d}$ with $r \equiv 1 \pmod 8$;*
2. *compute $p_r = \mathrm{ord}_r(2)/2$ and discard every r for which p_r is not prime;*

(3) *for each remaining r, test the exact condition $d \mid W_{p_r-2}$, equivalently the valuation-adjusted congruence system modulo L_d when d is admissible.*

If no factor survives, no compatible triple exists at this d.

Proof. In a compatible triple at d, Proposition 4.14 gives Pell rank $\rho(r) = 4d$, so r is a primitive divisor of U_{4d} and hence divides the primitive part Ψ_{4d}. The split residue condition is $r \equiv 1 \pmod 8$. Order-Pinning supplies the only possible Wagstaff exponent $p_r = \mathrm{ord}_r(2)/2$, which must be prime, and compatibility additionally requires $d \mid W_{p_r-2}$. Remark 6.8 gives the equivalent congruence formulation for the last condition when d is admissible; if d is not admissible, Theorem 6.7 already rules out the relevant divisibility at a non-Class-III prime factor. □

The band is effective but not uniform in d; the upper bound grows exponentially in $\varphi(d)$. Thus NCT-general is reduced to infinitely many finite searches, not to one finite computation. The algebraic reason the known identities stop here is the split realisation of Observation 8.1: after projecting to the μ_p-coordinate of $\mathbb{F}_r^\times$, the identities coming from $\sqrt{2} = \alpha - 1$ collapse to tautologies and do not determine the odd-order component that Conjecture R3 needs.

A.2. **Problem 2: PPPC.** PPPC (Conjecture 7.1) asserts $|T_p^>| \le 1$ for every prime p, equivalently $\Pi = 0$. By Theorem 6.15, this closes the all-inert branch once the mild Wieferich–Wagstaff residual $W = 0$ is also granted. The useful way to inspect the residual is to partition by the order parameter.

Proposition A.5 (Refined second-moment identity)**.** *For odd admissible $d > 43$, let*

$$P_d(p) = \#\{\, r \in T_p^> : d_r = d \,\}.$$

Then $|T_p^>| = \sum_d P_d(p)$ and

$$\binom{|T_p^>|}{2} = \sum_{d_1<d_2} P_{d_1}(p)P_{d_2}(p) + \sum_d \binom{P_d(p)}{2}.$$

After summing over p,

$$\Pi = \sum_{d_1<d_2} S(d_1,d_2) + \sum_d S_{\mathrm{diag}}(d),$$

where $S(d_1,d_2) = \sum_p P_{d_1}(p)P_{d_2}(p)$ and $S_{\mathrm{diag}}(d) = \sum_p \binom{P_d(p)}{2}$.

Proof. This is the multinomial identity for $\binom{\sum_d n_d}{2}$ with $n_d = P_d(p)$. All terms are non-negative, so Tonelli's theorem justifies the rearrangement after summing over p. □

The diagonal term matters: one Pell–Lucas number V_{2d} can have two distinct primitive divisors, so bounding only the off-diagonal terms does not bound Π. The diagonal is, however, exactly the part of Π where unconditional progress is possible, because both members of a diagonal pair divide the one fixed integer V_{2d}. The following package develops this.

Lemma A.6 (Half-pinning)**.** *Let d be odd and let $r > 3$ be a prime with $r \mid V_{2d}$ and $\operatorname{ord}_r(2) = 2m$, m odd (for $r \equiv 3 \pmod 8$ the parity of m is automatic). Then there is a unique sign $\varepsilon = \varepsilon_r \in \{\pm 1\}$ with*

$$V_d \;\equiv\; \varepsilon\, 2^{(m+1)/2} \pmod r;$$

equivalently, r divides exactly one of the two integers $H_\varepsilon = V_d - \varepsilon\, 2^{(m+1)/2}$.

Proof. $r \mid V_{2d} = V_d^2 + 2$ (Proposition 5.7) gives $V_d^2 \equiv -2 \pmod r$. From $\operatorname{ord}_r(2) = 2m$, the element 2^m has order 2 in $\mathbb{F}_r^\times$, so $2^m \equiv -1 \pmod r$. Hence $V_d^2 \equiv 2^m \cdot 2 = 2^{m+1} \pmod r$ with $m+1$ even, so $\big(V_d\, 2^{-(m+1)/2}\big)^2 \equiv 1$ and $V_d \equiv \pm 2^{(m+1)/2} \pmod r$. The sign is unique: if both signs held, then $r \mid 2^{(m+1)/2+1}$, impossible for odd r. For $r \equiv 3 \pmod 8$, $v_2(\operatorname{ord}_r(2)) = 1$ as in the proof of Lemma 2.4, so m is odd automatically. $\square$

For a fixed pair (d, p) with d odd and $p \geq 5$ prime, write

$$\mathcal{L}(d,p) \;=\; \{\, r > 3 \text{ prime} \;:\; r \mid V_{2d},\;\; 2^p \equiv -1 \pmod r \,\}$$

for the *order-divisibility locus* at (d, p). Every danger triple (d, r, p) (Definition 5.15) has $r \in \mathcal{L}(d, p)$: primitivity gives $r \mid V_{2d}$ and the pinned order gives $2^p \equiv -1$. The locus is in general strictly larger than the set of danger factors — it also captures non-primitive divisors ($\operatorname{ord}_r(\omega_3) = 4e$ for a proper divisor $e \mid d$) and split divisors $r \equiv 1 \pmod 8$ of V_{2d}. This over-capture is deliberate and harmless: the certificate below is used for *exclusion*, which is a superset test, and any positives are post-filtered by the mod-8, primitivity and size requirements.

Proposition A.7 (Corner localization)**.** *Let d be odd, $p \geq 5$ prime, and set*

$$G(\varepsilon, \varepsilon') \;=\; \gcd\!\Big(V_d - \varepsilon\, 2^{(p+1)/2},\;\; 2U_d - \varepsilon'\Big), \qquad \varepsilon, \varepsilon' \in \{\pm 1\}.$$

Then $\mathcal{L}(d, p)$ is exactly the set of prime divisors > 3 of the four integers $G(\varepsilon, \varepsilon')$, and each $r \in \mathcal{L}(d, p)$ divides $G(\varepsilon, \varepsilon')$ for exactly one sign pair $(\varepsilon_r, \varepsilon'_r)$. The four integers are computable from the residues $2^{(p+1)/2} \bmod (2U_d \mp 1)$ — at most four modular exponentiations and four gcds on integers of $O(d)$ digits — with no factorization of V_{2d}.

Proof. For odd d, the identities $V_{2d} = V_d^2 + 2$ (Proposition 5.7) and $V_d^2 - 8U_d^2 = -4$ (§1.5) combine to the rational halving

$$V_{2d} \;=\; 8U_d^2 - 2 \;=\; 2\,(2U_d - 1)(2U_d + 1).$$

Let $r \in \mathcal{L}(d, p)$. From $2^p \equiv -1 \pmod r$: $\operatorname{ord}_r(2) \mid 2p$ and $\nmid p$; since p is *prime*, $\operatorname{ord}_r(2) \in \{2, 2p\}$, and $\operatorname{ord}_r(2) = 2$ forces $r \mid 3$, excluded — this is the one point where the primality of p is used. So $\operatorname{ord}_r(2) = 2p$ with p odd, and Lemma A.6 pins $V_d \equiv \varepsilon_r\, 2^{(p+1)/2} \pmod r$ for a unique ε_r. By the halving, the odd prime r divides $(2U_d - 1)(2U_d + 1)$ and divides exactly one of the two factors (their difference is 2): $2U_d \equiv \varepsilon'_r \pmod r$ for a unique ε'_r. Hence $r \mid G(\varepsilon_r, \varepsilon'_r)$ and r divides no other G. Conversely, let $r > 3$ be a prime divisor of some $G(\varepsilon, \varepsilon')$. Then $r \mid 2U_d - \varepsilon'$, so $r \mid V_{2d}$ by the halving, whence $V_d^2 \equiv -2 \pmod r$; and $r \mid V_d - \varepsilon\, 2^{(p+1)/2}$ squares to $V_d^2 \equiv 2^{p+1} \pmod r$. Together $2^{p+1} \equiv -2$, i.e. $2^p \equiv -1 \pmod r$, so $r \in \mathcal{L}(d,p)$. Finally $\gcd(A,B) = \gcd(A \bmod B, B)$ reduces each G to the two residues $2^{(p+1)/2} \bmod (2U_d \mp 1)$. □

Remark A.8 (Pair caps: three of the four sign classes are product-capped). Fix (d,p) as above, $d \ge 3$, and let $r_1 \ne r_2$ be two primes in $\mathcal{L}(d,p)$ with sign data $(\varepsilon_i, \varepsilon'_i)$.

- If $\varepsilon'_1 = \varepsilon'_2 = \varepsilon'$, then $r_1 r_2 \mid 2U_d - \varepsilon'$, so $r_1 r_2 \le 2U_d + 1 \sim \sqrt{V_{2d}/2}$ — a cap valid for *all* p.
- If $\varepsilon_1 = \varepsilon_2 = \varepsilon$, then $r_1 r_2$ divides $H_\varepsilon = V_d - \varepsilon\, 2^{(p+1)/2} \ne 0$, so $r_1 r_2 \le |H_\varepsilon| \le V_d + 2^{(p+1)/2}$. ($H_\varepsilon \ne 0$ because $V_d/2$ is odd and > 1 for odd $d \ge 3$, by $(V_d/2)^2 = 2U_d^2 - 1$, so V_d is not a power of 2.) This cap improves on the trivial bound $r_1 r_2 \le V_{2d}$ precisely in the range $p < 4\log_2(1+\sqrt{2})\, d + O(1) \approx 5.0862\, d$.

Only *anti-aligned* pairs — $\varepsilon_1 = -\varepsilon_2$ *and* $\varepsilon'_1 = -\varepsilon'_2$ — escape both product caps; pairs in the three aligned sign classes must fit inside a single explicit integer.

These statements stratify and cap the diagonal; they do not by themselves empty it. What does empty it, in the certified range, is composing them with complete factorizations.

Proposition A.9 (Diagonal vanishing through $d = 124$)**.** *For every admissible odd d with $43 < d \le 124$ (the complete list:* 57, 59, 67, 81, 83, 99, 107, 121*), no prime p admits two distinct danger triples (d, r_1, p), (d, r_2, p) sharing the parameter d. Consequently $S_{\mathrm{diag}}(d) = 0$ for all p at every admissible $d \le 124$. The statement is k-free: it excludes diagonal pairs at these d for danger factors of every size, not only in a finite k-window.*

Verification. The six Pell–Lucas numbers V_{118}, V_{162}, V_{166}, V_{198}, V_{214}, V_{242} (46, 63, 64, 76, 82, 93 decimal digits) are completely factored, with 4, 11, 5, 12, 5, 11 distinct prime factors exceeding 3 respectively (factorizations assembled from the public factor database [19] and re-verified by exact product identities); the 42 distinct primes occurring across the six are each certified prime by APR-CL (PARI/GP `isprime` flag 2). For each such prime r the danger prerequisites of Definition 5.15 are decided exactly: the residue $r \equiv 3 \pmod 8$; primitivity $\mathrm{ord}_r(\omega_3) = 4d$, computed by order descent in $\mathbb{F}_{r^2}$; $v_2(\mathrm{ord}_r(2)) = 1$; and primality of the pinned exponent $p_r = \mathrm{ord}_r(2)/2$. At $d \in \{83, 99, 107, 121\}$ no prime factor survives all four tests; at $d = 59$ exactly one survives ($r = 88{,}499$, pinned to $p = 44{,}249$) and at $d = 81$ exactly one ($r = 5507$, pinned to $p = 2753$). For $d = 57$ and $d = 67$ the same certificate over the complete factorizations of V_{114} and V_{134} (Remark 5.17) leaves the surviving divisors $\{683,\ 171{,}369{,}731{,}867\}$ — pinned to the *distinct* exponents 11 and 85,684,865,933 — and $\{148{,}937{,}135{,}261{,}632{,}265{,}803\}$, pinned to 24,822,855,876,938,710,967. No two surviving divisors share a pinned exponent at any of the eight d (condition (v) was not even needed), so no diagonal pair exists at any p. Certificate scripts (§B.5): `scripts/cp354_samed_diagonal_certificates.py` and `scripts/cp354_factordb_v2d_diagonal.py`.

□

Corollary A.10 (Diagonal vanishing through $d = 400$; no pair with $k \leq 1604$)**.** *No prime p admits two distinct danger triples (d, r_1, p), (d, r_2, p) with the same admissible $d \leq 400$: the diagonal contribution to Π vanishes there for danger factors of every size. In particular no diagonal pair exists with both $k_i = (r_i - 1)/2p \leq 1604$, unconditionally.*

Proof. The census of Proposition 5.18 is exhaustive on $43 < d \leq 400$, and no two of its rows share both the parameter d and the pinned exponent (the two rows at $d = 57$ have exponents $11 \neq 85{,}684{,}865{,}933$). For the k-form: by Theorem 6.10(4), such a pair has $4d \mid k_2 - k_1$ with $1 \leq k_2 - k_1 \leq 1603$, so $d \leq 400$; d is admissible with $d > 43$ (Definition 5.15), and every such d is closed. □

Remark A.11 (Factorization-free corner census). Independently of the factorizations above, the corner certificate of Proposition A.7 was run over every admissible d with $43 < d \leq 124$ and every prime $p \leq 10^6$ satisfying the condition-(v) congruence for that d ($d \mid W_{p-2}$; non-admissible d admit no such p at all, by Theorem 6.7) — 29,930 pairs (d, p) in total, at four modular exponentiations each. The locus $\mathcal{L}(d, p)$ is empty at every one of them except $(d, p) = (57, 11)$, where it consists of the single prime $683 = W_{11}$ — the known *phantom* triple of Remark 5.17, at which W_p is prime. Because the census tests the locus itself, it is k-unbounded — it excludes danger factors of every size at the swept (d, p), which p-indexed scans of the progression $r = 2pk + 1$ cannot do — and it over-captures by design. This is finite-range empirical verification within the swept rectangle, not a proof for general p. Script: `scripts/cp354_diag_corner_gcd.py`. An extension of the census to every admissible $d \leq 1000$ and every condition-(v) prime $p \leq 10^8$ — 9,506,750 rectangles, about 320 times the original sweep — found exactly four corner hits, all of them the known phantom rows at *prime* W_p ($683 = W_{11}$ at $d = 171$; $2731 = W_{13}$ at $d = 683$; $43691 = W_{17}$ at $d = 331$ and $d = 993$): zero danger events at composite W_p and zero diagonal pairs (script `scripts/cp358_corner_sweep_d1000.py`).

Remark A.12 (Same-d triples force alignment). Three distinct primes in $\mathcal{L}(d, p)$ carry three sign pairs $(\varepsilon_i, \varepsilon_i') \in \{\pm 1\}^2$; by pigeonhole some two share ε *and* some two (possibly a different two) share ε'. A same-d *triple* therefore always contains an ε'-aligned pair — product-capped by $2U_d + 1$ at every p (Remark A.8; the ε-aligned cap also applies, though it is size-binding only for

$p \lesssim 5.0862\,d$) — so the anti-aligned escape of Remark A.8 is closed *for triples*; the pair-level anti-aligned corner question of §A.4 remains open. Moreover a same-d triple of danger-class factors contains a same-d danger pair, so same-d triples are excluded at every admissible $d \leq 400$ (all k; Propositions A.9 and 5.18), and whenever two members have $k_i \leq 1604$ (Corollary A.10).

Off the diagonal, and on the diagonal beyond the certified range, the obstructions below delimit what presently-known methods can reach.

Proposition A.13 (Frobenius cannot see the order of 2)**.** *Let $M/\mathbb{Q}$ be any finite Galois extension. There exist primes $r \neq r'$, unramified in M, with the same Frobenius conjugacy class in* $\mathrm{Gal}(M/\mathbb{Q})$ *but* $\mathrm{ord}_r(2) \neq \mathrm{ord}_{r'}(2)$.

Proof. Fix a conjugacy class $\mathcal{C} \subset \mathrm{Gal}(M/\mathbb{Q})$. Chebotarev gives infinitely many unramified primes with Frobenius class $\mathcal{C}$. For a fixed integer n, only finitely many primes satisfy $\mathrm{ord}_r(2) = n$, since each divides $2^n - 1$. Hence $\mathrm{ord}_r(2)$ is unbounded on that infinite Chebotarev set and assumes at least two values there. $\square$

Thus no invariant factoring through a fixed finite Galois extension can decide membership in $T_p^{>}$: the pinned exponent $p = \mathrm{ord}_r(2)/2$ is not finite-level Frobenius data. This is the precise face (C) of Observation 8.1.

A.3. **Why the AP-only analytic fallback does not close the gap.** A weaker target would be $\Pi < \infty$, which would imply only finitely many all-inert global passes after the corresponding finite form of the W residual. The standard second-moment route still fails for a structural reason. For a fixed d, every relevant r is a divisor of the fixed integer V_{2d}, hence

$$\sum_p P_d(p) \leq \omega(V_{2d}), \qquad S(d_1, d_2) \leq \omega(V_{2d_1})\omega(V_{2d_2}).$$

A sieve or effective-Chebotarev estimate bounds primes in arithmetic progressions; it does not bound how the prime divisors of the fixed integer V_{2d} distribute among the pinned orders $\mathrm{ord}_r(2)$. Feeding only the AP condition $d \mid W_{p-2}$ gives infinitely many p in one valuation-adjusted congruence class and no finiteness for Π.

The correct modulus for composite d is the valuation-adjusted L_d of Remark 6.8, not the squarefree shorthand M_d. Thus an attempted second-moment summation must control expressions of the shape

$$\frac{\omega(V_{2d_1})\omega(V_{2d_2})}{\varphi(L_{d_1})\varphi(L_{d_2})}$$

together with the distribution of the actual primitive divisors of V_{2d_i}. This is a sharper and more concrete target than a raw AP count, but it is not presently supplied by Brun–Titchmarsh, effective Chebotarev, or GRH alone.

Finding. *The AP-only fallback does not prove the weaker bound $\Pi < \infty$. The missing input must detect that a prime divides the specific integer V_{2d}*

while also satisfying the pinned Wagstaff order condition; a sharper prime-in-AP estimate, even under GRH, does not supply that information. A valuation-adjusted average over L_d remains a legitimate open direction, not a closed argument.

A.4. **Directions.** The remaining plausible routes are those that keep the two divisibilities coupled across $W_p+1 = 4W_{p-2}$: sharper divisor bounds for V_{2d} beyond Stewart–Tijdeman; an average distribution theorem for prime divisors of the non-standard sequence $\{V_{2d}\}$; a half-dimensional sieve for primes of the form a^2+2b^2 combined with information about the fixed integer V_{2d}; or a genuinely new congruence law for the linked pair $(\mathrm{ord}_r(2), \mathrm{ord}_r(\omega_3))$. Each would be new arithmetic input rather than a refinement of the existing reduction.

The diagonal after the half-pinning package. On the diagonal — the sub-sum $\sum_d S_{\mathrm{diag}}(d)$ of Π — the situation is sharper, in both directions. On the one hand, residue algebra on the diagonal is *complete.* At a prime $r \in \mathcal{L}(d,p)$ with $\mathrm{ord}_r(2) = 2p$, every natural residue datum the configuration supplies lies in the single cyclic subgroup $\langle 2\rangle \le \mathbb{F}_r^\times$ of order $2p$: $-1 = 2^p$; $V_d \equiv \varepsilon\, 2^{(p+1)/2}$ (Lemma A.6); $2U_d \equiv \varepsilon'$ (Proposition A.7); and, at a prime π of $\mathbb{Z}[\sqrt{-2}]$ above r chosen with $\pi \mid V_d + \sqrt{-2}$ (Remark 5.9), $\sqrt{-2} \equiv -V_d \pmod{\pi}$, while of the conjugate elements $\beta_\pm = 1 \pm (V_d/2)\sqrt{-2}$ of norm $V_{2d}/2$, the one not divisible by π reduces to the trace $\beta_+ + \beta_- = 2$ itself. The subgroup generated by all of these is therefore exactly $\langle 2\rangle$, so every μ_p-character evaluated on them is tautological — the inert-diagonal incarnation of face (A) of Observation 8.1 — and the only non-tautological residue data are the two sign bits $(\varepsilon, \varepsilon')$, both of which Lemma A.6 and Proposition A.7 already extract. There is no further residue-level information on the diagonal to be had; what remains is size (the caps of Remark A.8) and realizability. On the other hand, this localizes the entire open content of the diagonal in one named question.

Anti-aligned corner question. *Can the integers $G(\varepsilon, \varepsilon')$ and $G(-\varepsilon, -\varepsilon')$ of Proposition A.7 simultaneously contain primitive inert prime divisors exceeding 10^{12} at a single prime p satisfying condition (v)?*

The two corner integers are coprime — each is odd, and an odd common prime divisor would divide both $2V_d$ and $2^{(p+1)/2+1}$ — so the question concerns the joint large-prime content of two coprime integers attached to one (d,p). A diagonal pair at any admissible $d > 124$ (the range left open by Proposition A.9) either occupies two anti-aligned corners or is confined by Remark A.8 to the divisors of a single explicit integer of roughly half the size of V_{2d}. We record one honest calibration: every statement of the package — the identity, the corners, the caps — uses only the quadratic recurrence, the order of 2, and integer size, and transfers verbatim to the analogous Lucas data over $\mathbb{F}_q[t]$, a setting in which the corresponding pair statement is false (we use this nowhere; it is recorded as a guard against over-reading). The package can therefore localize the diagonal difficulty but cannot alone close it: closing the anti-aligned corner question requires input of the same fixed-divisor type as faces (C)/(A) above — a determinacy law for which corners of which (d, p) capture large primitive divisors.

Appendix B. Computational verification

This appendix keeps only the certification facts used in the paper. Full tables, scripts, and longer diagnostic logs are treated as supplementary material.

B.1. **Known Wagstaff values.** The current record and probable-prime status for Wagstaff numbers is maintained by PrimePages [17]; it changes over time and is not an input to any implication in this paper. The finite computations here use explicit exponent lists in the supplementary material. Three certified cases $p \in \{2617, 10501, 12391\}$ also have ECPP-independent certificates in the companion paper [22].

B.2. **Split-branch verification.** All 185 known prime factors $r \equiv 1 \pmod 8$ of composite W_p satisfy Conjecture R3. For each listed factor, r is independently certified prime and the order condition $p \mid \operatorname{ord}_{\mathbb{F}_r^\times}(1+s)$ (with $s^2 = 2$) is checked directly; the case labels record which structural shortcut explains the computation:

Configuration	Count	Reference
Case A ($r \mid U_p$)	1	Theorem 4.10
Case B ($r \mid V_p$)	6	Theorem 4.11
Case C, $4U_p^2 \equiv 1$	46	Theorem 4.12
Case C, inadmissible d	43	Proposition 4.14
Case C, general ($d > 200$)	89	direct R3 verification

Together these cover every known split factor.

B.3. **NCT and primitive-divisor scans.** For NCT, Theorem 4.18 gives the unconditional range $d \leq 99$. The parity-unblocked values in $100 \leq d \leq 200$ are

$$\{107, 121, 129, 131, 139, 163, 171, 177, 179\}.$$

Six of them are closed by complete fixed-d certificates in the sense of Corollary A.4. For $d = 107$:

$$\begin{aligned}\Psi_{428} = {}& 161522909729 \cdot 1262880768427 \cdot 17449564219651 \\ & \cdot 2906272939091099 \cdot 1321684963208293564613613536 83.\end{aligned}$$

The only split factor is $r = 161522909729$, and

$$\operatorname{ord}_r(2)/2 = 10095181858 = 2 \cdot 107 \cdot 47173747$$

is composite. For $d = 121$:

$$\begin{aligned}\Psi_{484} = {}& 1451 \cdot 568699 \cdot 1197595081 \cdot 1614230507 \cdot 135855589627 \\ & \cdot 7499151961531 \cdot 249507118668323 \cdot 4005463110187137017.\end{aligned}$$

The split factors are 1197595081 and 4005463110187137017. For the first:

$$\mathrm{ord}_{1197595081}(2)/2 = 9979959 = 3 \cdot 11^2 \cdot 19 \cdot 1447,$$

and for $r = 4005463110187137017$:

$$\begin{aligned}\mathrm{ord}_r(2)/2 &= 500682888773392127\\ &= 11^2 \cdot 23 \cdot 3739247 \cdot 48113327.\end{aligned}$$

Both half-orders are composite, so no compatible triple exists at $d = 107$ or $d = 121$. The script `scripts/cp350_nct_107_121_verify.py` verifies the displayed primitive-part factorizations, primality of each listed factor by SymPy and PARI/GP `isprime`, and the split half-order decompositions.

The four further values $d = 129, 139, 171, 177$ are closed by the same certificate over the complete factorizations of their primitive parts (65, 106, 83 and 89 decimal digits respectively):

$$\begin{aligned}\Psi_{516} &= 75390435818587404514891047039001\\ &\quad \cdot 276431598001487177038457370080041,\\ \Psi_{556} &= 6568027 \cdot 574593323881\\ &\quad \cdot 1976717870190267288567284435734058447888899\\ &\quad \cdot 5764333243492029733503944458218529327657872457,\\ \Psi_{684} &= 8209 \cdot 10259 \cdot 1186057 \cdot 82658931465203\\ &\quad \cdot 300531150542299499\\ &\quad \cdot 1926987883277478331480345370673300299,\\ \Psi_{708} &= 6362089 \cdot 2101543793110577486131976361925047345 7\\ &\quad \cdot 4902409838027929629661924666159500844585 11337.\end{aligned}$$

For $d = 129$ both factors are split. The 32-digit factor has odd order,

$$\mathrm{ord}_r(2) = 94238044773234255643613808798 75,$$

and the 33-digit factor has composite half-order

$$\begin{aligned}\mathrm{ord}_r(2)/2 &= 691078995003717942596143425200 1\\ &= 3 \cdot 13^2 \cdot 43 \cdot 89 \cdot 11483 \cdot 14449 \cdot 45697 \cdot 469764750691.\end{aligned}$$

For $d = 139$ the two factors $\equiv 3 \pmod 8$ are inert and irrelevant to the split search; of the two split factors, $r = 574593323881$ has odd order $\mathrm{ord}_r(2) = 23941388495$, and the 46-digit factor has composite half-order

$$\begin{aligned}\mathrm{ord}_r(2)/2 &= 7205416554365037166879930572773161659572340 57\\ &= 3 \cdot 139 \cdot 157 \cdot 20305590747839\\ &\quad \cdot 542010570914390872795421827.\end{aligned}$$

For $d = 171$ four factors are inert; the split factor $r = 8209$ has half-order $1026 = 2 \cdot 3^3 \cdot 19$ (even, hence composite), and the split factor $r = 1186057$

has odd order $\mathrm{ord}_r(2) = 148257$. For $d = 177$ all three factors are split, with composite half-orders

$$\begin{aligned}\mathrm{ord}_{6362089}(2)/2 &= 265087 = 59 \cdot 4493,\\ \mathrm{ord}_r(2)/2 &= 2626929741388221857664970452406309182\\ &= 2 \cdot 3 \cdot 7 \cdot 59 \cdot 257671 \cdot 36652148601361 \cdot 112248920701199\end{aligned}$$

for the 38-digit factor, and

$$\begin{aligned}\mathrm{ord}_r(2)/2 &= 61280122975349120370774058326993760557313917\\ &= 3^2 \cdot 11 \cdot 19 \cdot 59 \cdot 601 \cdot 2281 \cdot 18218257\\ &\qquad \cdot 44560482149 \cdot 496160696018531\end{aligned}$$

for the 45-digit factor. No split factor of any of the four primitive parts has prime half-order, so step 2 of the certificate already excludes every candidate (step 3 is never reached), and no compatible triple exists at $d \in \{129, 139, 171, 177\}$. The script `scripts/cp351_nct_certificate_129_139_171_177.py` verifies the displayed primitive-part factorizations (exact product identities against Ψ_{4d}) and the split half-order decompositions; primality of all fifteen listed factors is certified by APR-CL (PARI/GP `isprime` with flag 2) in the independent end-to-end recomputation `scripts/cp352_verify_nct_recompute.py`, which also re-derives the primitive parts from scratch, recomputes every rank of apparition and every order with minimality witnesses, and re-tests the exact condition $d \mid W_{p_r-2}$.

For $d = 163$ the primitive part Ψ_{652} (125 decimal digits) is completely factored:

$$\begin{aligned}\Psi_{652} = {} & 1124699 \cdot 1096864817 \cdot 13182839507\\ & \cdot 12036824328615957881 \cdot 221024872680238792171\\ & \cdot 65609788974347359786729\\ & \cdot 3577684812700647698297361563656018739,\end{aligned}$$

all seven factors certified prime by APR-CL. Of the three split factors, two have composite half-orders,

$$\begin{aligned}\mathrm{ord}_{1096864817}(2)/2 &= 68554051 = 31 \cdot 163 \cdot 13567,\\ \mathrm{ord}_{12036824328615957881}(2)/2 &= 9230693503539845\\ &= 5 \cdot 431 \cdot 4283384456399,\end{aligned}$$

and the 23-digit factor has odd $\mathrm{ord}_r(2)$; so no compatible triple exists at $d = 163$. (The 20-digit split factor is the prime of Remark 4.21.) The certificate is `scripts/cp353_nct_certificate_163.py`.

For $d = 131$ the primitive part Ψ_{524} (100 decimal digits) is completely factored (GNFS):

$$\begin{aligned}\Psi_{524} = {} & 523 \cdot 1049 \cdot 67073 \cdot 135193 \cdot 1235593 \cdot 6948763 \\ & \cdot 3525458899 \cdot 319560279309632944107942971 \\ & \cdot 67103907069340766277660216218294177.\end{aligned}$$

Of the five split factors, three have odd $\mathrm{ord}_r(2)$ or half-order divisible by 4, and the 35-digit factor has composite half-order $2^2{\cdot}131{\cdot}842505864169103634462387206437$. The factor $r = 1049$ is the only certificate instance in this paper reaching step 3: its half-order is the *prime* $p_r = 131 = d$, and the exact condition $d \mid W_{p_r-2}$ fails because $p \nmid W_{p-2}$ (Lemma 2.10) — the $d = p$ diagonal. No compatible triple exists at $d = 131$. Certificate: `scripts/cp353_nct_certificate_131.py`.

For $d = 179$ the primitive part Ψ_{716} (137 decimal digits) is completely factored (ECM, $B_1 \leq 11{\times}10^6$, final 58-digit prime cofactor):

$$\begin{aligned}\Psi_{716} = {} & 1433 \cdot 4297 \cdot 3627971 \cdot 28260363913 \\ & \cdot 58136103167853442603 \\ & \cdot 98733110510495882688859990311113569 \\ & \cdot 4970296102561833027001182370298510775841518525241649807641.\end{aligned}$$

All five split factors are excluded at step 2: four have odd $\mathrm{ord}_r(2)$, and the 35-digit factor has composite half-order $2^2 \cdot 179 \cdot 937 \cdot 373453 \cdot 2525800903 \cdot 6500756482369$. No compatible triple exists at $d = 179$. Certificate: `scripts/cp353_nct_certificate_179.py`. All sixteen listed prime factors of Ψ_{524} and Ψ_{716} are certified by APR-CL (PARI `isprime` flag 2).

A separate finite-window sieve searched all nine parity-unblocked values above: all primes $r = 8dk + 1$ with $r < 10^{11}$ were enumerated, rank $\rho(r) = 4d$ was tested by the Pell recurrence, and surviving primitive divisors were checked for a compatible prime p with $\mathrm{ord}_r(2) = 2p$ and $d \mid W_{p-2}$. The run tested about 1.07×10^8 primes, found 12 primitive Pell divisors, and found no compatible triple in this window. Since the structural bound of Proposition A.3 is far larger than 10^{11} for some of these d, this is evidence only, not a complete verification of $100 \leq d \leq 200$. After the nine complete closures at $d = 107, 121, 129, 131, 139, 163, 171, 177, 179$, no parity-unblocked value of $100 \leq d \leq 200$ remains open: NCT holds unconditionally for every odd $d \leq 200$.

The window $200 < d \leq 300$ contains ten parity-unblocked values: $d = 201$, 209, 211, 227, 243, 249, 251, 281, 283, 297. At six of them the primitive part Ψ_{4d} is completely factored in the public factorization tables (sizes 102–215 decimal digits; the products were re-verified exactly against locally recomputed Ψ_{4d}, and all 37 prime factors were re-certified by APR-CL, PARI `isprime` flag 2), and the fixed-d certificate closes each one: at $d = 201$, 209, 249, 251, 281, 297 every split factor is excluded at step 2 — odd $\mathrm{ord}_r(2)$ or composite half-order; none reaches the condition-(v) test.

The multiplicative orders were computed by explicit descent from completely factored $r-1$ (four stubborn cofactors of 37–70 digits were factored by ECM for this purpose; every piece was certified prime and the products verified). *No compatible triple exists at* $d = 201$*,* 209*,* 249*,* 251*,* 281*, or* 297. Certificates: `scripts/cp356_nct_certificates_200_300.py`, `scripts/cp356_aprcl_recert.py`. The value $d = 243$ was closed separately: its 125-digit Ψ_{972} fell entirely to ECM ($B_1 \leq 11{\times}10^6$, about half an hour of a 12-thread workstation), $\Psi_{972} = 3 \cdot 971 \cdot 15114601 \cdot 2068037891 \cdot 935133871378249 \cdot p_{35} \cdot p_{55}$, all seven factors APR-CL-certified; the three split factors all have odd $\mathrm{ord}_r(2)$, so no compatible triple exists at $d = 243$ (certificate `scripts/cp358_nct_certificate_243.py`). The value $d = 211$ fell the same way: its 161-digit Ψ_{844} was completely factored by ECM in about twenty minutes ($146857 \cdot 338779913 \cdot 52213127964665809 \cdot 307892103084433531 \cdot p_{33} \cdot p_{81}$, all APR-CL-certified; every split factor excluded at step 2; certificate `scripts/cp358_nct_certificate_211.py`). The window $(300, 400]$ contains nine parity-unblocked values. Six — $d = 307$, 321, 347, 361, 387, 393, with Ψ_{4d} of 163 to 290 digits — are completely factored in the public tables; the same re-verification (exact products against locally recomputed Ψ_{4d}, 46 factors APR-CL, orders by descent from completely factored $r-1$) and the same certificate close all six; and $d = 363$'s 169-digit Ψ_{1452} fell to a single ECM round ($11 \cdot 2352241 \cdot 69796187 \cdot p_{17} \cdot p_{25} \cdot p_{36} \cdot p_{78}$, seven factors APR-CL, every split factor with composite half-order; certificate `scripts/cp358_nct_certificate_363.py`): *no compatible triple exists at any of these fifteen values* (certificates `scripts/cp358_nct_certificates_range.py`). Two further values fell to a Pell half-split: for odd d the Pell–Lucas number factors as $V_{2d} = 8P_d^2 - 2 = 2\,(2P_d - 1)(2P_d + 1)$ with coprime odd halves, so any cofactor of $\Psi_{4d} \mid V_{2d}$ splits by a gcd with the halves into two pieces of roughly half the digit size. At $d = 227$ the ECM-stalled $C143$ cofactor split as $p_{84} \times p_{60}$, completing $\Psi_{908} = 907 \cdot p_{84} \cdot 57203 \cdot 5580569 \cdot 16552486912925363 \cdot p_{60}$ (six factors APR-CL; the two split factors have odd $\mathrm{ord}_r(2)$, excluded at step 2): *no compatible triple exists at* $d = 227$ (certificate `scripts/cp367_nct_certificate_227.py`). At $d = 379$ the halves brought the $C290$ primitive part down to ECM scale, completing $\Psi_{1516} = 4547 \cdot p_{141} \cdot 351030947611 \cdot p_{29} \cdot p_{30} \cdot p_{75}$ (six factors APR-CL, exact product identity; both split factors, p_{30} and p_{141}, have odd $\mathrm{ord}_r(2)$): *no compatible triple exists at* $d = 379$ (certificate `scripts/cp372_nct_certificate_379.py`). The last two parity-unblocked values of $(200, 400]$ fell to the same half-split followed by GNFS on the single remaining half-cofactor (`cado-nfs`, ECM to $t35$ first). At $d = 283$ the L-half residual $C104$ factored as $p_{45} \cdot p_{59}$, completing $\Psi_{1132} = 12451 \cdot p_{45} \cdot p_{59} \cdot 6793 \cdot 11489975459 \cdot 21850712277340062427 \cdot p_{75}$ (seven factors APR-CL, exact product identity; the three split factors are excluded — 6793 by composite half-order, p_{45} by $v_2(\mathrm{ord}_r(2)) \geq 2$, and p_{59} by odd $\mathrm{ord}_r(2)$): *no compatible triple exists at* $d = 283$ (certificate `scripts/cp375_nct_certificate_283.py`). At $d = 331$ the M-half residual $C106$ factored as $p_{36} \cdot p_{71}$, completing $\Psi_{1324} = 6619 \cdot 43691 \cdot$

$4510867 \cdot 224936385721 \cdot p_{22} \cdot p_{78} \cdot p'_{22} \cdot p_{36} \cdot p_{71}$ (nine factors APR-CL, exact product identity; all five split factors excluded — odd $\mathrm{ord}_r(2)$, $v_2 \geq 2$, or composite half-order): *no compatible triple exists at $d = 331$* (certificate `scripts/cp376_nct_certificate_331.py`). With these, every parity-unblocked value of $(200, 400]$ is closed: *NCT holds unconditionally for every odd $d \leq 400$.*

A $d \leq 1000$ primitive-divisor survey on the inert side finds five danger triples; three are phantom because the corresponding W_p is prime, and the two real triples are closed at secondary factors as described below. The survey is empirical context only; Theorem 7.2 uses the Platinum enumeration rather than this catalog.

B.4. **Inert enumeration and local passes.** Among 869 known inert factors $r \equiv 3 \pmod 8$ with $r < 2 \times 10^5$, Condition (II) fails at every factor. In the extended Platinum enumeration, every inert prime factor $r \leq 10^{12}$ of every composite W_p with $p < 5 \times 10^{11}$ was enumerated by the candidate form $r = 2pk + 1$. The exponent-indexed scan over k reaches $r = 10^{12}$ at each p; the small-exponent band $p < 5 \times 10^5$, where this requires k as large as $\sim 10^{12}/2p$, was completed by a dedicated sub-scan (41,498 primes, no additional local pass), so the coverage $r \leq 10^{12}$ is exhaustive across the full exponent range. The run recorded 684,965,381 pairs (p, r):

Outcome	Count	Fraction
$G_r \leq 43$	669,378,360	97.72%
order-excess exclusion	15,587,018	2.28%
local-pass witnesses	3	$< 10^{-6}$%

The three local-pass witnesses occur at distinct exponents:

$$\begin{array}{rll} 1{,}251{,}488{,}009 & r = 2p + 1 & d = 41{,}716{,}267, \\ 10{,}916{,}765{,}939 & r = 152{,}834{,}723{,}147 & d = 38{,}208{,}680{,}787, \\ 85{,}684{,}865{,}933 & r = 2p + 1 & d = 57. \end{array}$$

Of the three Platinum local-pass witnesses, two are now discharged at secondary factors. The $d = 57$ witness, at $p = 85{,}684{,}865{,}933$, overlaps the $d \leq 1000$ primitive-divisor catalog and is closed by the secondary factor below. The $d = 41{,}716{,}267$ witness, at $p = 1{,}251{,}488{,}009$, is closed by a secondary factor found by a directed scan of the progression $r = 2pk + 1$ with $k \leq 10^{11}$. The third witness exponent $p = 10{,}916{,}765{,}939$ remains undischarged — the analogous directed scan at this exponent, complete for $k \leq 10^{12}$, finds no second factor (the witness factor itself, at $k = 7$, is the only hit in the progression $r = 2pk + 1$), its full W_p cofactor is not resolved here, and the expected failure is recorded as the at-a-factor statement X1 (Conjecture 6.29). Separately, the $d \leq 1000$ primitive-divisor catalog contains a second real triple at $d = 67$, also closed at a secondary factor:

$$\begin{aligned} d = 57: \quad & r_2 = 7{,}675{,}646{,}348{,}774{,}307{,}083, \\ & \text{inert; parity obstruction} \\ & (d_{r_2} = 3 \cdot 487 \cdot 1{,}313{,}423{,}399{,}858{,}711), \\ d = 67: \quad & r_2 = 5{,}474{,}333{,}343{,}676{,}555{,}561{,}818{,}313, \\ & \text{split; R3 verified directly,} \\ d = 41{,}716{,}267: \quad & r_2 = 33{,}829{,}735{,}778{,}964{,}491 \ (k = 13{,}515{,}805), \\ & \text{inert; parity obstruction} \\ & (d_{r_2} = 3 \cdot 41 \cdot 68{,}759{,}625{,}567{,}001). \end{aligned}$$

These secondary closures are certified computational facts (Proposition 6.30 records the two witness-exponent discharges and their verification); they are not counted as unconditional entries in Table 1. The local-pass witness rows above remain part of the enumeration record; their discharge status is as stated. The census of Proposition 5.18 is the d-indexed complement of this r-indexed record: at composite W_p, the only locally passing inert factors with $d_r \leq 400$ — for r of *any* size — are the $d = 57$ Platinum witness and the $d = 67$ catalog row, both closed above.

B.5. **Software and supplementary material.** Integer factorization uses SymPy/gmpy2 routines, Cunningham tables [5], and the online factor database [19] only to locate candidate factors; every asserted prime factor used in finite exhaustion is independently certified by APR-CL in PARI/GP [9], and every assembled factorization is re-verified by an exact product identity. BPSW and deterministic Miller–Rabin are preliminary screens only. The supplementary deposit records factor tables, Platinum enumeration outputs, direct Condition-(II) checks, and the longer closed-approaches log omitted from this lean draft.

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